Silicon-nitride photonic circuits for steering optical tweezers in a neutral-atom processor: simulation study

Volume I: synthesised report

Simulation study for proposal ANRF/ARG/2026/009937 (Phase A, Objective 3)

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Abstract

We simulate the silicon-nitride (SiN) photonic circuits that are to steer the transport tweezers (1064.4 nm) and switch the addressing beams (894.6 nm) of a Cs neutral-atom processor, and the system built from them, up to arrays of 10^6 sites. The 400 nm SiN platform meets the proposal's component targets; two targets need restating: the steering update rate should be ≥ 200 kHz for moves at 1.2 mK, and position noise should be specified as a spectral density at the trap frequency. Independent motion of about 20 tweezers along a row requires a sliding-window focal-plane phased array. Its optimised design (128 ports, 192 emitters at 1.5 µm pitch) needs about 57 mW per 1.2 mK tweezer once facet loss and a glued collimator are included. Chips stacked in a 16-row anamorphic relay give independent rows on a 4 µm lattice without per-chip frequency shifters, using common phase noise on the seed laser and fibre delays. In three-dimensional transport with handoff, 3,000 of 3,000 simulated atoms are delivered along lanes between the rows, with 0.12 µK of heating per handoff (about 0.3 µK per move) at a moving depth of 0.3 mK when the ramps last 200 µs. At 10^6 sites, a block of 128 engines holds a time-averaged defect fraction of about 2×10^{-4}, set by vacuum and imaging losses. The main limitation found is the cylindrical collimator: light that forms tweezers away from the row centre detunes it, so in its present refractive form one engine serves only the central half (±45 µm) of its 180 µm segment. All values are simulation outputs or derivations within stated models and should be reviewed before use.

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1 Introduction

A neutral-atom processor holds its qubits in an array of optical tweezers. Loading is stochastic, so the array must be sorted after every load and repaired after every loss, and the operations themselves need atoms moved between sites. Proposal ANRF/ARG/2026/009937 [1] plans to do the steering of the transport tweezers, and the switching of the addressing beams, with silicon-nitride (SiN) photonic integrated circuits (PICs) instead of acousto-optic deflectors and spatial light modulators. The proposal fixes the working points. Transport tweezers at 1064.4 nm (a double-magic wavelength for the Cs $45S_{1/2}$ Rydberg state) and addressing at 894.6 nm (the Cs D1 line) are focused by a meta-objective of numerical aperture ≥ 0.5 onto a 4 µm lattice. The transport depth is about 1.2 mK, with about 6.6 mW per tweezer at the atoms. The targets for the circuits are a steering update rate of ≥ 100 kHz, switch extinction of ≥ 50 dB at 894.6 nm, a position noise of ≤ 10 nm rms, and a corrected field of ± 100 µm. The waveguide platform is the 400 nm stoichiometric LPCVD SiN available to the project.

Whether such circuits can do the job is not obvious, for three reasons. (i) An optical phased array (OPA) steers all of its tweezers together. Independent motion of many tweezers along a row, as with the tones of an acousto-optic deflector, needs a different engine. (ii) A chip forms one row. Rows of independent engines must be stacked and relayed onto a lattice of 4 µm period without the rows disturbing each other. (iii) The proposal's array is small, while tweezer arrays of 2,088 and 6,100 sites have been reported [2, 3]. An architecture built on chips must therefore also be judged at 10^4 to 10^6 sites. Each question also carries a hard physical limit: the handoff of atoms between static and moving traps, heating per move, laser power per tweezer, and losses that compete with repair.

In this work, we simulate the PIC elements and the system around them, from waveguide modes to the steady state of a 10^6-site array. The models are scalar or vector wave optics, beam propagation, 2D FDTD, ray tracing, classical and quantum transport, and stochastic scheduling of moves and losses. Every number carries its provenance. Values from the proposal are marked USER, values from sources read during the work SOURCE, simulation outputs SIM, calculations DERIVED, stated assumptions CHOICE, and values recalled but not verified RECALLED. All simulated values are valid in the regime simulated and should be reviewed before they enter the proposal. The code, results and module-by-module documents are in the repository. Volume II compiles those documents without rewriting, and Appendix B maps every result to its script.

The report is organised by physical question. Section 2 covers the waveguide platform and the passive and active components. Section 3 develops the steering engine for independent tweezers. Section 4 stacks engines into rows and treats their mutual independence. Section 5 asks how far the tweezer count per engine and per row can grow. Section 6 simulates transport and handoff in three dimensions, and Section 7 schedules rearrangement and repair. Section 8 scales the architecture to 10^4, 10^5 and 10^6 sites. Section 9 combines chip stacks into larger blocks and replaces the ideal collimator by the real one, which exposes a limit of the cylindrical collimator for tweezers away from the row centre. Section 10 separates what was shown from what is model-dependent, and states the main limitations and next steps.

2 Platform and passive components

This chapter reports the Phase-A simulations of the individual SiN elements named in the proposal [1], from the waveguide cross-section to the system-level behaviour of the optical phased array (OPA), and the error-budget links that follow from them.

2.1 Scope and methods

The waveguide platform is stoichiometric LPCVD Si_3N_4, 400 nm thick (the CoE-CPPICS limit), with 200 and 300 nm films as foundry alternatives, clad in SiO_2. Material dispersion is taken

from Luke et al. [4] and Malitson [5], and the thermo-optic coefficients (2.45×10^{-5} /K for Si_3N_4, 1×10^{-5} /K for SiO_2) from [6]. The working wavelengths are 894.6 nm (Raman addressing on the Cs D1 line) and 1064.4 nm (transport tweezers).

Each element is modelled with the cheapest method that captures its physics: (i) vector FEM mode solving (femwell 0.1.12, second-order Nédélec elements) for straight, bent, coupled and leaky guides; (ii) effective-index reduction with guided-mode eigenmode expansion for MMIs; (iii) scalar overlap with Gaussian fibre modes for edge couplers; (iv) 2D FDTD (MEEP 1.35, 40 px/µm) for grating antennas, with the Si substrate included; (v) 2×2 transfer matrices with Monte Carlo sampling for switches; (vi) 2D thermal FEM for heaters; (vii) 1D scalar Fourier optics for the OPA (emitter near field, Fourier-plane slit, aplanatic objective); and (viii) an exact harmonic-oscillator result for transport heating. The mode, slab, MZI, OPA and transport models are checked against analytic limits or published examples (16 unit tests); the FDTD closes its energy balance to within 3% at the centre wavelength (7% across the band), and the heater result changes by 1.4% when the domain is doubled.

The principal limits are as follows. The MMI and grating models are 2D. Edge-coupler numbers are mode overlaps without Fresnel or taper conversion loss. The OPA model is scalar and 1D along the steering axis, with an aberration-free objective. Sidewall scattering loss and fabrication roughness are not modelled. The AlN actuator is a single mechanical mode with assumed resonance frequency and quality factor. Every simulated value below is valid only within the regime simulated; the provenance of each input (proposal value, source, derived, design choice or recalled) is listed in Appendix A.

2.2 Waveguide platform

We choose the design width as the widest width on a 0.1 µm grid for which (i) the TE1 mode is cut off or lies within 0.015 of the cladding index, and (ii) the TE0/TM0 effective-index splitting is at least 0.015. The second condition matters on the thick film: at 894.6 nm a 0.4×0.4 µm^2 guide is polarisation-degenerate (TE0 = TM0 = 1.693), which would couple the polarisations in bends. Hence we use 0.5 µm there. Table 1 lists the resulting designs.

Table 1: Single-mode design guides (simulated, vector FEM). Here, n_{eff} and n_g are the effective and group indices of TE0, $\mathrm{d}n_{\text{eff}}/\mathrm{d}T$ is for uniform heating of core and cladding, and R is the smallest swept bend radius with less than 0.01 dB per 90° bend (20 µm was the smallest swept, hence ≤20).

film (nm)	λ (nm)	width (µm)	n_{eff}	n_g	$\mathrm{d}n_{\text{eff}}/\mathrm{d}T$ (10^{-5}/K)	R, TE / TM (µm)
400	1064.4	0.6	1.708	2.14	2.36	8 / 10
400	894.6	0.5	1.745	2.17	2.41	6 / 6
200	1064.4	0.9	1.613	1.98	2.02	≤20 / 60
200	894.6	0.7	1.651	2.05	2.16	≤20 / 30

Substrate leakage on the 400 nm film is below 10^{-5} dB/cm for TE with a buried oxide of 2 µm or more at both wavelengths; TM at 1064.4 nm leaks 7×10^{-5} dB/cm at 2 µm, and 3 µm keeps both polarisations below 10^{-5} dB/cm. On the 200 nm film at 1064.4 nm the oxide must be at least 3 µm (TE leakage 7×10^{-8} dB/cm); at 2 µm, TM leaks 0.33 dB/cm. The coupling length of two 1064.4 nm guides on the 400 nm film is 1.8 mm at 1.6 µm pitch and 29 mm at 2.1 µm pitch, which sets how closely routing guides can run in parallel. At 1.4 W in one guide the peak in-guide intensity is 420 MW/cm^2 and the Kerr phase 0.05 rad/cm (with a Kerr index of 2.5×10^{-19} m^2/W, recalled, to be verified).

The thick film confines both polarisations well. On 400 nm film the TM bend loss is only about ten times the TE loss at the same radius, hence bends cannot strip TM light. On 200 nm film at 1064.4 nm, in contrast, a 20 µm bend radiates 1.5 dB per 90° of TM against 5×10^{-4} dB of TE,

which makes a compact TM filter possible; at 894.6 nm the same film gives only 0.03 dB per 90° of TM. This matters for the switch extinction (Sec. 2.4).

2.3 Edge couplers and splitters

Inverse tapers on the 400 nm film reach a mode overlap of −0.37 dB with a 2.5 µm lensed fibre at 1064.4 nm (150 nm tip, ±0.6 µm lateral tolerance at −1 dB) and −0.68 dB with HI1060 fibre (125 nm tip). At 894.6 nm the best values are −0.46 dB (lensed) and −1.1 dB (PM780). Near cutoff the 100 nm tip mode at 1064.4 nm decays over 4.8 µm, so the 7 µm computational window of the sweep clipped it; with a converged window it gives −0.87 dB with HI1060 (not −0.48 dB), while the other optima change by less than 0.04 dB. The fibre mode-field diameters used (HI1060 6.2 µm, PM780 5.3 µm) are recalled, to be verified against datasheets. Several optima lie at the narrowest tip swept (100 nm), so narrower tips may do better if lithography allows. These overlaps exclude Fresnel and taper conversion loss and are therefore lower bounds on the 1 to 2 dB per facet assumed in the proposal. At 1.4 W through one 400 nm facet the intensity is 45 MW/cm^2; three input fibres keep it below 20 MW/cm^2. We have no CW facet-damage threshold for SiN at 1 µm ([VALUE?]).

A 4 µm wide 1×2 MMI has 0.044 dB excess loss at both wavelengths (lengths 14.8 µm at 1064.4 nm and 17.7 µm at 894.6 nm), so a 6-stage, 64-way tree costs 0.27 dB, against the ∼1 dB assumed. A symmetric 1×2 MMI is balanced by symmetry; imbalance arises only from asymmetric errors, 0.13 dB for a 20 nm lateral offset of the input guide. With 1σ errors of 10 nm (width), 5 nm (thickness) and 5 nm (offset), the 2×2 split ratio varies by 0.0024 (894.6 nm) and 0.0034 (1064.4 nm). Our first-pass 894.6 nm 2×2 design sits at 0.490 rather than 0.500; this offset is removable by redesign and is carried separately below.

2.4 Switches, router and phase shifters

The proposal asks for ≥ 50 dB extinction at 894.6 nm. A single balanced MZI with a phase error δ has an extinction $\mathrm{ER} = -20\log_{10}(\delta/2)$, so 50 dB needs $\delta \leq 6.3$ mrad. In practice the split-ratio spread of the MMIs dominates. Table 2 and Fig. 1 compare three architectures with the simulated fabrication spread (0.0024) and a 0.5% rms arm amplitude imbalance.

Table 2: Simulated MZI switch extinction at 894.6 nm (Monte Carlo, split-ratio spread 0.0024, arm amplitude imbalance 0.5% rms). Here, P is the fraction of devices reaching ≥ 50 dB; "trimmed" means an input splitter trimmed to match the output coupler.

Architecture	phase error 3 mrad			phase error 10 mrad		
	median (dB)	5th pct (dB)	P	median (dB)	5th pct (dB)	P
single MZI	49.8	41.4	0.49	45.3	38.8	0.21
single MZI, trimmed	55.1	48.8	0.89	44.6	38.3	0.18
two cascaded MZIs	100	88	1.00	92	81	1.00

A cascaded pair, as in [7], reaches ≥ 50 dB in every trial, with a 5th percentile of 88 dB. The 0.010 design offset of the first-pass 2×2 does not change this: it is shared by both couplers of an MZI, and two identical couplers still null exactly, so with the offset as a shifted mean the figures are unchanged ($P = 0.49$ for a single MZI, 88 dB for the cascade). An incoherent stray-light floor of −70 dB caps the cascade at 70 dB. For the AlN half-wave voltage at 894.6 nm, which we bracket between 84 V (λ-scaled from 74.7 V at 795 nm [7]) and 132 V (interpolated to 200 V at 1013 nm [7]), the 6.3 mrad limit of a single MZI requires driver noise below 0.17 to 0.27 V rms. The measured V_π at 894.6 nm remains open ([VALUE?]).

Hence the binding constraints are polarisation and free-space crosstalk, not the interferometer. TM light is not nulled when the TE path is. For an input polarisation extinction of 20 dB, 50 dB

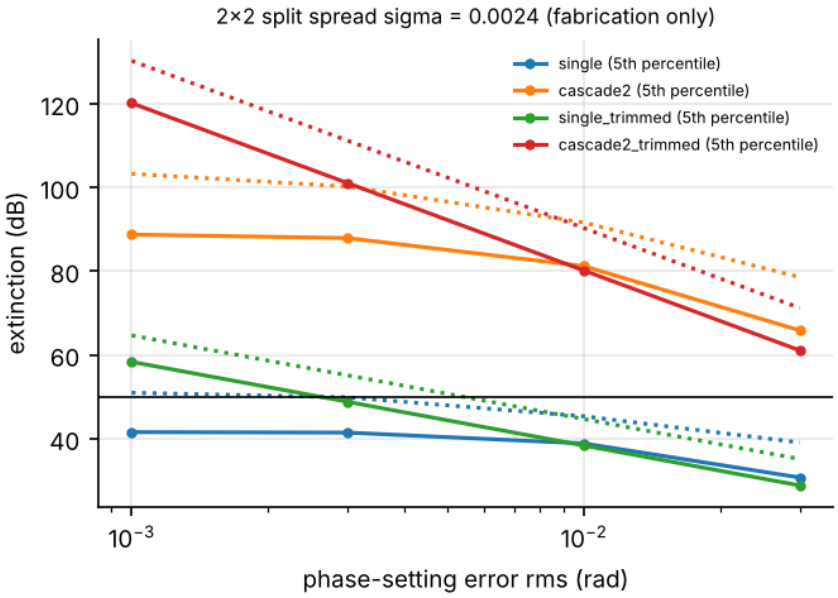


Figure 1: Extinction of single, trimmed single, cascaded and trimmed cascaded MZI switches at 894.6 nm against the rms phase-setting error, from 2×2 transfer matrices with 4000 Monte Carlo trials per point. Solid lines with markers: 5th percentile; dotted lines: median. The 2×2 split ratio varies with the simulated fabrication spread $\sigma = 0.0024$ only (no design offset); arm amplitude imbalance 0.5% rms. The horizontal line marks the 50 dB target.

at the output needs 24 to 30 dB of TM suppression after the chip; for 25 dB input, 19 to 25 dB. On 400 nm film bends cannot provide this (Sec. 2.2), hence a polariser in the free-space delivery path or a dedicated on-chip TE-pass element is required.

The settling of the switch is set by mechanical ringing of the AlN actuator, not by its 26 ns optical rise [7]. For an assumed single mode at $f_0 = 20$ MHz and $Q = 10$ ([VALUE?] for the real values), an unshaped step needs 0.98 µs to reach −50 dB after switch-off. A two-impulse (ZV) shaper gives 24 ns, but 0.43 µs with a 2% error in f_0; a three-impulse (ZVD) shaper gives 49 ns with a 2% error and 0.19 µs with 5%, at a 28 ns optical rise. Thus settling within 100 ns needs ZVD shaping with f_0 known to about 2%.

The 16-output router (four stages of thermo-optically tuned MZI splitters) leaks at most 1.4×10^{-3} into the worst unaddressed output (95th percentile) for a 30 mrad trim error, so with the cascaded output switch the idle-site extinction exceeds 110 dB (116 dB at 30 mrad). Its insertion loss is 5.0 dB, of which 3 dB is the assumed free-space delivery, so 16 sites at 20 µW need about 1 mW at the input.

Two families of phase shifter serve different time scales. Thermo-optic shifters with a 2 µm wide heater 2 µm above the core need 84 mW for π at 1064.4 nm (68 mW at 894.6 nm), with a 34 µs rise and 16% thermal crosstalk to a guide 10 µm away. Isolation trenches reduce these to 54 mW and 2%, and an additional undercut to 3 mW, at a 1.5 ms rise. The thermal conductivities and heat capacities in this model are recalled values, to be verified. Thermo-optic shifters suit slow router trims but not OPA updates at ≥ 100 kHz. For the OPA, the PZT shifters of [8] (V_π of 2.8 V at 532 nm, about 5.6 V at 1064.4 nm by wavelength scaling, 0.4 MHz bandwidth) settle in about 2 µs, within a 5 µs update period; their hysteresis is not known ([VALUE?]).

2.5 System-level behaviour of the OPA

Range and resolution. The atom-plane field is a scaled copy of the slit-filtered OPA far field. The number of resolvable positions along the steering axis equals the channel number N, so the range at the atoms is $N\lambda/(4\,\mathrm{NA}\,F)$. Here, NA is the objective numerical aperture and F the pupil fill. Table 3 gives the consequences at NA 0.5 and 1064.4 nm. At full fill, 64 channels steer over

±33 µm (17 sites at 4 µm pitch), not ±100 µm, and cannot place more than about 15 tweezers on one row. More than 20 simultaneous tweezers on one row need 128 channels; at 0.75 fill they cover ±89 µm at 1.33 times the power per tweezer. Uniform channel amplitudes give 4.7% first sidelobes along the steering axis; a Gaussian channel apodisation ($1/e^2$ at the aperture edge) lowers them to 0.9% at a FWHM of 1.09 µm. Section 3 builds on this count.

Table 3: Simulated steering field of the OPA (1D scalar Fourier optics, NA 0.5, 1064.4 nm). Here, the half-range is the usable one, in which the steered peak keeps at least half its on-axis value and sits within 0.1 µm of its target; "sites" is the number of 4 µm sites on one row within it; and "power" is the relative power for equal depth.

N	fill F	half-range (µm)	sites	FWHM along x (µm)	power
32	1.0	16	9	0.95	1.00
64	1.0	33	17	0.95	1.00
64	0.5	67	33	1.89	1.99
128	1.0	67	33	0.94	1.00
128	0.75	89	45	1.26	1.33
256	1.0	134	67	0.94	1.00

Emitter type and pitch. A Fourier-plane slit passes only the central diffraction zone, so the efficiency into it falls with emitter pitch: 0.65 at 1.5 µm, 0.52 at 2 µm and 0.34 at 3 µm for the 0.6 µm emitter guide (a wider emitter mode at the facet raises these, Sec. 3). The minimum pitch is set by coupling between neighbouring emitters over their parallel length, for less than −20 dB power transfer: 1.46 µm for an end-fire facet with an assumed 50 µm parallel section, against 1.30, 1.56 and 1.82 µm for gratings with 200, 100 and 50 nm etch (parallel lengths 20, 90 and 384 µm). The gratings radiate 0.76, 0.72 and 0.68 upward at 8.9 to 10.0°, but their directionality oscillates with oxide thickness through the substrate reflection (0.42 to 0.76 over 2 to 4 µm), the emission angle moves by −0.10°/nm, and the 200 nm etch reflects 6% back into the guide. Our choice of end-fire emitters is a compromise between two considerations. First, end-fire allows a dense pitch with short parallel sections, has no directionality loss and points independently of wavelength. However, it emits in plane and needs a relay to turn the beam towards the objective. The grating array becomes attractive only if a bottom reflector or two-layer grating is available.

Simultaneous tweezers by phase synthesis. With weighted Gerchberg-Saxton alone ([REF?], recalled reference to be verified), light at untargeted lattice sites reaches 11 to 21% of the tweezer intensity. A refinement that also penalises stray light and off-centre spots holds it at 3 to 7%: for $N = 64$ with 8 or 12 tweezers the maximum stray is 0.056 and 0.053 and the maximum peak non-uniformity $(I_{\max} - I_{\min})/(I_{\max} + I_{\min})$ is 0.062 and 0.061; for $N = 128$ with 16 or 21 tweezers the stray is 0.054 and 0.057 and the non-uniformity 0.106. About 0.45 of the emitted power reaches the tweezers in all cases (four random target sets per case, one row along the steering axis). A 5.7% stray at a static site corresponds to about 70 µK under a 1.2 mK transport tweezer. This stray floor is one reason the steering engine of Section 3 avoids shared-aperture synthesis.

Phase accuracy. For independent channel phase errors of rms σ, the position jitter at the atoms is $\sigma\sqrt{12/N}\,\lambda/(4\pi\,\mathrm{NA})$, which Monte Carlo reproduces to 2%. Hence a 10 nm position error needs $\sigma \le 0.14$ rad at $N = 64$ (0.10 rad at 32, 0.19 rad at 128). Thus 8-bit phase resolution (7 mrad) is sufficient, and a PZT hysteresis error must stay below about 4% of the applied phase at $N = 64$.

Update rate. A tweezer moved by a staircase of OPA updates is excited because the staircase aliases the move spectrum to $|f_{\mathrm{update}} - f_{\mathrm{trap}}|$, where it can overlap the trap frequency. At the

1.2 mK transport depth the radial trap frequency is 84 kHz for $w_0 = 1.04$ µm. As can be seen from Fig. 2, a 100 kHz update then adds about 40 motional quanta to a 100 µs, 4 µm move and about 2,900 quanta to a 50 µs move, against 0.005 and 0.2 for a smooth trajectory. At 200 kHz and above the staircase adds nothing measurable. Hence the ≥ 100 kHz target of the proposal holds for 0.3 mK tweezers (42 kHz) but should read ≥ 200 kHz for moves at 1.2 mK. Alternatively, lowering the depth to ≤ 0.4 mK during moves (48 kHz) with move times of ≥ 100 µs keeps 100 kHz: a 4 µm move then adds 0.025 quanta (0.008 for a smooth trajectory).

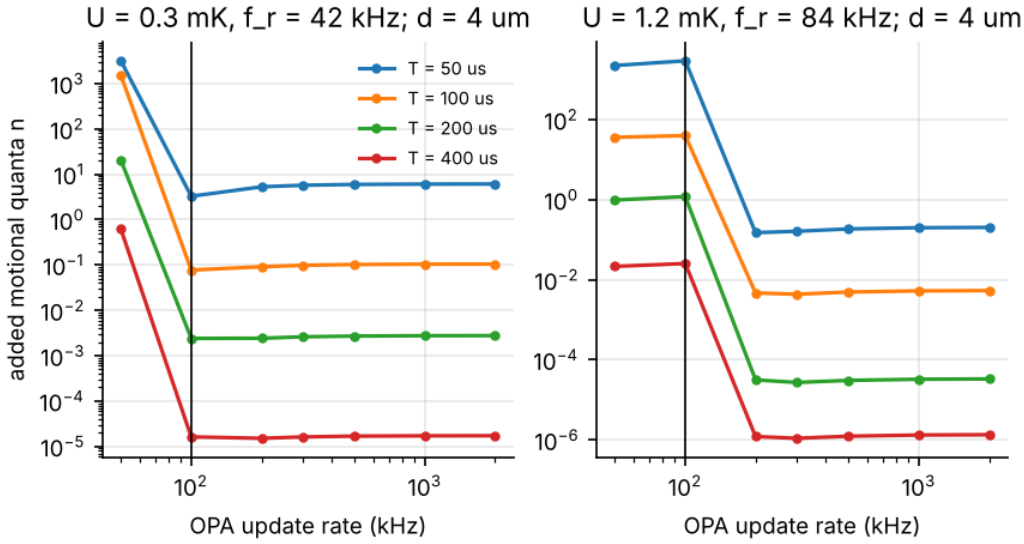


Figure 2: Motional quanta added to an atom by a 4 µm OPA move against the update rate, for move durations T of 50 to 400 µs, at 0.3 mK ($f_r = 42$ kHz, left) and 1.2 mK ($f_r = 84$ kHz, right). Exact harmonic-oscillator result on the radial axis, atom initially in the ground state, minimum-jerk trajectory sampled and held at the update rate, $w_0 = 1.04$ µm, polarisability 1160 a.u. The vertical line marks the 100 kHz target of the proposal. The curves are nearly flat from 200 kHz at 1.2 mK and from 100 kHz at 0.3 mK.

Loss budget. For an end-fire OPA with $N = 128$, 21 tweezers and 2 µm pitch, the chain from fibre to atoms is: edge coupler −0.37 dB, 7-stage MMI tree −0.31 dB, propagation over 1 cm at 0.3 dB/cm (assumed) −0.30 dB, emitters with slit, pupil and synthesis −3.43 dB, and meta-objective transmission 0.7 (assumed) −1.55 dB, in total −5.95 dB. The grating case adds −1.2 dB of directionality (−7.15 dB for the best grating emitters). At 1.5 µm pitch the slit term improves by about 1 dB. Against the ∼10 dB assumed in the proposal, the per-tweezer on-chip power falls from 66 mW to about 26 mW. This estimate excludes sidewall scattering.

2.6 Phase-A error budget

Idle-qubit error and trap depth. The proposal's idle-qubit error for a switch of extinction ER is $(\pi^2/4)\,10^{-\mathrm{ER}/10}$, which gives 2.5×10^{-5} at 50 dB when one Raman beam is addressed; we reproduce it. The proposal's trap depth is not reproduced: 6.6 mW in a Gaussian of $w_0 = 1.04$ µm with a polarisability of 1160 a.u. gives 1.01 mK, not 1.2 mK; 1.2 mK needs 7.8 mW, or $w_0 = 0.96$ µm. The OPA spot along the steering axis is 22% brighter per watt than that Gaussian, which brings 6.6 mW to about 1.24 mK.

Addressing crosstalk. Neighbour crosstalk of the 894.6 nm beams is set by the objective point-spread function, not by the chip. With a uniformly filled NA 0.5 pupil, the spot puts 1.0×10^{-3} of its peak intensity on the 4 µm neighbour, which gives a neighbour error of 2.5×10^{-3}, above the 10^{-3} budget. As can be seen from Fig. 3, a Gaussian pupil fill truncated at 0.7 of the pupil radius

brings this to 1.2×10^{-4} (error 2.9×10^{-4}) at a spot FWHM of 1.13 µm instead of 0.92 µm. The model is aberration-free; a meta-objective with a Strehl ratio of 0.8 adds a halo that these figures do not include.

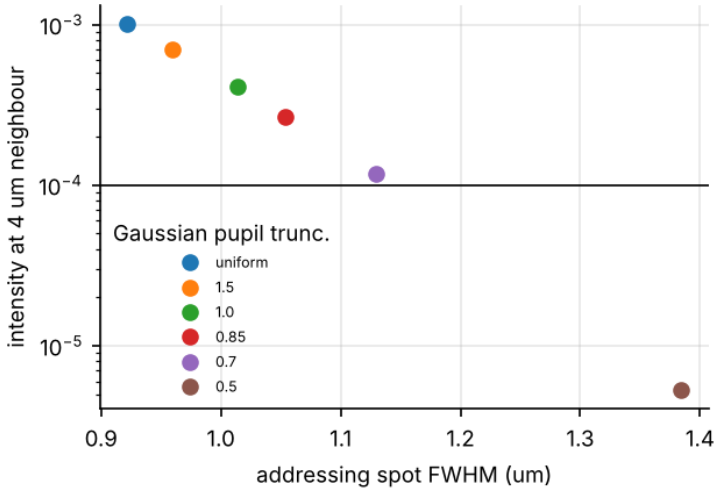


Figure 3: Intensity at the 4 µm neighbour against the FWHM of the 894.6 nm addressing spot, for a uniform pupil and for Gaussian pupil fills truncated at 1.5, 1.0, 0.85, 0.7 and 0.5 of the pupil radius. Scalar Debye point-spread function, NA 0.5, aberration-free. The horizontal line marks 10^{-4}.

Position and intensity noise. The budget statement "position noise $\leq$ 10 nm rms keeps transport survival $\geq$ 99%" does not fix the heating rate, because heating depends on the noise spectral density at the trap frequency [9]: $\dot{Q} = 4\pi^4 M \nu^4 S_x(\nu)$. Here, M is the Cs mass (recalled, to be verified), ν the trap frequency and S_x the position-noise spectral density. If 10 nm rms is white over 1 MHz, a 1.2 mK tweezer heats at about 7,700 quanta/s, about 0.8 quanta per 100 µs move against a depth of about 300 quanta. Heating below 10 quanta/s needs $S_x(84\ \text{kHz}) \leq 1.3 \times 10^{-25}$ m^2/Hz. In the same way, from $\Gamma = \pi^2 \nu^2 S_I(2\nu)$ [9], the relative intensity noise must satisfy $S_I(2\nu) \leq 1.4 \times 10^{-11}$ /Hz at 1.2 mK for a parametric heating time of 1 s.

Power handling. At 1.4 W into the input guide, with 0.3 dB/cm fully absorbed, the guide warms by 3.8 K and accrues 5.3 rad/cm of common-mode phase before the split. After the 64-way split each channel accrues at most 0.08 rad/cm, which the per-channel calibration absorbs.

Consequences for the specification. The simulations imply six changes to the proposal's targets: (i) an OPA update rate of $\geq$ 200 kHz for transport at 1.2 mK, or $\leq$ 0.4 mK during moves of $\geq$ 100 µs; (ii) a tweezer count tied to the channel count, with the steering field (±33 µm at $N = 64$, ±68 µm at $N = 128$, $F = 1$) stated separately from the ±100 µm meta-objective field; (iii) about 6 dB (end-fire) to 7 dB (grating) of modelled chip and OPA loss, with 10 dB kept as the conservative value; (iv) position noise specified as $S_x(\nu_{\text{trap}}) \leq 1.3 \times 10^{-25}$ m^2/Hz, together with the 894.6 nm pupil apodisation needed for neighbour crosstalk $\leq 10^{-4}$; (v) 7.8 mW, or the OPA spot, for 1.2 mK; and (vi) cascaded MZIs (or trimmed single MZIs with $\leq$ 3 mrad phase accuracy), an output polariser of $\geq$ 30 dB on the 894.6 nm path, and ZVD-shaped drive for $\leq$ 100 ns settling. Item (ii) is superseded in its architectural part by the steering engine of Section 3.

3 The steering engine

This chapter derives the on-chip engine that moves twenty or more tweezers independently along one row, compares it with the switched-port star coupler, and gives the beam-propagation design D1 with its simulated errors, power cost and limits.

3.1 The requirement and what an AOD supplies

Section 2.5 showed that the number of resolvable positions of an OPA along its steering axis equals its channel number N: 64 channels place at most about 15 tweezers on a row of 4 µm sites, and more than 20 need 128 channels. Count alone is not the requirement, however. Rearrangement needs each tweezer to move on its own, as the tones of an acousto-optic deflector (AOD) do. An AOD owes this to two physical ingredients (standard AOD physics, [REF?] for a citable source): (i) frequency tagging, since each RF tone shifts its diffracted light by its own frequency, so that tweezers from different tones add incoherently on average and each tone's amplitude sets one tweezer's depth; and (ii) a time-delay medium, since the acoustic transit time across the aperture converts each tone's frequency into its own angle, and the number of resolvable spots is the RF bandwidth times the transit time. The independence is per axis. A single AOD moves its spots independently along one line, and crossed AODs give a grid of row and column tones rather than independent 2D positions (ref. [17] of the proposal [1]). Hence the fair target for an integrated engine is a 1D AOD along a row.

A shared-arm OPA, in which every tweezer uses every arm, has neither ingredient, and phase-only synthesis couples all tweezers. We tested this directly on a 128-channel OPA with 20 synthesised tweezers. Removing one of them by warm-started re-synthesis (that site added to the dark points) leaves 1.9 to 3.3% of the tweezer intensity at its site (-15 to -17 dB), changes the 19 kept tweezers by up to 12%, and along a linear phase path between the two patterns the kept tweezers deviate by up to 19%. Stray light at untargeted sites stays at about 2%. Such an engine cannot gate 894.6 nm light at 50 dB, and it perturbs every other atom whenever one tweezer is switched.

Table 4 lists the routes we considered for supplying an equivalent of at least one AOD ingredient. Route E, the sliding-window focal-plane OPA, is the only one that works at a single optical frequency, uses only components already in the proposal (SiN star coupler, MZI trees, PZT or AlN shifters) and needs no new physics. Complex synthesis on the arms (A′) is a fallback with fewer optical elements, but needs amplitude modulation on every arm and power headroom, because moving one tweezer changes the optimum. The wavelength-addressed route (D) is excluded by the chromatic focal shift of a diffractive meta-objective unless that objective is made achromatic over several nanometres around 1064 nm.

3.2 The switched-port star coupler and its limits

Before route E we designed an engine that keeps per-tweezer switching but gives up independent motion. Each tweezer has its own input port and MZI switch; the ports illuminate a common 128-arm phased array through a free-propagation region (FPR) of radius R; and a phase ramp on the arms moves all lit tweezers together. One chip per wavelength carries (i) edge-coupled input, (ii) a 1×32 tunable-splitter tree that sends power only to lit ports, so that the power per tweezer does not depend on how many are lit, (iii) 20 port switches (thermo-optic at 1064.4 nm, two cascaded AlN MZIs at 894.6 nm), (iv) the FPR, (v) 128 arms, 1.7 µm wide at the FPR edge (0.3 µm gaps), each with a phase shifter, and (vi) 128 end-fire emitters at 1.5 µm pitch, followed by the slit-filtered relay and NA 0.5 meta-objective; a dichroic combines the two chips. Port j imposes a linear phase $k_s d_j p_a / R$ per arm, which steers its beam to $\sin\theta_j = n_s d_j p_a / (R\, p_{\mathrm{out}})$. Here, k_s and n_s are the slab wavenumber and index, d_j is the port offset, p_a the arm pitch at the FPR and p_{out} the emitter pitch. The relay maps $x = \sin\theta\, N p_{\mathrm{out}} / (2\,\mathrm{NA})$, so a 4 µm step is $\sin\theta = 0.0208$ for $N = 128$ and $p_{\mathrm{out}} = 1.5$ µm.

Each port illuminates all arms with the far field of its taper mode, which apodises the array. The tails of this envelope set the light a lit tweezer leaves on its neighbours and the interference between lit ports, whose relative phases drift with the router temperature. Table 5 shows the trade-off: a standard star coupler ($1/e^2$ at the outer arms) leaves 1.2×10^{-3} on the neighbour and 19% peak spread, an order of magnitude above the 10^{-4} target. A narrower envelope needs a smaller FPR radius, which brings the ports closer together; widening the input taper restores the radius, and its mode, closer to Gaussian, lowers the tails further. The final designs use 1.4 µm

Table 4: Routes to independently steered tweezers on one row (20 tweezers). Efficiencies are simulated (E, A′, F) or derived (D, G, T). The surface-acoustic-wave velocity used for route G (∼5 km/s) is recalled, to be verified for an AlN/SiN/SiO_2 stack.

Route	Source of independence	Efficiency	Show-stopper or cost
E: sliding-window focal-plane OPA	tweezers on separate ports (position basis), linear superposition	0.40 at centre, 0.19 at field edge	127 fast splitters and 128 phase shifters; ∼1% per-port accuracy
A′: complex synthesis on the arms	linear superposition of 20 phase ramps (amplitude and phase per arm)	0.75 (peak-to-mean 1.34)	128 amplitude and phase modulators; power headroom
F: AOD emulation on the arms	one RF tone per tweezer: frequency tags and per-tone phase ramps	0.44 (peak-to-mean 2.28)	256 RF-driven modulators; intermodulation ghosts from modulator nonlinearity
D: wavelength-addressed dispersive OPA	optical frequency tags and delay lines	high (passive)	focal shift 14 µm/nm allows only ±71 pm (±19 GHz); 5 GHz per site needs 6.7 m of delay lines
G: integrated surface-acoustic-wave deflector	a literal AOD in the slab	unknown	128 spots need 640 MHz of RF bandwidth on 1 mm, 128 MHz on 5 mm; photoelastic response of SiN not established
T: time-multiplexed OPA	time-averaged potentials	1 per slot	17 MHz pattern rate, 60 ns slots, 20 times the peak intensity

tapers at 1.89 µm pitch with $R = 223$ µm at 1064.4 nm, and 1.0 µm tapers at 1.64 µm pitch with $R = 197$ µm at 894.6 nm, both at a nominal e^{-6} envelope (realised 2.7×10^{-5} and 7.2×10^{-6}, because the wide-taper modes are not Gaussian). Two 1.4 µm tapers 0.49 µm apart have a coupling length of 1.33 mm, so their ∼20 µm convergence transfers well below −20 dB. We keep this cleaner design at 1064.4 nm although an e^{-5} envelope gives 9% more depth per watt, because its 3% peak spread and 4×10^{-5} neighbour light would make tweezer depths depend on drifting port phases. With $N = 96$ the half-range would be 51 µm (1064.4 nm) or 43 µm (894.6 nm), leaving only 5 to 13 µm of collective travel for a 20-site row (±38 µm); $N = 128$ gives ±68 and ±57 µm. Table 6 summarises the final designs.

Table 5: Effect of the arm-power envelope of the star coupler at 1064.4 nm (simulated, 2D paraxial FPR, 0.6 µm input taper). "Spread" is the largest variation of peak intensity with all 20 ports lit at random relative phases.

outer-arm envelope (nominal)	FWHM (µm)	neighbour / peak	spread, 20 lit
e^{-2}	1.10	1.2×10^{-3}	19%
e^{-4}	1.24	4.2×10^{-4}	11%
e^{-6}	1.38	9.6×10^{-5}	5.8%
e^{-8}	1.52	1.3×10^{-5}	2.0%

The collective move is one linear ramp: a single analog waveform drives the 128 arm shifters with fixed gains $n = 0 \ldots 127$, plus slow per-arm calibration offsets, and a continuous ramp has no staircase, which removes the update-rate heating of Sec. 2.5. With all 20 lit, collective shifts of 4 and 24 µm change the peaks by −2.5/+2.3% and −16.7/+10.3%, following the emitter element factor; this is pre-compensated through the port powers. With equal port phases the outer spots carry a static offset of up to 25 nm at 1064.4 nm (5 nm at 894.6 nm), which changes by less than 2 nm over a 24 µm sweep. Its cause is the tails of the other ports' fields, not the element factor

Table 6: Simulated performance of the switched-port star-coupler OPA (128 emitters, 20 ports on a row of 4 µm sites). Fibre power assumes a 1.04 µm Gaussian profile along the cross axis, 0.3 dB/cm propagation and meta-objective transmission 0.7 (both assumed).

	1064.4 nm (transport)	894.6 nm (addressing)
spot FWHM along the row	1.52 µm	1.27 µm
intensity at the 4 µm neighbour / peak	1.0×10^{-6}	3.5×10^{-6} (error 8.5×10^{-6})
peak spread, all 20 lit, random port phases	$\leq 0.73\%$	$\leq 0.89\%$
stray at unlit sites (10 of 20 lit)	$\leq 1.2 \times 10^{-5}$	$\leq 2.1 \times 10^{-5}$
static position error: one port lit / all 20 lit, random port phases (rms, max)	$\leq$ 6 nm / 16, 42 nm	$\leq$ 5 nm / 8, 20 nm
switch per port	one trimmed thermo-optic MZI	two cascaded AlN MZIs
extinction needed	$\geq$ 31 dB (1 µK residual at 1.2 mK)	$\geq$ 50 dB (idle error 2.5×10^{-5})
fibre power	33 to 43 mW per tweezer; 0.74 W for 20	$\leq$ 0.14 mW per site; 2.4 mW for 20

(one port lit alone is off by at most 6 nm), so it depends on the relative port phases: with random phases every spot moves, by 16 nm rms and up to 42 nm at 1064.4 nm (8 nm rms, 20 nm at 894.6 nm; 50 draws). The port phases must therefore be stabilised, or the error calibrated per phase state; unstabilised, the engine exceeds the 10 nm position budget at 1064.4 nm. At 894.6 nm the switch leak equals the 3.5×10^{-6} optical tail of a lit neighbour at 54.6 dB, well inside the 88 dB (5th percentile) of the cascaded pair; at 1064.4 nm a trimmed thermo-optic MZI exceeds the 31 dB needed (5th percentile 38 dB at 10 mrad, computed at 894.6 nm and to be repeated at 1064.4 nm).

The limits of this engine are structural. All lit tweezers move by the same displacement, so a sorting sequence must be built from parallel moves, as with AODs, although at one optical frequency, so that two tweezers brought together do not beat. The two chips need about 380 electrical channels, against the 32 to 64-channel OPA and 16-channel router of the proposal, and the 51 heaters of the 1064.4 nm chip draw about 2.1 W without trenches (about 0.1 W with trenches and undercut). The FPR model is 2D and paraxial. We retain this design for the 894.6 nm addressing beams (Sec. 3.3), where its per-port cascaded switch sets the off-state.

3.3 Sliding-window focal-plane OPA (route E)

Route E replaces the few switched ports by a dense row of 128 input ports, one per resolvable spot. The FPR radius is $R = Np_a p_{\mathrm{port}}/\lambda_s$, which makes port k form a spot at $x_k = k\lambda/(2\,\mathrm{NA})$, i.e. 1.064 µm per port at the atoms. Here, p_{port} is the port pitch and λ_s the wavelength in the slab. A tweezer is a Gaussian-weighted group of about five ports and moves by sliding this window. In this position basis different tweezers use different ports, so their fields add linearly and interfere only through their tails. In addition, two tweezers can approach without interference if their relative phase is set to $\pi/2$, since the cross term of two real fields then vanishes; this is a control an AOD lacks, where approaching tones beat at their frequency difference. The first, paraxial geometry had 128 ports 1.7 µm wide at 2.0 µm pitch, $R = 886$ µm, 128 arms (1.7 µm wide at 2.0 µm pitch) and 128 end-fire emitters at 1.5 µm pitch. Adjacent ports at 0.3 µm gaps have a coupling length of 625 µm, so the $\sim$20 µm over which they converge transfers below -20 dB.

The window width σ (in ports) sets the spot apodisation. Our choice of $\sigma = 0.9$ is a compromise between two considerations. First, a narrow window gives a small spot and hence more depth per watt. However, it leaves a larger position nonlinearity and peak ripple as the window slides between ports (Fig. 4): 18 nm and 3.4% at $\sigma = 0.8$, 8.6 nm and 1.5% at 0.9, 3.7 nm and 0.55% at 1.0, and 0.5 nm and 0.07% at 1.2, at spot FWHMs of 1.51, 1.66, 1.82 and 2.16 µm and neighbour intensities

at 4 µm of 1.1×10^{-4}, 2.0×10^{-5}, 2×10^{-6} and 7.6×10^{-5}. Both residuals are deterministic in the window position and are removed by a lookup table, together with the fall-off of delivered power from 0.40 at the centre to 0.19 at ±64 µm.

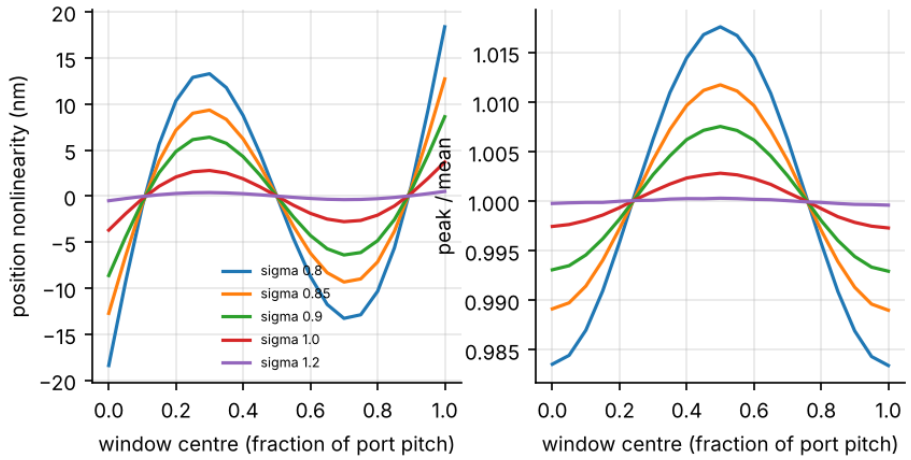


Figure 4: Uncalibrated response of a sliding Gaussian window in the first (paraxial) route-E design, as the window centre moves across one port pitch, for window widths $\sigma = 0.8$ to 1.2 ports. Left: deviation of the spot position from a linear map. Right: peak intensity relative to its mean. 2D paraxial FPR with isolated port and arm modes, $N = 128$ ports at 2.0 µm pitch (1.7 µm wide), $R = 886$ µm, 128 arms, emitter pitch 1.5 µm, NA 0.5, 1064.4 nm. Both curves are periodic in the port pitch and are removed by a lookup table.

With 20 tweezers at random positions over ±60 µm and $\sigma = 0.9$ (calibrated), a tweezer moved by 0.1 to 4 µm lands within 0.9 nm of its target with a peak change of at most 0.10%, while the other 19 shift by at most 1.5 nm and 0.19%. Removing a window leaves 8.9×10^{-6} of the tweezer intensity at its site and changes the others by at most 0.08%. The static peak spread of the 20 is 1.4% with depth equalised by the tree. Two tweezers in quadrature at 2.5 µm pull each other by 3.5 nm with a 0.16% peak change, against 146 nm and 9.5% in phase.

The cost lies in the actuators and their accuracy: 127 fast tunable splitters and 128 port phase shifters per chip. Because each tweezer is set by only about five ports, per-port errors average little. Per-port amplitude and phase errors of 0.3%/3 mrad, 1%/10 mrad and 3%/30 mrad rms give position noise of 3.1, 9.4 and 28 nm and peak noise of 0.5, 1.9 and 5.6%. Thus 10 nm needs about 1% and 10 mrad, about 15 times tighter than the arm-side OPA (0.14 rad at $N = 64$, Sec. 2.5), and PZT hysteresis ([VALUE?]) must be corrected, for example by closed-loop calibration against on-chip taps. A moving tweezer is updated through the tree, so the staircase result applies: ≥ 200 kHz updates at 1.2 mK, which PZT shifters (0.4 MHz [8]) meet and thermo-optic shifters (34 µs rise) do not. The node settings follow in closed form from the desired port amplitudes, since each node splits in the ratio of the powers of its two subtrees, so an FPGA can compute them in real time. A tweezer is switched by ramping its window through the tree adiabatically (≥ 100 µs), and its off-state is then set by the extinction of the tree nodes.

This last point decides the 894.6 nm engine. Raman gates act on atoms in static traps, and the addressed Raman leg must keep a fixed frequency relative to the global leg, so frequency tagging is unavailable at 894.6 nm. Route E there would give steerable addressing with an off-state set by the tree nodes, rather than by a cascaded switch per beam. The switched-port star coupler of Sec. 3.2 therefore remains the addressing engine, and route E the transport engine at 1064.4 nm.

3.4 Design D1 by beam propagation

The paraxial model above treats ports and arms as isolated and the FPR as aberration-free. Dense ports at 0.3 µm gaps couple, and an AWG-type star coupler needs a full propagation design. Since the 400 nm film is uniform and fully etched, every guide and the slab share one vertical mode, and

a lateral effective-index model carries the star-coupler physics. The plain effective-index model overestimates lateral confinement: its coupling lengths are 31 to 36% short of the 3D vector-FEM values. We therefore calibrate the lateral core and cladding indices (1.8423 and 1.3551) to the vector-FEM effective indices of 0.6, 1.0 and 1.7 µm guides (residuals $\leq 4.3 \times 10^{-4}$) and to the coupling length of the dense ports (622 against 625 µm); a held-out check on 1.4 µm guides at 0.49 µm gap agrees to 3%, and narrow, widely spaced guides (off by 29%) couple negligibly in the design. Propagation uses a 2D wide-angle (Padé (1,1)) Crank-Nicolson beam-propagation method (BPM) with the TM-type lateral operator that matches the chip TE mode, all ports propagated together ($\Delta x = 0.04$ µm, $\Delta z = 0.125$ µm; capture converged to 0.5%). On a reduced 8-port, 8-arm coupler, 2D FDTD and BPM agree in arm field with fidelity 0.997 (centre port) to 0.93 (edge port, rising to 0.981 at finer FDTD resolution) and in capture to 2 to 5%, the residual being dominated by FDTD discretisation. For fast optimisation an exact-path Rayleigh-Sommerfeld model reproduces the full 128×128 BPM to within 1% in capture and with arm-field fidelity 0.996 to 0.9996 (0.96 at the two extreme ports, where taper coupling matters); on the optimised design both give the same field-averaged efficiency to 0.6%.

The propagation revealed two effects absent from the paraxial model. (i) The confocal coupler (ports on an arc of radius R) is aberrated: each off-axis port's arm phases deviate from a straight line by up to 0.90 rad peak-to-peak (0.22 rad rms) at the edge port in the beam propagation (0.41 rad half-way out); exact path lengths, which ignore the finite acceptance of the arms, give 1.3 rad at the edge. Arm trims cannot remove this because it depends on the port. Placing the ports on an arc of radius $0.555R$, close to the Rowland circle, halves it over the field of use, and with 192 arms the worst residual is 0.08 rad rms; the different piston phase this gives each port is removed by static port trims. (ii) Neighbouring ports exchange light in the converging tapers. With fixed Gaussian windows this broadens the spot tails, and moving one of 20 tweezers then changes the others by up to 7 nm and 1.1%. Both effects are linear and static, which the synthesis of Sec. 3.5 exploits.

We then optimised the geometry for the field-averaged laser power per tweezer at equal depth, i.e. the harmonic mean of the peak intensity per unit port power over $|x| \leq 60$ µm, under the constraints of a neighbour intensity at 4 µm of $\leq 10^{-4}$, gaps ≥ 0.3 µm, and 128 ports (the fast actuators) fixed. The roll-off across the field has one origin on both sides of the chip, scan loss: the arms capture tilted light from off-axis ports, and the emitters radiate tilted beams, each with an element pattern that falls across its angular range. Two levers act on it (Fig. 5). Oversampling, i.e. more arms and emitters than ports, confines the ports to the central part of the arms' angular range and raises the edge capture from 0.44 to 0.54 with 192 arms. Arms and emitters carry only static trims in route E, so this costs area and slow heaters, not fast actuators; 256 arms would add 7% for 64 more trims and a 1.8 mm slab. Wider emitter facets narrow the element pattern and raise the relay transmission at the centre from 0.64 (0.6 µm) to 0.84 (1.2 µm at 1.5 µm pitch). With 0.5 µm gaps the transmission saturates near 0.73 to 0.76 for any pitch from 1.5 to 2.2 µm, so the 0.3 µm gap of the 1.2 µm facet buys the last 10%; its coupling length of 227 µm limits the wide facet section to about 15 µm. At the slab, the widest ports and arms (1.7 µm at 2.0 µm pitch) give a field-averaged efficiency of 0.261 against 0.18 to 0.24 for 1.2 to 1.5 µm widths, because fill factor outweighs the slightly better angular acceptance of narrow arms. Narrowing the window from 0.9 to 0.85 ports gains 5% but raises the neighbour intensity to 4.8×10^{-5}; we adopt least-squares synthesis with a 1.68 µm target instead.

The resulting design D1 has 128 input ports and 192 arms, both 1.7 µm wide at 2.0 µm pitch at the slab (0.3 µm gaps); a slab radius $R = 1329$ µm; ports on an arc of radius $0.555R$ through the slab vertex, pointing at the arm-array centre; arms on an arc of radius R with 30 µm tapers to 0.6 µm guides; 192 end-fire emitters at 1.5 µm pitch with 1.2 µm facets (facet taper ≤ 15 µm); static phase trims on all arms and ports; and port amplitudes from least-squares synthesis.

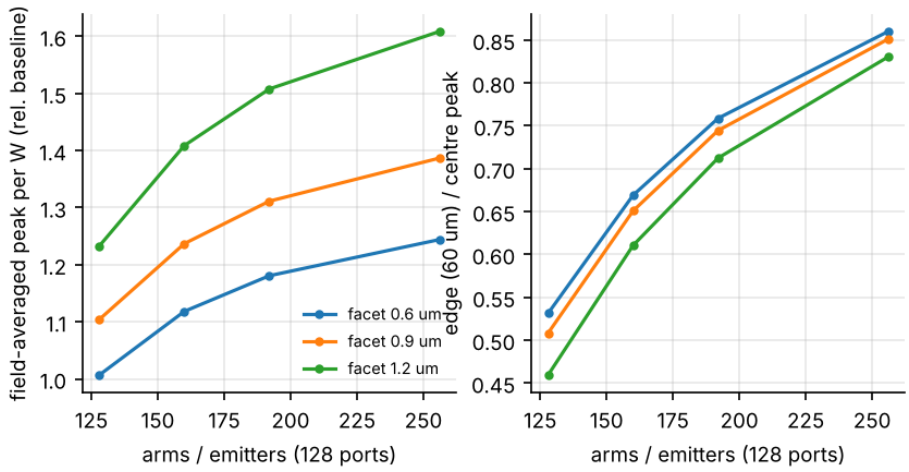


Figure 5: Geometry optimisation of the route-E star coupler with 128 ports, against the number of arms and emitters, for emitter facets of 0.6, 0.9 and 1.2 µm at 1.5 µm pitch. Left: field-averaged peak intensity per unit port power (harmonic mean over $|x| \leq 60$ µm), relative to the confocal 128×128 design with 0.6 µm facets. Right: peak at the field edge ($x = 60$ µm) relative to the centre. Exact-path (Rayleigh-Sommerfeld) star-coupler model, ports on a Rowland-type arc of radius $0.555R$, ports and arms 1.7 µm wide at 2.0 µm pitch, Gaussian window $\sigma = 0.9$ ports, NA 0.5. D1 is the 192-arm, 1.2 µm-facet point (gain 1.51, edge ratio 0.71).

3.5 Least-squares port synthesis and residual errors

Instead of fixed Gaussian windows, we compute the port amplitudes by least squares on the calibrated port-to-atom response, with the target spots as the objective. Because port coupling and aberration are linear and static, the synthesis removes them, at a cost of 1 to 2% in efficiency, and it equalises depth over the field without a lookup table. Each update is one product with a precomputed $128 \times n$ matrix. Table 7 compares D1 with the first (confocal 128×128) design, both verified by BPM of the full structure with the calibrated effective-index model and least-squares synthesis.

Table 7: Simulated performance of the route-E engine: first confocal design against the optimised design D1 (full BPM, least-squares port synthesis, 20 tweezers over ±60 µm). Laser power uses facet −0.37 dB, 7-stage tunable tree −0.39 dB, 1 cm at 0.3 dB/cm and meta-objective transmission 0.7 (both assumed), and a 1.04 µm Gaussian profile across the row; see the text for the revised value.

quantity	first design (128×128, confocal)	D1
field-averaged peak per unit port power (relative)	1.00	1.50
edge (60 µm) / centre peak	0.52	0.70
residual arm-phase aberration, worst port	≤ 1.3 rad pk-pk	≤ 0.08 rad rms
others' change when one of 20 moves by up to 4 µm	≤ 3.5 nm, $\leq 0.12\%$	≤ 3.5 nm, $\leq 0.11\%$
residual at a switched-off site	1.3×10^{-6}	1.6×10^{-6}
static peak spread of 20 tweezers	0.56%	0.52%
spot FWHM along the row	1.71 µm	1.71 µm
laser power per 1.2 mK tweezer, field mean (fibre)	∼65 mW	∼43 mW
20 tweezers over ±60 µm	∼1.3 W	∼0.86 W

Under full propagation with synthesis the others' displacement is up to 3.5 nm rather than the 1.5 nm of the paraxial windows (Sec. 3.3), while the peak change, switch-off residual and peak

spread are smaller. The per-port actuator accuracy, about 1% and 10 mrad for 10 nm of position noise, is not relaxed by the synthesis and remains the main requirement on the hardware.

3.6 Power per tweezer

The loss chain from fibre to atoms is the facet (−0.37 dB), the 7-stage tunable tree (−0.39 dB), 1 cm of propagation at 0.3 dB/cm (assumed, −0.3 dB), the simulated FPR, arms, emitters, slit and pupil, and a meta-objective transmission of 0.7 (assumed). In the first paraxial design a 1.2 mK tweezer needs 48 mW at the fibre at the field centre and 91 mW at the edge (60 µm); equalising depth costs a mean factor of 1.27, so 20 tweezers spread over ±60 µm need about 1.2 W, and restricting operation to ±40 µm lowers the mean factor. D1 brings the field mean to about 43 mW (0.86 W for 20), a factor 1.4 in efficiency. However, both figures omit the cross axis and the emitter facet loss (about 21% into air). With both included and a collimator glued to the facets, a 1.2 mK tweezer needs about 57 mW (`docs/stacked_relay.md`, `docs/collimator.md`); this is the current value. For comparison, the switched-port star coupler, computed with the same Gaussian cross-axis profile, needs 33 to 43 mW per tweezer, and the proposal assumes 1.4 W for 21 tweezers.

3.7 Limits

The model is 2D with a calibrated effective index. The residual 3D effects, polarisation coupling at sidewalls and slab-edge scattering, are not computed, and the 3D excess loss of the dense-port FPR is open ([VALUE?]). Fabrication variation of the 0.3 µm gaps and of the taper geometry has not been simulated; static errors are absorbed by the trims and the measured response matrix, but the arm phase errors after fabrication, which set the trim range, are not known ([VALUE?]). The relay and objective are scalar and aberration-free, so a meta-objective with a Strehl ratio of 0.8 adds a halo not included in the crosstalk figures. Steering is along one row; the second axis needs a second engine or an OPA-AOD hybrid. The hardware load of D1 is 255 fast actuators (127 tree splitters and 128 port phase shifters, which also carry the port trims) and 192 static arm trims, with PZT shifter footprint, hysteresis and long-term drift after closed-loop calibration still open ([VALUE?]). The procedure that measures the port-to-atom response on the device, on which the least-squares synthesis rests, has yet to be defined.

4 Rows of engines: the stacked-chip relay

One focal-plane engine steers one row of tweezers along x. This section describes how 16 chips are stacked and imaged onto the atoms as 16 independent rows, what the collimator in front of each chip must do, and how the rows are made mutually incoherent without one frequency shifter per chip.

Sixteen chips of design D1 (128 input ports, 192 emitters, Sec. 3), stacked at 0.5 mm pitch, each with its own fast-axis collimator (FAC), are imaged as 16 rows 4 µm apart through the NA 0.5 objective at 1064.4 nm. With the adopted collimator, a glued Weierstrass block followed by a two-asphere aplanat, a 1.2 mK tweezer needs 57 mW. Rows are as independent as the tweezers within a row provided their relative optical phase is randomised much faster than the trap period, which one noise-driven phase modulator on the common seed plus different fibre delays per chip achieves. The costs are a collimator alignment of ±0.1 µm in height, a diffraction-limited collimator at NA 0.9, and two rules on filters and reflections in the laser chain. All values are simulated (scalar wave optics; ray tracing and 2D FDTD where stated) or derived from those simulations.

4.1 Anamorphic layout

In x, the relay of Sec. 3 images the emitter plane onto the objective pupil, so the atoms see the far field of the chip. In y, each facet sits in the front focal plane of its own cylindrical collimator.

The collimators' back focal planes form a plane S, which a common y lens group of effective focal length F_y Fourier transforms onto the pupil. Thus S is imaged onto the atoms with magnification $f_{\mathrm{obj}}/F_y = 1/D$, and chip m lands at $y_m = m\,d_c/D$. Here, d_c is the chip pitch, D the demagnification of the stack, and $f_{\mathrm{obj}} = 17.3$ mm the objective focal length (working distance 15 mm). At the same time, the pupil is an image of the facet, with magnification 60 in x and 12,400 in y. Because each chip has its own collimator, the 16 facet images overlap in the pupil and the rows are separated only by the tilt y_m/F_y; a single common collimator would place the facet images side by side. The free parameter of the y design is the pupil radius referred back to the facet, $\beta = \mathrm{NA}\, f_{\mathrm{FAC}}/D$. Here, f_{FAC} is the collimator focal length and NA = 0.5. The row pitch fixes D, and β then fixes f_{FAC} (Table 8).

Table 8: First-order layout (thin lenses, ABCD matrices of both axes checked) for 16 chips at 0.5 mm pitch, $\beta = 0.7$ µm, FAC NA 0.9. Derived. The design point is 4 µm.

row pitch	D	F_y	f_{FAC}	collimated half-height	facet to pupil
3 µm	167	2.89 m	233 µm	210 µm	439 mm
4 µm	**125**	**2.17 m**	**175 µm**	**158 µm**	**382 mm**
5 µm	100	1.73 m	140 µm	126 µm	347 mm
6 µm	83	1.44 m	117 µm	105 µm	324 mm

At 4 µm, F_y is realised as a 25× demagnifying y telescope (100 mm and 4 mm) followed by an 86.6 mm Fourier lens. The x relay is a 4f pair (3.12 mm and 187.7 mm) with its Fourier slit 6.2 mm from the facets, and both pupils coincide 382 mm from the facets. The Lagrange invariant matches (stack half-height 4 mm times collimated NA 0.0022 equals atom half-field 30 µm times 0.5 times the fill), and the collimated beams fit within the 250 µm half-pitch. This layout is first order; a lens design of the anamorphic relay over its 8 mm field has not been made.

4.2 Facet mode, pupil fill and crosstalk floor

Validity of the y results. All y-axis results of this section (pupil fill, collimator, row crosstalk, tolerances) were computed for light with no x direction component, $s_x = 0$. However, a tweezer at x_a along the row is made of light with $s_x = x_a/288$ µm (D1), and the cylindrical collimator then acts with effective indices $\sqrt{n^2 - s_x^2}$. The adopted collimator therefore defocuses the facet by $-99\,s_x^2$ µm. The y figures below hold near the row centre only: the peak falls to 0.97 at $x_a = 30$ µm and to 0.48 at 60 µm, and with the best common focus offset a row is usable over $|x_a| \le 45$ µm, shorter than the ±60 µm row of D1. Section 9 gives the details.

The facet. The D1 emitter facet (1.2×0.4 µm SiN in oxide) has a vertical mode of $1/e^2$ equivalent radius 0.39 µm and radiates into air with a far-field $1/e^2$ half-angle of $\sin\theta = 0.86$. A collimator of NA 0.8 captures 96% of this light, one of NA 0.9 captures 98.8%. Thinning the SiN at the facet to 100 nm lowers the far-field NA to 0.51, but it also expands the lateral mode: the coupling length of adjacent emitters (0.3 µm gap) falls from 228 µm to 47 µm, which leaves no room for an adiabatic taper. We therefore keep the 400 nm facet and accept the alignment tolerances it sets, which scale with the vertical mode radius (decentre) and its square (focus).

Pupil fill. The y profile at the atoms is the facet's far field, clipped by the collimator aperture, scaled by β/NA and low-pass filtered by the pupil. Crosstalk between rows has two sources: (i) at small β, the pupil clips the sharply peaked facet image; (ii) at large β, the hard edge of the collimator aperture rings through the pupil. As can be seen from Table 9 and Fig. 6, the two balance near $\beta = 0.7$ µm, which we adopt. A collimator of NA 0.95 lowers the crosstalk at 4 µm to 7.4×10^{-5}; a graded collimator edge costs 7 to 8% of the light and reaches only 9.4×10^{-5}. None of these options gets below about 5×10^{-5} at 4 µm, which is adequate for mutually incoherent

rows but not for coherent ones (Sec. 4.4). At $\beta = 0.7$ µm the y spot is 1.31 µm FWHM, against 1.22 µm for the Gaussian cross axis assumed before. With the 2.7% cross-axis loss and a 3% loss to the circular pupil, the power per tweezer rises by a factor 1.10, from the 43 mW of the engine design to 47 mW at 1.2 mK.

Table 9: Pupil fill scan, ideal FAC of NA 0.9, single tweezer, 1D in y, $s_x = 0$. Power relative to a Gaussian cross axis of $w_0 = 1.04$ µm. Simulated.

β	cross-axis transmission	y FWHM	power vs Gaussian	row crosstalk at 3 / 4 µm	row crosstalk at 6 / 8 µm
0.5 µm	0.946	1.09 µm	0.92	3.5×10^{-3} / 1.8×10^{-3}	7.2×10^{-4} / 3.9×10^{-4}
0.6 µm	0.968	1.18 µm	0.97	1.2×10^{-3} / 6.2×10^{-4}	2.5×10^{-4} / 1.3×10^{-4}
0.7 µm	**0.973**	**1.31 µm**	**1.07**	$\mathbf{2.8 \times 10^{-4}}$ / $\mathbf{1.1 \times 10^{-4}}$	$\mathbf{3.2 \times 10^{-5}}$ / $\mathbf{1.5 \times 10^{-5}}$
0.8 µm	0.973	1.49 µm	1.22	1.1×10^{-3} / 2.6×10^{-4}	4.9×10^{-5} / 1.9×10^{-5}
1.0 µm	0.977	2.08 µm	1.65	2.1×10^{-3} / 6.4×10^{-4}	1.9×10^{-4} / 9.3×10^{-5}

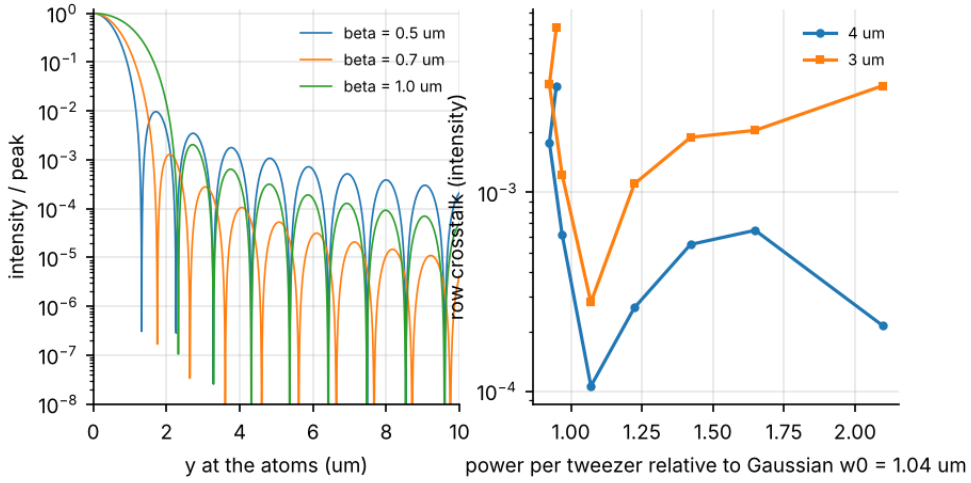


Figure 6: Pupil fill and row crosstalk for an ideal FAC (NA 0.9, $D = 125$), scalar wave model, 1D in y, $s_x = 0$ (valid near the row centre, Sec. 9). Left: y intensity at the atoms relative to its peak for $\beta = 0.5$, 0.7 and 1.0 µm. Right: crosstalk at the neighbouring row (3 and 4 µm pitch) against the power per tweezer relative to a Gaussian cross axis ($w_0 = 1.04$ µm), for β from 0.5 to 1.0 µm; the minimum at 1.07 is $\beta = 0.7$ µm.

4.3 The adopted collimator and the facet loss

The collimator must bend rays from $\sin\theta = 0.9$ (64° in air) to the axis within a lens no taller than the 0.5 mm chip pitch. Single refracting surfaces fail: a plano-hyperbola collimates only up to $\sqrt{1 - 1/n^2}$ (0.72 in fused silica, 0.83 at $n = 1.8$); a plano-asphere reaches NA 0.9 but meets rays at up to 43° (fused silica) or 64° ($n = 1.8$); and an ellipse on a glued block collimates exactly but violates the sine condition (the ray at $s = 0.9$ exits at 68% of the aplanatic height). Here, n is the lens index and $s = \sin\theta$ at the facet. The Weierstrass construction resolves this. A circular surface of radius R on a block glued to the facet, with the facet at R/n from its centre, is aplanatic and lowers the NA by n^2 (0.9 to 0.43 in fused silica). A two-asphere aplanat of focal length $f_{\mathrm{FAC}} n^2 = 368$ µm (Wassermann-Wolf construction) then collimates. A scan of 96 combinations (29 reaching NA 0.9) selected $R = 80$ µm, a 10 µm gap, a 250 µm aplanat and $n = 1.8$, with incidence angles of at most 25°. The traced lens is 395 µm long, ±158 µm tall, and meets the sine condition to 4×10^{-5} with no residual wavefront error (Table 10).

Facet loss. The engine model treated the emitter facet as lossless. A 2D FDTD (MEEP, resolution 40/µm, core index 1.965 chosen to match the 3D mode index 1.7995) shows that the facet reflects 12.8% into the chip and transmits 79.1% into air, of which 77.2% of the guided power falls within the collimator aperture (Kirchhoff estimate 72.3%). Into fused silica, as for the glued block, it reflects 2.3% and transmits 95.9%, with 85.2% within the aperture (Kirchhoff 82.9%). The emitter facet thus loses about a quarter of the light into air, a loss absent from all earlier power figures; gluing the collimator recovers about 10% and cuts the back-reflection sixfold. The FDTD is 2D, with lateral confinement folded into the core index.

Table 10: Collimator designs in the wave model (FDTD-scaled facet, AR-coated surfaces, NA 0.5 pupil, $s_x = 0$). Peak is per unit guided power relative to an ideal lens in air at $\beta = 0.7$ µm; power is per 1.2 mK tweezer, field mean. Simulated.

design	sine condition at $s = 0.3$ / 0.6 / 0.9	β (µm)	peak	crosstalk at 3 / 4 µm (10^{-4})	power
ideal lens, air	1 / 1 / 1	0.7	1.00	2.8 / 1.1	61 mW
Weierstrass + aplanat	1.000 / 1.000 / 1.000	0.7	**1.10**	4.0 / 0.80	**57 mW**
ellipse, printed	0.954 / 0.834 / 0.675	0.9	1.03	6.6 / 1.8	61 mW
ellipse, printed	0.954 / 0.834 / 0.675	0.7	1.22	21 / 10	
plano-asphere $n = 1.8$, air	0.988 / 0.960 / 0.971	0.7	1.03	4.0 / 1.7	59 mW

The aplanatic design reproduces the ideal lens in glass (peak 1.10; a 2D check gives 1.107; y FWHM 1.35 µm). The ellipse narrows the spot but its distortion raises the crosstalk, and opening its fill to $\beta = 0.9$ µm costs 15% in peak. Without AR coatings all designs lose 10 to 30%. Thus the power per 1.2 mK tweezer is 57 mW with the adopted collimator, against 47 mW with the facet loss omitted and 61 mW for an ideal lens in air. The first x lens of the relay ($f = 3.12$ mm, field ±144 µm) needs no such care: its static aberration is a fixed linear map of the chip's response, which the least-squares synthesis absorbs. After recalibration an aplanat (0.15 waves rms at the field edge) gives an edge-tweezer peak of 1.001 with a shift of at most 1.5 nm, and even a lens without the sine condition (0.60 waves rms) loses only 3.6%. The collimator gets no such help, because the y axis has no actuators.

4.4 Crosstalk between rows

For the 2D test, chip m forms a full row of 41 tweezers at 3 µm over ±60 µm (alternating phases 0 and $\pi/2$), and the neighbouring chip forms 11 tweezers offset by 1.5 µm in x (ideal FAC, $s_x = 0$). "Coherent" means one laser frequency for both chips (worst of eight relative phases); "incoherent" means their relative phase is averaged out (Table 11, Fig. 7).

Table 11: Crosstalk between neighbouring rows in 2D (ideal FAC, $\beta = 0.7$ µm, $s_x = 0$). Full row: intensity of 41 tweezers at the neighbouring row relative to the peak, maximum over x. Static: change of a neighbouring tweezer's depth / position when the full row is on. Moving: change when one tweezer of row m moves by 0.25 µm. Simulated.

row separation	full row	incoherent rows		coherent rows	
		static	moving	static	moving
3 µm	4.3×10^{-4}	2.0×10^{-4} / 0.13 nm	6.2×10^{-5} / 0.04 nm	2.9% / 18 nm	6.5×10^{-3} / 5.6 nm
4 µm	1.8×10^{-4}	1.7×10^{-4} / 0.002 nm	2.4×10^{-5} / 0.04 nm	2.6% / 11 nm	2.8×10^{-3} / 4.5 nm
6 µm	6.3×10^{-5}	4×10^{-6} / 0.02 nm	3×10^{-6} / 0.01 nm	0.41% / 8.9 nm	1.7×10^{-3} / 2.1 nm
8 µm	4.3×10^{-5}	$< 10^{-6}$	$< 10^{-6}$	0.08% / 1.8 nm	1.1×10^{-3} / 1.7 nm
16 µm	1.3×10^{-5}	2×10^{-6}	$< 10^{-6}$	0.27% / 1.6 nm	4×10^{-4} / 0.6 nm

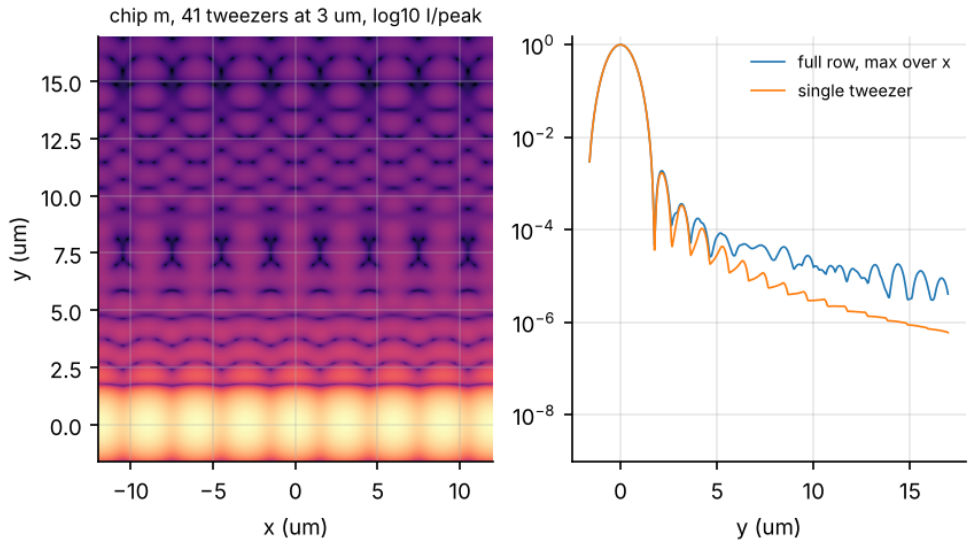


Figure 7: Light from one chip at the neighbouring rows: 2D scalar model with D1's beam-propagation response in x and the facet mode through an ideal NA 0.9 FAC in y ($\beta = 0.7$ µm, $s_x = 0$, valid near the row centre). Left: $\log_{10}$ of the intensity relative to the peak for a full row of 41 tweezers at 3 µm, alternating phases 0 and $\pi/2$ ($|x| \leq 12$ µm shown). Right: maximum over x against y, compared with a single tweezer. The full row leaves a tail plateau of about 10^{-5} out to 16 µm.

With incoherent rows, a moving tweezer changes the next row by at most 6×10^{-5} in depth and 0.04 nm in position, even at 3 µm pitch, two orders of magnitude below the in-row figures of D1 (0.11%, 3.5 nm). With coherent rows, the crosstalk intensity ϵ interferes with the neighbour's own field and changes its depth by about $2\sqrt{\epsilon}\cos\phi$. Here, ϕ is the relative optical phase of the two chip feeds, which drifts with temperature and acoustics in the separate fibres. The 2.6% and 11 nm at 4 µm thus become slow, uncontrolled noise, and the tail plateau keeps it present at all simulated distances, so alternating two frequencies is not enough. The original plan therefore gave every chip its own frequency, with pairwise differences of at least 1 MHz, far above twice the 84 kHz trap frequency: one acousto-optic shifter per chip feed at $80 + 1.13\,m$ MHz ($m = 0$ to 15), each handling 2.6 to 3.4 W [VALUE?]. This plan is superseded by the scheme below.

4.5 Decorrelating the rows without frequency shifters

A frequency offset only serves to randomise ϕ faster than the trap period, so that the atom sees the time average; a common random phase seen with different delays does the same. If the seed carries a phase $\varphi(t)$ and chip m receives it after a delay τ_m, the residual interference between chips m and n is the field coherence $|\langle\exp\{i[\varphi(t) - \varphi(t-\tau)]\}\rangle|$ at $\tau = \tau_m - \tau_n$.

Our choice of drive is a compromise between residual coherence and tolerance to real hardware. A binary $0/\pi$ m-sequence ($N = 1023$) leaves $9.8 \times 10^{-4} = 1/N$ at exact integer delays, but every transition passes through $+\pi/2$, so a modulator with a 0.3-bit rise leaves a carrier of 0.12 (coherence 4.1×10^{-2}). A Zadoff-Chu polyphase code also needs exact delays (6.2×10^{-2} at non-integer ones). Gaussian phase noise of $\sigma = 3$ rad rms needs neither: the carrier is $\exp(-\sigma^2/2)$, any delay longer than about three correlation times decorrelates, and the ensemble coherence is $\exp(-9) = 1.2 \times 10^{-4}$ (4.2 to 7.2×10^{-3} in our 32,768-bit records, their statistical floor). What the atom sees is the low-frequency density of the interference term, $0.40/B$, with B the correlation rate. For $2\sqrt{\epsilon} = 2.1 \times 10^{-2}$ this gives a relative intensity noise $S_I(2\nu_{\mathrm{trap}}) = 8.8 \times 10^{-14}$/Hz at $B = 2$ GHz and a parametric heating rate $\Gamma = \pi^2\nu^2 S_I(2\nu) = 6 \times 10^{-3}$ s^{-1} at $\nu = 84$ kHz [9], an energy e-folding time above 2 minutes (2.4×10^{-2} s^{-1} at $B = 0.5$ GHz).

The adopted hardware is one fibre-coupled $LiNbO_3$ phase modulator on the low-power seed, ahead of the 1:16 split, driven by Gaussian noise of about $1\,V_\pi$ rms (3 rad) over 1 to 2 GHz,

and extra fibre of 0, 0.3, …, 4.5 m on the 16 chip feeds. Within one chip, path differences of at most about 1 mm (5 ps) cost $\sigma^2(\tau/\tau_c)^2/2 \approx 5 \times 10^{-4}$ of coherence, so the synthesis sees a coherent source. The optical spectrum widens to 1.35 GHz rms (5 pm) at $B = 2$ GHz; through the objective's axial chromatic shift (16 µm/nm, derived for a diffractive lens) this smears the focus by 0.08 µm rms (peak 0.9998) and the position at 60 µm by 0.29 nm rms. The mean light shift at the double-magic wavelength is unchanged to first order; the second-order term needs the curvature of the differential polarisability [VALUE?]. A seed broadened to about 1 GHz also raises the stimulated-Brillouin threshold of the high-power amplifier and the delivery fibres (3 W each) ([REF?]; recalled that this is standard practice), and the scheme removes 16 RF drivers and the shifters' insertion loss (about 15% each [VALUE?]). Table 12 compares the options.

Table 12: Perturbation of a tweezer by all other rows within ±16 µm (rms over drifting relative phases; 4 µm rows of 41 tweezers; $s_x = 0$). In-row criteria: others' depth change $< 3 \times 10^{-3}$ and position change < 5 nm when a tweezer moves; proposal budget 10 nm rms. Simulated (vector focusing for the polarisation options) and derived.

option (added hardware)	power	static: depth / position	tweezer moves 0.25 µm
all chips coherent (none)	1	2.3% / 9.2 nm	2.3×10^{-3} / 5.1 nm
per-chip frequency offsets (16 shifters)	1 + loss	1.8×10^{-4} / 0.09 nm	2×10^{-5} / 0.05 nm
phase noise + delays (1 modulator, ≤ 5 m fibre)	**1**	$\mathbf{4.0 \times 10^{-4}}$ **/ 0.17 nm**	$\mathbf{4 \times 10^{-5}}$ **/ 0.10 nm**
alternating x/y polarisation (wave-plate strips)	1	4.2×10^{-3} / 2.5 nm	1.1×10^{-3} / 1.6 nm
y apodiser $a = 0.65$ (graded mirror)	1.84	5.2×10^{-3} / 4.1 nm	1.0×10^{-3} / 1.8 nm
polarisation + apodiser $a = 0.8$	1.55	2.8×10^{-3} / 1.8 nm	5.7×10^{-4} / 0.7 nm

Leaving all chips coherent breaks the position budget (9.2 nm rms of slow wander plus 5.1 nm when neighbours move). Of the passive options, orthogonal polarisation on alternate chips lowers the nearest-row interference by a factor of 26 at NA 0.5 (the residual comes from depolarised high-angle rays, mainly E_z). It meets the moving-tweezer criteria and the position budget, but rows two apart share a polarisation, and the static depth wanders by 0.4% rms at Hz to kHz rates. It has three further costs: (i) a polarisation-insensitive meta-objective; (ii) a vector light shift where the fields overlap, about $2 \times 10^{-3}\, g_F m_F$ of the depth at 4 µm (from a vector-to-scalar ratio of about $0.09\, g_F m_F$ per unit ellipticity, recalled, to be verified [10]), i.e. up to tens of kHz for stretched states and zero to first order for clock states; (iii) only two orthogonal states. A Gaussian y apodiser costs a factor 1.84 in power, widens the spot from 1.35 to 1.73 µm and still leaves 4.7×10^{-3} of static depth change in 2D. We adopt phase noise with fibre delays, and keep alternating polarisation as a free backup that gives an independent factor of about 26 on the nearest rows if the modulation fails.

4.6 Conversion of phase noise to intensity noise

Pure phase noise leaves the intensity constant (the simulated control sits at the numerical floor, −376 dB/Hz). Intensity noise appears only where an element filters the spectrum. We simulated the elements of a laser chain one at a time on the modulated field (3 rad rms, $B = 2$ GHz; record 131 µs at 16 GS/s). The criterion is $\Gamma \leq 0.1\ \mathrm{s}^{-1}$ at $\nu = 84$ kHz, i.e. a relative intensity noise (RIN) density at 168 kHz of $S \leq 1.4 \times 10^{-12}$/Hz (−118 dB/Hz). Gain tilt and fibre dispersion respond to the instantaneous frequency, whose spectrum vanishes at low frequency, so their RIN lies at GHz: −216 dB/Hz at 168 kHz for 0.5 dB/nm tilt, below −280 dB/Hz for dispersion. A delayed copy of amplitude r adds $2r\cos[\varphi(t) - \varphi(t-\tau)]$; beyond the correlation time (about 0.3 ns) this is a telegraph-like signal of variance $2r^2$ spread over B, so $S \approx 4r^2 \times 0.40/B$ (Fig. 8). A band-pass filter narrower than the spectrum acts as a delay distributed over its impulse response, about $1/(\pi\,\mathrm{FWHM}) = 40$ ps for 30 pm. Table 13 lists the elements that matter.

Table 13: Phase-to-intensity conversion at 168 kHz $= 2\nu_{\mathrm{trap}}$ for Gaussian phase noise of 3 rad rms, $B = 2$ GHz (worse of two relative phases for copies). Element parameters are chosen ranges, not component data; the PM-fibre group delay of 1.2 ps/m is recalled, to be verified. Simulated.

element	RIN at 168 kHz	Γ (s^{-1})
band-pass 1 nm FWHM, offset 250 pm (ASE filter)	-173 dB/Hz	$< 10^{-6}$
band-pass 100 pm, centred / offset 25 pm	-130 / -133 dB/Hz	7×10^{-3} / 3×10^{-3}
band-pass 30 pm, centred / offset 7.5 pm	**-110** / **-113** dB/Hz	**0.70** / **0.37**
two reflections $R_1 = R_2 = 10^{-2}$, delay ≥ 0.3 ns	-130 dB/Hz	7×10^{-3}
two reflections $R_1 = R_2 = 10^{-2}$, delay 0.1 / 0.01 ns	-133 / -171 dB/Hz	3×10^{-3} / 10^{-6}
two reflections $R_1 = R_2 = 10^{-3}$ or 10^{-4}, any delay	≤ -149 dB/Hz	$\leq 8 \times 10^{-5}$
PM fibre, -20 dB crosstalk, 5 / 20 m, then TE guide	-161 / -137 dB/Hz	6×10^{-6} / 1.4×10^{-3}
representative chain	-134.5 dB/Hz	2.5×10^{-3}

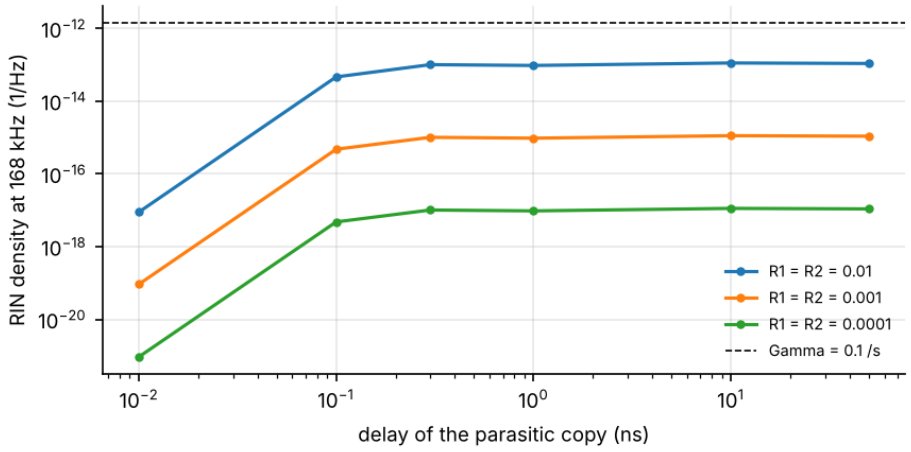


Figure 8: RIN density at 168 kHz from one parasitic copy made by two reflections $R_1 = R_2$, against its delay, for Gaussian phase noise of 3 rad rms at $B = 2$ GHz (simulated, record 131 µs at 16 GS/s, worse of two relative phases). Dashed: the criterion $\Gamma = 0.1$ s^{-1} at $\nu = 84$ kHz (1.4×10^{-12}/Hz). Beyond about 0.3 ns the density saturates at $\approx 4r^2 \times 0.40/B$, $r^2 = R_1 R_2$.

Two design rules follow: (i) no filter narrower than about 100 pm (20 times the 1.35 GHz rms width) in the light path; (ii) the product of any two reflections below $R_1 R_2 = 1.3 \times 10^{-3}$ ($r \leq 0.036$, set by the simulated multipath rows; the analytic estimate gave 1.7×10^{-3}), so that reflecting surfaces in series should be AR coated (an uncoated fused-silica pair, 1.2×10^{-3}, sits at the limit). A representative chain (0.5 dB/nm tilt, 1 nm ASE filter, 5 m of PM fibre, two −30 dB connector reflections 5 m apart, an on-chip etalon between a −30 dB input facet and the 2.3% glued emitter facet, 5 m of dispersion) gives −134.5 dB/Hz and $\Gamma = 2.5 \times 10^{-3}$ s^{-1}, 16 dB inside the criterion, dominated by the etalon and the connector pair. The choice $B = 2$ GHz is a middle value: a higher B helps against long-delay copies (−125, −130, −138 dB/Hz at $B = 0.5$, 2, 8 GHz for $r = 10^{-2}$, $\tau = 1$ ns) and hurts against short ones. Without modulation the same reflections give a slow offset $2r \cos \Phi$ that follows the path phase Φ (2% for $r = 10^{-2}$); the modulation spreads it over GHz, so the phase noise also helps against parasitic etalons provided rule (ii) holds. Amplifier gain dynamics, self-phase modulation and Brillouin scattering are not modelled. The prediction to test is a RIN floor below −130 dB/Hz at 10 kHz to 1 MHz after the amplified chain, with the modulation on and off.

4.7 Alignment, assembly and drift

Table 14: Alignment tolerances of one chip and its FAC (ideal collimator, 2D, single tweezer, $s_x = 0$). Peak at the nominal atom plane relative to no error; worst crosstalk at the 4 µm neighbour (no error: 1.1×10^{-4}); FAC errors except the last row. Simulated.

error	value	peak	crosstalk (10^{-4})
decentre in y	0.05 / 0.1 / 0.2 / 0.3 µm	0.992 / 0.969 / 0.881 / 0.750	2 / 4 / 12 / 28
focus along z	0.25 / 0.5 / 1 / 2 µm	0.993 / 0.973 / 0.893 / 0.641	3 / 9 / 28 / 43
roll about z	0.5 / 1 / 2 mrad	0.998 / 0.992 / 0.968	1 / 1 / 2
smile (p-v)	0.1 / 0.2 / 0.4 µm	0.998 / 0.992 / 0.967	1 / 2 / 4
wavefront, spherical	0.05 / 0.1 waves rms	0.942 / 0.783	3 / 12
wavefront, coma	0.05 / 0.1 waves rms	0.937 / 0.767	5 / 20
chip piston	10 / 100 / 300 µm	≥ 0.999 / ≥ 0.992 / ≥ 0.908	

Table 14 lists the per-chip errors; none moves the row in y by more than 0.01 nm. The collimator decentre is the tightest, ±0.1 µm for 3% loss. A decentre δ shifts the facet image in the pupil by δ/β of its radius, so the tolerance is a property of the facet, not of the lens: the ray-traced designs all give ±0.1 µm (adopted collimator: 0.973 at 0.1 µm). Focus is looser for the glued design, about ±1 µm for 3% in the 1D line model, because the facet radiates into glass; the 2D model with the circular pupil is stricter (peak 0.988 at 0.5 µm and 0.953 at 1 µm, and 0.968 at 0.1 µm decentre). The wavefront requirement is about 0.03 waves rms for ≤ 3% loss; for the adopted collimator, the $n = 1.8$ surfaces need a form error of at most 20 nm rms for ≤ 2% loss. A chip piston of 300 µm shifts tweezers in x by up to 6.8 nm, so facets must be flush to about 100 µm; stair-step stacking for wire bonds is excluded, and fibre and electrical access (not addressed here) must come from the chip sides or back.

Row positions follow from the conjugations (derived). A height error of chip plus collimator moves its row by $1/D = 8$ nm per µm, a chip tilt of 1 mrad about x by 1.4 nm, and a chip piston $\mathrm{d}z$ defocuses the row by $\mathrm{d}z/D^2$ (6 nm per 100 µm). A stack built to ±2 µm places rows to ±16 nm, a static offset that the static 1075 nm array must match. For a collimator glued to its own facet, differential expansion acts only over the lens height, about 0.2 nm/K (CTE of Si 2.6×10^{-6}/K and fused silica 0.55×10^{-6}/K, recalled, to be verified); a monolithic collimator array referenced at the base of the 8 mm stack would see 16 nm/K at the top chip. Hence one collimator per chip, or a stack held to ±1 K. Thermal expansion of the Si stack moves the outer rows by about 0.2 nm/K at the atoms. The glue line lies along z, so cure shrinkage mainly changes the focus, but the height must still be held to ±0.1 µm during cure [REF?]. The alignment signal is strong: a

decentre moves the facet image across the pupil by $\delta \times 12{,}400$ (0.1 µm becomes 1.2 mm), so a pupil camera with 10 µm pixels resolves 1 nm in principle, and we recommend active alignment against it before and during cure. The placement accuracy of two-photon-printed collimators [REF?], the achievable wavefront error of NA 0.9 micro-collimators [REF?] and the long-term stability of bonded collimators [REF?] still need sources.

4.8 Power and laser plan

Table 15 gives the plan for 16 chips of 192 ports (Sec. 5). Our choice of one input fibre per chip is a compromise between facet intensity and phase stability. The lensed-fibre inverse taper has a 3.1 µm^2 mode, so a 20 MW/cm^2 cap (a design choice; no damage threshold was found) allows 0.62 W per fibre and would need 5 to 6 fibres per chip. However, several inputs split the 128 ports into sub-trees whose relative phases drift with their fibres, and a tweezer window straddling a boundary would fluctuate. The HI1060-matched taper with a 100 nm tip (−0.87 dB, 0.50 dB worse) has a 37 µm^2 mode and carries 7.4 W at the same cap, at the price of a 0.1 µm tip at the assumed minimum feature size. With 16 rows the stack offers 720 (4 µm) to 960 (3 µm) independent tweezers and needs 41 to 55 W of single-frequency 1064.4 nm light at 1.2 mK, or 17 to 23 W at 0.5 mK; the availability of 40 to 55 W is open [VALUE?]. Power scales linearly with depth, so a lower transport depth is the cheapest lever (Secs. 5 and 6).

Table 15: Laser power for 16 chips with the adopted collimator (57 mW per 1.2 mK tweezer) and input fibres per chip at a facet cap of 20 MW/cm^2. Derived.

tweezers per row	depth	per chip	16 chips	lensed-tip fibres	HI1060-matched fibres
45 (4 µm)	1.2 mK	2.6 W	41 W	5	1 (7 MW/cm^2)
60 (3 µm)	1.2 mK	3.4 W	55 W	6	1 (9 MW/cm^2)
45	0.5 mK	1.1 W	17 W	2	1
60	0.5 mK	1.4 W	23 W	3	1

The limits are as follows. The relay is modelled with ideal thin cylindrical lenses in scalar optics, without the objective's apodisation and polarisation (except in the vector calculation of the polarisation option). The collimator trace and the facet FDTD are 2D, the $n = 1.8$ glass is a class rather than a catalogue glass, and tolerances are evaluated one at a time. All y results assume $s_x = 0$ and hold near the row centre only (Sec. 9).

5 More tweezers per engine and per row

This section asks how many independent tweezers one engine can carry, what limits the number, and how one stack of engines can serve more sites than it has tweezers. Packing is simulated on the beam-propagation matrix of design D1, the star-coupler aberration with exact path lengths, and transport depths with a 1D classical Monte Carlo; the scaling tables are derived from these.

The useful number of tweezers per row is the row length divided by the site pitch, and the row length is set by the diffraction-limited field of the objective. A 192-port chip filling the ±100 µm field of the proposal carries 45 tweezers on the 4 µm site lattice with 383 fast actuators. Sixteen such chips in the stacked relay (Sec. 4) give 720 simultaneous tweezers, and a row-select tilt lets this stack serve all 50 rows of the field, 2,250 sites, in four passes. The per-tweezer costs, 8.5 fast actuators and about 57 mW at 1.2 mK, do not fall with scale. For comparison, the 1,024-atom assembly with crossed acousto-optic deflectors cited in the proposal moved a median of 13 (maximum 38) atoms in parallel [1].

5.1 Packing within a row

We filled the ±60 µm row of D1 with tweezers at spacing s, neighbours in phase quadrature, using least-squares synthesis (Table 16, Fig. 9). The peak per tweezer per unit port power stays at 0.256 from 6 to 2.5 µm spacing, so the power per tweezer does not depend on how many are lit. At 3 µm the barrier between neighbours is 78% of the depth; at 2.5 µm the independence degrades to the 1% level, and at 2 µm the spots merge. We take 3 µm (2.8 resolution cells, 1.75 spot widths) as the packing limit. It is set by the 1.71 µm FWHM spot, hence by the NA: a narrower spot at fixed NA needs a less apodised window, which raises the tails.

Table 16: Packing a ±60 µm row of D1 (beam-propagation matrix, least-squares synthesis, neighbours in phase quadrature). Simulated.

spacing	tweezers	intensity midway between neighbours	others' change when one moves 0.25 µm	residual after switch-off
6 µm	21	5×10^{-4}	0.11%, 1.6 nm	$< 10^{-5}$
4 µm	31	3.8%	0.10%, 2.6 nm	10^{-5}
3 µm	41	22%	0.11%, 3.3 nm	9×10^{-4}
2.5 µm	49	43%	0.79%, 10 nm	9×10^{-3}
2 µm	61	76% (spots merge)	5.2%, 74 nm	7×10^{-2}

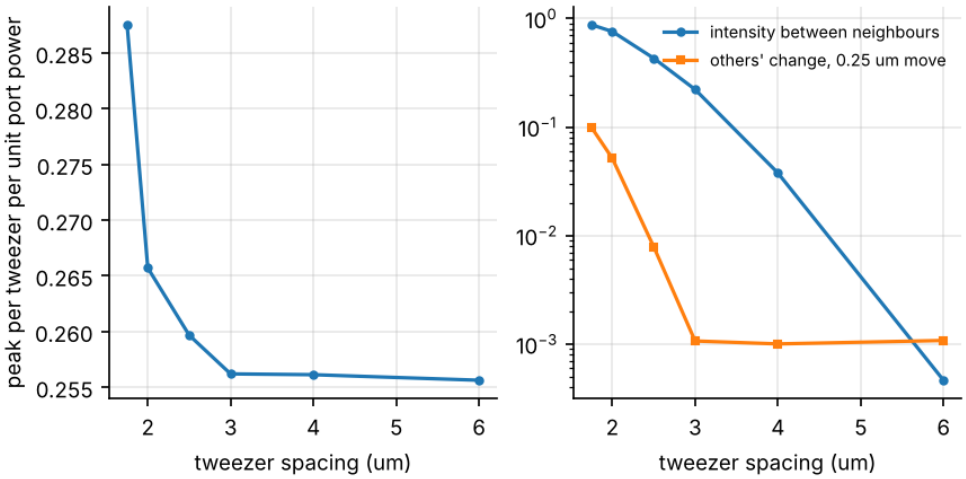


Figure 9: Packing limit of one engine: beam-propagation response matrix of D1 (128 ports, NA 0.5, 1064.4 nm), least-squares synthesis of a ±60 µm row. Left: peak per tweezer per unit port power against spacing. Right: intensity midway between neighbours relative to the peak, and the largest change of the other tweezers' depth when one tweezer moves by 0.25 µm. Below 3 µm the independence degrades rapidly.

However, the proposal places atoms on a 4 µm site lattice. The 41 tweezers of D1 then reduce to 30 useful ones over the same row, and in general a row of length L holds $L/4 + 1$ tweezers. The number per row cannot be raised by packing more densely, so scaling means longer rows and more rows.

5.2 Ports per engine and the 192-port chip

Each input port adds one resolution cell (1.06 µm at NA 0.5) to the row, so a row needs $L/1.06$ ports: 3.8 ports and 7.5 to 8.7 fast actuators per tweezer, independent of L. Table 17 scales the engine with 1.5× oversampled arms, 2.0 µm slab pitch and 1.5 µm emitter pitch. The binding limits, in order, are (i) the objective field, (ii) the number, speed and accuracy of the fast actuators, (iii) the laser power per chip and (iv) the slab size. Star-coupler aberration is not a limit below about 512 ports.

Table 17: Scaling of one engine (row field 88% of the port field; two fast actuators per port; data rate at 200 kHz and 16 bit; aberration is the worst-port rms arm-phase error on the Rowland circle, a geometric lower bound). Derived from the simulated packing and aberration results.

ports	row field	tweezers at 4 / 3 µm	slab radius	fast actuators	static trims	data rate	aberration
128	136 µm	30 / 40	1.3 mm	255	192	0.8 Gbit/s	0.03 rad
192	**204 µm**	**45 / 60**	**2.0 mm**	**383**	**288**	**1.2 Gbit/s**	**0.05 rad**
256	272 µm	60 / 80	2.7 mm	511	384	1.6 Gbit/s	0.07 rad
512	545 µm	120 / 160	5.3 mm	1023	768	3.3 Gbit/s	0.13 rad
1024	1090 µm	240 / 320	10.6 mm	2047	1536	6.6 Gbit/s	0.27 rad

Objective field. The row can only be as long as the objective's diffraction-limited field. The ±100 µm meta-objective target corresponds to about 192 ports, and longer rows need a larger corrected field at NA 0.5, the hardest optical requirement in the system. Refractive plan objectives of NA 0.5 have flat fields of several hundred micrometres (recalled, to be verified); a ±250 µm field would allow 480 ports and 110 tweezers per row, with a slab radius of about 5 mm and about 0.12 rad of aberration.

Fast actuators. Two per port, each updated at $\geq$ 200 kHz (the transport-heating limit at 1.2 mK) and accurate to about 1% in amplitude and 10 mrad in phase for 10 nm position noise. Beyond a few hundred channels per chip, this needs drivers flip-chip bonded or co-integrated with the photonics and closed-loop calibration (on-chip power taps or camera feedback) against PZT hysteresis and drift. This is the main engineering limit.

Slab and aberration. The slab radius grows linearly with the port count. At 512 ports (5.3 mm) it still fits one die but exceeds a single electron-beam field, so stitching errors must be absorbed by the static trims, or the slab written by DUV lithography. The worst-port phase error grows as about 0.033 rad × (ports/128) on the Rowland circle; a quartic correction of the port curve gains nothing, so an aspheric arm curve would be the next step beyond about 512 ports. These geometric values are about half the beam-propagation figure for D1 (0.079 rad), hence lower bounds, but even doubled they keep the Strehl ratio above 0.93 up to 512 ports.

The 192-port chip. We choose 192 ports because this matches the row to the ±100 µm field (±90 µm of 4 µm sites) while staying well inside the slab and aberration limits (2.0 mm, 0.05 rad). It carries 45 tweezers (60 at the 3 µm packing limit) with 383 fast actuators, 288 static trims and 1.2 Gbit/s of data. With the 57 mW per 1.2 mK tweezer of the adopted collimator (43 mW in the engine design, factor 1.34 from the cross axis and the facet loss, Sec. 4.3), a full row needs 2.6 W, which one HI1060-matched fibre carries; an earlier plan to feed a row through 2 to 3 fibres into separate sub-trees is superseded (Sec. 4.8). The row length assumed here exceeds what the adopted cylindrical collimator serves in the stacked relay: for D1, skew rays limit a row to $|x_a| \leq 45$ µm with the best common focus offset (Sec. 4.2). The counts below assume full rows; Sec. 9 treats the skew-ray limit.

5.3 Stacks, row select and transport depth

One chip makes one row of the stacked relay. We take 8 to 16 rows as the practical stack size, an engineering judgement by analogy with laser-diode stacks rather than a result. Sixteen chips make an 8 mm stack and 64 µm of rows, against a 200 µm field.

Row select. The pupil of the y relay is conjugate to every facet, and the atom plane to the collimators' focal plane S. A tilt at a plane conjugate to the pupil therefore translates all rows

of the stack rigidly in y, without aberration in the paraxial limit. Shifting 16 rows by 64 µm is a 3.7 mrad tilt at the objective pupil, or 80 mrad optical (40 mrad mechanical) at the internal Fourier plane of the y telescope, where the beam is 21.6 times smaller. Placing rows to 10 nm needs about 6.5 µrad of mechanical repeatability, galvanometer-scanner class (recalled, to be verified), on a mirror about 1 mm tall and 20 mm wide. Translating the stack by 8 mm with 1 µm accuracy (8 nm at the atoms) is a slower alternative. With atoms waiting in the static 1075 nm array between passes, one 16-chip stack serves all 50 rows of the field, 2,250 sites, in four passes, and the hardware per site drops by a factor 3.1. The same tilt moves all rows together, so the stack carries atoms in 2D: independently along x within each row and in common along y (Sec. 7). A second 16-chip stack, orthogonally polarised, offset by 64 µm and combined by a polarising beam splitter, doubles the simultaneous capacity at twice the hardware and needs a polarisation-insensitive objective (Table 18).

Table 18: Configurations at 4 µm site pitch (57 mW per 1.2 mK tweezer). Derived. Lower transport depths reduce the power, see text.

configuration	simultaneous	sites	passes	chips	fast actuators	power, 1.2 mK
one D1 chip (128 ports)	30	30	1	1	255	1.7 W
one 192-port chip	45	45	1	1	383	2.6 W
stack of 16	720	720	1	16	6,128	41 W
stack of 16 + row select	**720**	**2,250**	**4**	**16**	**6,128**	**41 W**
two stacks + row select	1,440	2,250	2	32	12,256	83 W

Transport depth. Power is linear in depth, and the 1.2 mK of the proposal is what fast long moves need: a 40 µm move in 100 µs survives only at 1.2 mK, and with a 200 kHz staircase at no simulated depth, because each update step (up to 3.75 µm) exceeds the 1.43 µm waist. In the 1D Monte Carlo (4,000 atoms at 10 µK, minimum-jerk trajectory, 200 kHz staircase; loss $\leq 10^{-3}$ and energy gain at most one trap quantum), a three-site move (12 µm) in 200 µs survives at 0.2 mK and a ten-site move in 200 µs at 0.5 mK. This first estimate cut the power of 720 tweezers from 41 W to 6.9 to 17 W. The 3D simulation with handoff to and from the static array (Sec. 6) has since revised it: the moving trap must be at least three times the static hold depth, the hold about $20\,k_BT$, and moves must run along lanes between the rows. The minimum moving depth is therefore 0.3 mK for 5 µK atoms (0.6 mK for 10 µK), which gives 10 to 21 W for 720 tweezers (17 to 34 W for 1 to 2 µK heating per move). With short moves, a cycle of four passes at about 0.6 ms each (100 µs ramps, 200 µs move, 200 µs for the tilt to settle) takes 2 to 3 ms.

5.4 What does not work

Three routes to more tweezers per chip fail: (i) packing tweezers below the site pitch adds nothing on a 4 µm lattice; (ii) several emitter rows per chip cannot be separated at the atoms, because rows a few µm apart on the facet land within nanometres of each other and a collimator per emitter row cannot be fitted; (iii) spectral multiplexing of rows through the dispersive star coupler is excluded by the objective's chromatic focal shift of 16 µm/nm. One route that would remove the stack and the collimators is planar tiling of surface-emitting chips (speculative, not simulated): chips side by side, each with a 2 mm weak grating emitter, imaged in y with $D = 1500$ for a 6 mm tile pitch. The object-side NA is then 3.3×10^{-4}, chip placement errors shrink by D (10 µm becomes 7 nm), and the 0.1 µm decentre tolerance becomes an emission-angle tolerance of about 30 µrad ($\delta n_{\text{eff}} \sim 3 \times 10^{-5}$, trimmable by a heater at about 1.5 K). Grating efficiency and fitting the engine within one tile are open. Scaling to 10^4 to 10^6 sites with blocks of engines is treated in Sec. 8.

6 Transport and handoff in three dimensions

This chapter tests whether a moving tweezer of the adopted design can lift an atom out of a static trap, carry it across the array and hand it to an empty static trap without losing it, heating it, or disturbing the atoms it passes. With the moving-trap field computed through the real optical chain and a static array of Gaussian traps, a 12 or 40 µm move delivers 3,000 of 3,000 simulated atoms (loss $< 10^{-3}$ at 95% confidence) under three conditions: (i) the move runs along a lane between the rows, not along a row; (ii) the moving depth is at least three times the static hold depth; (iii) the hold depth is about $20\,k_BT$. Here, k_BT is the thermal energy of the atoms. With slow handoff steps (200 µs ramps, 400 µs lane steps) the heating per handoff falls to 0.12 µK (about 0.3 µK per move with the x move; 0.16 µK in the 3D classical run) at a moving depth of only 0.3 mK, in both a one-dimensional quantum and a three-dimensional classical model, so the shallow, low-power operating point is also the low-heating point. The one tight alignment is the axial registration of the 1064.4 and 1075 nm foci. A lattice with 3 µm pitch along the rows and 4 µm between rows transports as cleanly as the 4×4 µm lattice.

These results revise two earlier estimates. The one-dimensional transport model of the scaling study, which left out the handoff, the static array and the third dimension, gave a minimum moving depth of 0.2 mK; once handoff and waiting atoms are included this is too optimistic. The first three-dimensional operating points, with 100 µs steps, heated the atom by 1 to 10 µK per move; the slower steps of Sec. 6.4 remove most of this heating.

6.1 Model

The moving trap is the intensity of one tweezer of design D1 (beam-propagation response, least-squares synthesis), passed through the Weierstrass and aplanat collimator and the circular NA 0.5 pupil, and computed by scalar focusing with defocus on a grid of $7 \times 6 \times 20$ µm. Its full widths at half maximum are 1.73 µm along the row (x), 1.37 µm across it (y) and 9.8 µm along the axis (z); at 1 mK its trap frequencies are 54, 67 and 22 kHz. We hold its shape fixed while it moves and sample its position and depth at 200 kHz. The static array consists of Gaussian beams at 1075 nm with a waist of 0.85 µm (our choice) and a Rayleigh range of 2.1 µm, at 4 µm pitch, in three rows of 15 sites. All sites share one global hold depth, since a fixed meta-optic array cannot switch single sites. The two wavelengths add incoherently.

The atoms are classical Cs atoms, integrated by velocity Verlet with 0.25 µs steps and gravity along y, with 1,500 to 3,000 atoms per case. Each starts in a thermal distribution sampled in the harmonic approximation and truncated to atoms bound in the real Gaussian site. The sequence is: ramp the moving trap up on the source site ($\sin^2$, 100 µs); step 2 µm in y into the lane (100 µs); move in x with a minimum-jerk profile; step back to the row (100 µs); ramp down on the target site (100 µs); hold for 100 µs. An atom counts as delivered if it is bound in the target static trap after the moving trap is off. The lane lies halfway between rows 4 µm apart, which puts it as far as possible from the waiting atoms on both sides.

6.2 Lanes between the rows

A tweezer that moves along its own row passes over empty static sites. The total potential then develops a dip that travels with the moving trap and kicks the atom. Losses depend strongly on the move time, with resonances where the transit time over one site matches the trap period (Table 19). At most settings 45 to 95% of the atoms are lost. Only a moving trap five times deeper than the static traps, moving slowly (400 µs), recovers 99.1 to 99.7%, and then at 14 to 24 µK of heating.

Along a lane 2 µm off the row, reached with the row-select y tilt, the static traps have fallen to 1.6×10^{-5} of their depth and transport is clean. The lane costs one shared y tilt per pass (100 µs each way), which the row-select mirror already provides. Because the tilt is shared, all tweezers of

Table 19: Fraction of atoms delivered when the 12 µm move runs along the row, over empty static sites, for two move times (simulated, classical, 1,500 atoms per case).

hold / moving depth	move in 200 µs	move in 400 µs
0.1 / 0.3 mK	5 to 7%	54 to 57%
0.1 / 0.5 mK	42 to 48%	99.1 to 99.7% (heating 14 µK)
0.2 / 0.5 mK	11%	99.3 to 99.6% (heating 24 µK)

a stack enter and leave the lanes together; this is the main constraint the scheduler of Sec. 7 must respect.

6.3 Depth ratio, hold depth and temperature

The ratio of moving to hold depth decides whether the handoff completes. At a ratio of 1.5, 5% of the atoms stay behind in the source site; at 2, 3 to 5% are lost; at 3, 99.5 to 99.9% are delivered with about 10 µK of heating per move; at 5, 99.6 to 100% are delivered with 1 to 2.6 µK (all with 100 µs steps).

The hold depth is set by the atoms that wait next to a lane. They feel the out-of-focus light of the passing tweezer, which at 1.5 to 2.5 µm from its axis and a few micrometres from focus is 6 to 15% of its peak intensity. At a hold depth of $10\,k_BT$, 1 to 2% of these waiting atoms are lost from the hot tail of the distribution; at $20\,k_BT$, none of 3,000. Table 20 lists the resulting operating points. The required depth is set by the atom temperature, not by the move: about $60\,k_BT$ for the moving trap, or $100\,k_BT$ for low heating with fast steps. Cooling the atoms from 10 to 5 µK therefore halves the laser power.

Table 20: Operating points for a 12 µm move in 400 µs along a 2 µm lane, with 100 µs ramps and lane steps and 200 kHz updates (simulated, classical, 3,000 atoms). Power per tweezer is derived by scaling 57 mW at 1.2 mK with depth; the second number is for 720 tweezers.

atoms	hold / moving	delivered	heating per move	waiting lost	power, 1 / 720 tweezers
5 µK	0.1 / 0.3 mK	3000/3000	9.9 µK	0/3000	14 mW / 10 W
5 µK	0.1 / 0.5 mK	3000/3000	1.8 µK	≤2/3000	24 mW / 17 W
10 µK	0.2 / 0.6 mK	3000/3000	10.0 µK	0/3000	29 mW / 21 W
10 µK	0.2 / 1.0 mK	3000/3000	0.9 µK	≤3/3000	48 mW / 34 W

6.4 Step timing and heating per move

The classical model is not reliable when the energy gain is near one trap quantum (about 1 to 3 µK here). We therefore check the handoff with a one-dimensional split-step propagation along y, the handoff and lane axis. The atom starts in the ground state of the static site; the y profile of the moving tweezer is taken from the three-dimensional grid, and gravity is included. The x move is treated separately. We compare with classical trajectories from a Wigner sample of the same ground state (Table 21).

The x move adds 0.08 to 0.37 µK in both models, with 0.98 to 0.99 of the ground state kept. Two findings follow. First, the lane step, not the x move, sets the heating, and the heating falls steeply with the step duration: from 100 to 400 µs the gain at 0.1/0.3 mK drops 80-fold. Second, above about one trap quantum the classical model agrees with the quantum one to within a factor of 2, in either direction (at 100/200 µs it gives 1.4 against 2.0 µK); below it, Wigner sampling overstates the heating by up to a factor of about 12, as expected for zero-point motion. A three-dimensional classical run with the slow steps confirms delivery of 3,000 of 3,000 atoms, with 0.16 µK

Table 21: Energy gain per handoff along y (source ramp, lane step out and back, target ramp) from 1D split-step propagation, the fraction of atoms left in the ground state, and the classical gain from a Wigner sample of the same initial state (simulated). The trap quantum of the static site is 1.4 µK at 0.1 mK and 2.0 µK at 0.2 mK.

hold / moving	ramp / lane step	gain, quantum	ground state kept	gain, classical
0.1 / 0.3 mK	100 / 100 µs	9.7 µK	0.08	11.6 µK
0.1 / 0.3 mK	100 / 200 µs	2.0 µK	0.21	1.4 µK
0.1 / 0.3 mK	200 / 400 µs	0.12 µK	0.98	0.94 µK
0.1 / 0.5 mK	100 / 100 µs	0.84 µK	0.84	1.53 µK
0.1 / 0.5 mK	200 / 400 µs	0.13 µK	0.99	0.87 µK
0.2 / 0.6 mK	200 / 400 µs	0.15 µK	0.99	1.72 µK
0.2 / 1.0 mK	100 / 100 µs	1.58 µK	0.66	1.58 µK

of heating for 5 µK atoms at 0.1/0.3 mK and 0.59 µK for 10 µK atoms at 0.2/0.6 mK, and no loss of waiting atoms.

Hence three step timings are available, and the choice between them is a trade between heating and cycle time (Sec. 7). The slow steps (200 µs ramps, 400 µs lane steps) heat by 0.12 to 0.16 µK per handoff at 0.3 mK, or 0.2 to 0.3 µK per move once the x move is included. The intermediate steps (100 µs ramps, 200 µs lane steps) heat by about 2 µK per move. The fast steps (100 µs lane steps) heat by about 10 µK at a moving depth of 0.3 mK, or 1.2 µK at 0.5 mK. The slow, cold regime needs the least laser power; the fast regime at 0.5 mK shortens the cycle at the cost of 24 instead of 14 mW per tweezer (Table 20).

6.5 Alignment and depth tolerances

Table 22 lists the fraction delivered under misalignments and timing changes. Three points follow. (i) Axial registration of the two foci is the one tight alignment, to about ±1 µm for delivery. A diffractive objective shifts focus by about 170 µm between 1064.4 and 1075 nm (16 µm/nm, derived), so the two paths must be focused independently before they are combined, and then held. (ii) Lateral registration is loose: 0.4 µm costs nothing, so the 16 nm row-placement budget of the stacked relay is not a constraint for transport. (iii) Speed is not limiting in this regime: 40 µm in 400 µs is as safe as 12 µm.

Table 22: Tolerances of the handoff: fraction of 10 µK atoms delivered at a hold depth of 0.1 mK, 12 µm in 400 µs along the lane, 100 µs steps, unless the row states otherwise (simulated, classical).

parameter	values	moving 0.3 mK	moving 0.5 mK
nominal		0.9993	0.9993
axial focus offset	0.5 / 1 / 2 / 4 µm	0.998 / 0.995 / 0.939 / 0.455	0.995 / 0.997 / 0.957 / 0.902
lateral offset, y	0.1 / 0.2 / 0.4 µm	1.000 / 0.998 / 0.999	1.000 / 1.000 / 0.998
lateral offset, x	0.1 / 0.2 / 0.4 µm	0.998 / 1.000 / 0.998	1.000 / 1.000 / 0.999
ramp time	25 / 200 µs	0.994 / 0.998	0.993 / 1.000
move	40 µm in 400 / 800 µs	0.998 / 0.998	0.998 / 1.000
updates	smooth, not 200 kHz	0.999	0.999
gravity	off	0.999	0.999

Delivery is not the tightest criterion for the axial offset; heating is. We propagate the axial motion in one dimension from the ground state of the static site (axial trap quantum 0.40 µK at 0.1 mK), with the static depth following the lateral separation of the moving-trap centre and the moving trap's axial profile taken from the three-dimensional grid. Here, δz is the axial offset between the 1075 and 1064 nm foci. With coincident foci the axial motion is not excited, because

the change of axial trap frequency during the handoff is adiabatic. A focus offset displaces the axial equilibrium twice per move, and that is what heats (Table 23). With the slow steps, keeping the axial heating below the lateral 0.12 to 0.15 µK needs $|\delta z| \leq 0.5$ µm; 1 µm costs about 2 µK per move and 2 µm costs 4 to 8 µK. This tightens the ±1 µm delivery tolerance into a heating tolerance of ±0.5 µm, or 0.3% of the chromatic focal offset. A slow stage on the last y and x lenses of the relay, calibrated on the atoms, would provide the adjustment; an active focus lock between the two paths is the corresponding hardware requirement.

Table 23: Axial energy gain per move versus the axial focus offset δz (1D split-step propagation; the classical Wigner values agree to within 35% above 1 µK, on either side, and overstate the gain below).

hold / moving	ramp / lane step	$\delta z = 0$	0.5 µm	1 µm	2 µm
0.1 / 0.3 mK	100 / 100 µs	0.06 µK	4.1 µK	16.6 µK	40.7 µK
0.1 / 0.3 mK	200 / 400 µs	< 0.01 µK	0.19 µK	2.0 µK	8.3 µK
0.1 / 0.5 mK	100 / 100 µs	0.02 µK	6.4 µK	31.0 µK	28.3 µK
0.1 / 0.5 mK	200 / 400 µs	< 0.01 µK	0.22 µK	1.8 µK	4.4 µK

The depth uniformity of the static array, in contrast, is not a constraint. With lane transport, 2,000 atoms per case and the three operating points 5 µK at 0.1/0.3 mK, 5 µK at 0.1/0.5 mK and 10 µK at 0.2/0.6 mK, a source site deeper by 20, 40 or 60% still delivers at least 0.996, a target site shallower by a factor 0.8, 0.6 or 0.4 at least 0.998, and random depth errors of 5, 10 or 20% rms deliver 1.000 in all cases. This is far beyond the errors a meta-optic or SLM array would show.

6.6 Denser lattices: 3 µm pitch

At 3 µm pitch in both directions, the lane between rows is only 1.5 µm from the waiting atoms. The moving tweezer, 1.37 µm wide across the row and with strong out-of-focus light, then captures them (Table 24, upper part; slow steps, 3,000 atoms). The transported atom is always delivered, but at 0.1/0.3 mK only 0.335 of the waiting atoms in the same row stay and 0.584 are lost. A square 3 µm lattice would therefore need empty corridor rows, which cost most of the density gain: a repeating layout of computation row, corridor and reservoir row gives $(1/3)/9$ µm^{-2} = 0.037 µm^{-2} computation sites, only about 20% above 0.031 µm^{-2} at 4 µm with a 1:1 reservoir layout.

However, only the closest approach to the waiting atoms matters, and that is set by the lane, not by the pitch along the row. A 3×4 µm lattice (3 µm along the rows, rows 4 µm apart, lane 2 µm) therefore transports as cleanly as 4×4 µm (Table 24, lower part). It has 33% more sites per area: 0.063 against 0.047 computation sites per µm^2 with a 3:1 reservoir layout. This supersedes the earlier conclusion that 3 µm pitch needs corridors; that conclusion holds only for a 3 µm row spacing. Corridor rows 6 µm apart with a 3 µm lane are also safe (3000/3000 delivered, 0.3 to 1.5 µK, no waiting atom lost at all three operating points), but less dense.

The intermediate steps are also clean on the dense lattice. With 100 µs ramps, 200 µs lane steps and 300 µs moves over 9 µm along the lane, 3,000 of 3,000 atoms are delivered with 3.0 µK of heating per move at 0.1/0.3 mK (5 µK atoms) and 0.8 µK at 0.2/0.6 mK (10 µK atoms), and no bystander at 3 µm, at 6 µm or in the next row is lost or displaced. This is consistent with the 2.0 µK of the quantum check (Table 21), and we use 2 µK per move for this timing in the steady-state analysis of Sec. 8.

6.7 Limits

The three-dimensional model is classical, and the quantum check is one-dimensional per axis, with the atom assumed to sit at the moving-trap centre laterally for the axial check. Light-assisted processes, photon scattering by the trapping light (a few photons per second at 0.3 to 0.5 mK, estimated from the far-detuned rate, not simulated) and the vector light shift are not included.

Table 24: Transport at 3 µm pitch (simulated, classical, 3,000 atoms per case, 200 µs ramps and 400 µs lane steps). Upper part: square 3 µm lattice with a 1.5 µm lane; fraction of waiting atoms that stay in the same row next to the lane and in the next row beside the drop. Lower part: 3×4 µm lattice with a 2 µm lane; waiting atoms lost among the in-row neighbour at 3 µm, the atoms passed by the lane and the next row beside the drop, out of 3,000.

hold / moving (atoms)	delivered	heating	waiting atoms
square 3 µm lattice, lane 1.5 µm: fraction that stays, same row / next row			
0.1 / 0.3 mK	1.000		0.335 / 0.149 (0.584 / 0.851 lost)
0.1 / 0.5 mK	1.000		0.001 / 0.006
0.2 / 0.6 mK	1.000		0.225 / 0.117
0.2 / 1.0 mK	1.000		0.014 / 0.012
3×4 *µm lattice, lane 2 µm: number lost, in-row neighbour / passed by lane / next row*			
0.1 / 0.3 mK (5 µK)	3000/3000	0.16 µK	0 / 0 / 0
0.1 / 0.5 mK (5 µK)	3000/3000	0.05 µK	0 / 0 / 3 of 3000
0.2 / 0.6 mK (10 µK)	3000/3000	0.59 µK	0 / 0 / 0

The moving trap's shape is assumed invariant along the row; least-squares synthesis keeps it within 0.1%, but the traced collimator of the later design study defocuses tweezers away from the row centre, which this model does not include. Neighbouring tweezers in the same row (independent tweezers 4 µm away) are not included, and the static traps are ideal Gaussian beams. The loss per move is not computed: 0 of 3,000 atoms bounds it below about 10^{-3}, and smaller values used in Sec. 7 are assumptions.

7 Rearrangement and repair

This chapter turns the single moves of Sec. 6 into a full rearrangement cycle, counts its passes, time and moves per atom, and asks what limits the probability of a defect-free block. A 16-chip stack with row select assembles a defect-free 32×32 block from a 50×45 array loaded at 50% in 17.2 ms on average over 40 loadings (worst 23.0 ms), with 5.8 balancing and 2 compaction passes. Every loading succeeded before losses were applied. The average atom in the block moves 1.12 times and none moves more than twice, so with the slow handoff steps the finished block is no more than about 0.6 µK hotter than at the start. One round of imaging and repair then takes 10 to 13 ms. The outcome is set by losses, not by the scheduler: by the loss per move, the loss per image and the vacuum lifetime. The photonic chip contributes 1.1 moves per atom and 17 ms per cycle; reaching a 99% defect-free block requires each loss term near 10^{-5}, which is a property of the platform.

7.1 Hardware rules and scheduler

The hardware sets three rules: (i) a stack addresses 16 consecutive rows at a time; (ii) every tweezer of the stack shares the y motion (into the lane 2 µm off the row, optionally along mid-column lanes over j rows, and back), while each tweezer moves independently in x; (iii) tweezers in one row keep their order and so stay at least 4 µm apart. The scheduler works in two stages. In vertical balancing, each pass chooses a stack position k and a common row shift j that move the most surplus atoms into target rows lacking atoms. Arriving atoms go straight to empty target columns where possible, and atoms that arrive in a pass are not picked again in the same pass. In horizontal compaction, one pass per group of 16 target rows matches each row's atoms in order to the target columns and pushes the extras just outside the block.

A pass takes two ramps, two lane steps, the x moves (minimum jerk, 400 µs per 40 µm, scaled as $\sqrt{d}$ with the distance d, at least 200 µs), the y travel for j rows, and 0.3 ms of scanner settling

(our choice). Of the 2.2 ms of a typical pass, ramps, lane steps and settling make up 1.3 ms. The schedule is greedy and not optimal: a planner that combines balancing and compaction, or shares passes between stack positions, would need fewer passes. Section 7.3 shows how much better pass selection gains.

7.2 Cycle time and moves per atom

Table 25 lists the cycle for several targets and loadings, and Fig. 10 shows the mean cycle time. A larger target from the same 50×45 array costs more balancing passes (36×36 at 60% needs 8.3 + 3 and 25.1 ms on average), while a higher loading saves them (32×32 at 60%: 14.8 ms). A second stack, combined with the first to address 32 rows, does not change the number of balancing passes but saves one compaction pass (15.5 against 17.2 ms for the 32×32 block at 50%). The number of moves per atom stays near 1.1 in every case: about 22% of the atoms in the 32×32 block at 50% loading move twice, the rest once or not at all.

Table 25: One rearrangement cycle from a 50×45 array, mean over 40 random loadings, base timing (100 µs ramps, 400 µs lane steps, 0.3 ms settling); passes are vertical (V, balancing) plus horizontal (H, compaction). Defects are expected numbers in the block, from the assumed loss per move (moves) and from vacuum loss during the cycle at lifetimes of 30 and 300 s (vacuum). Simulated; loss rates assumed.

target (loading)	rows	passes V + H	cycle (ms) mean / max	moves/atom mean / max	defects, moves 10^{-3} / 10^{-4}	defects, vacuum 30 / 300 s
30×30 (50%)	16	4.3 + 2	13.3 / 17.1	1.11 / 2	1.00 / 0.10	0.40 / 0.04
32×32 (50%)	16	5.8 + 2	17.2 / 23.0	1.12 / 2	1.15 / 0.12	0.59 / 0.06
32×32 (60%)	16	5.0 + 2	14.8 / 17.1	1.03 / 2	1.06 / 0.11	0.50 / 0.05
36×36 (60%)	16	8.3 + 3	25.1 / 35.6	1.07 / 2	1.38 / 0.14	1.08 / 0.11
32×32 (50%)	32	5.8 + 1	15.5 / 21.3	1.12 / 2	1.15 / 0.12	0.53 / 0.05

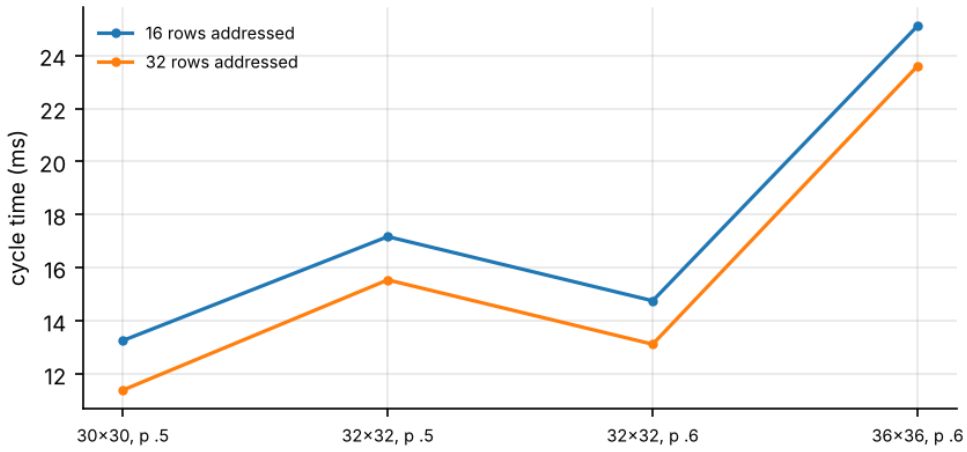


Figure 10: Mean cycle time of one rearrangement cycle (greedy scheduler, 40 random loadings of a 50×45 array at 4 µm pitch) for four target blocks and loading probabilities p, with one 16-chip stack (16 rows addressed) and two combined stacks (32 rows addressed). Timing: 100 µs ramps, 400 µs lane steps, minimum-jerk x moves of 400 µs per 40 µm scaled as $\sqrt{d}$ (at least 200 µs), 0.3 ms scanner settling per pass. The second stack saves one compaction pass in each case.

The cycle time depends mainly on the fixed costs of a pass (Table 26). The low-heating steps of Sec. 6.4 raise it from 17.2 to 18.8 ms. Shortening the lane steps to 100 µs gives 12.5 ms, and making everything fast gives 8.2 ms, while slower scanner settling (1 ms) gives 22.7 ms. Because vacuum loss grows with cycle time, the all-fast setting halves it (0.28 defects at 30 s). It costs heating: about 10 µK per move at a moving depth of 0.3 mK, or 1.2 µK at 0.5 mK. Thus two

regimes remain: slow and cold at 0.3 mK, with low power, or fast at 0.5 mK. With the slow steps (0.29 µK per move, lateral handoff plus x move) the block gains 0.33 µK on average and 0.58 µK for an atom moved twice; with fast steps at 0.5 mK the corresponding figures are 1.36 and 2.42 µK.

Table 26: Sensitivity of the cycle time to the step timing (32×32 block, 50% loading, one stack, mean of 40 loadings; simulated).

timing	cycle time
base (ramps 100 µs, lane steps 400 µs)	17.2 ms
low heating (ramps 200 µs, lane steps 400 µs)	18.8 ms
fast lane steps (100 µs)	12.5 ms
all fast (ramps 50 µs, lane steps 100 µs, x 200 µs per 40 µm)	8.2 ms
scanner settling 1 ms instead of 0.3 ms	22.7 ms

7.3 Pass selection and the pass floor

We compared three ways of selecting the balancing passes on 40 loadings (32×32, 50%, one stack): greedy (most atoms per pass), rate (most atoms per unit pass time) and lookahead (best two-pass sequence among the 8 best first passes). They need 8.05, 8.05 and 7.98 passes in all, with mean cycle times of 17.8, 17.6 and 17.6 ms (maxima 23.1, 22.8 and 22.9 ms). The three modes agree to within 1%. This is because the balancing stage has a geometric floor of about four passes. In one pass the 16-row window can hold at most 9 non-target rows above or below the 32-row target band, so it can feed at most 9 target rows; feeding all 32 therefore takes at least four passes, plus two compaction passes. The greedy schedule (5.8 + 2) is within about 30% of that floor. Further gains would come from shorter passes, not from fewer.

7.4 Stochastic losses and defects

The scheduler fills every loading; defects come from losses. The loss per move is not computed by the transport simulations, which bound it below about 10^{-3} (0 of 3,000 atoms lost); we take 10^{-3} and 10^{-4}, the latter an assumption. Vacuum lifetimes of 30 and 300 s are also assumed. At 10^{-3} per move the 32×32 cycle leaves 1.15 defects on average and at 10^{-4} 0.12; vacuum loss during the 17 ms adds 0.59 defects at 30 s and 0.06 at 300 s (Table 25). The probability that the block is defect-free after one cycle is 0.84 at 10^{-4} per move and 300 s. Reaching 0.99 in one cycle would need about 5×10^{-6} loss per move and a vacuum lifetime above about one hour. A repair round after imaging is the realistic route.

7.5 Image-and-repair rounds

After the cycle, the array is imaged (10 ms, our choice) and each hole in the block is filled directly by one reservoir atom, moved through the lanes from the nearest rows under the same pass rules. This takes 0.1 to 1.3 repair passes, so a round lasts 10 to 13 ms. The repair works, but it cannot go below a floor set by the image itself: atoms lost because of the image are not seen by it, and every atom is exposed to vacuum loss for the duration of the round. The expected number of defects after a round is

$$n_{\mathrm{d}} \approx N \left(\epsilon_{\mathrm{img}} + \frac{t_{\mathrm{round}}}{\tau} \right) + p_{\mathrm{move}}\, n_{\mathrm{rep}}. \tag{1}$$

Here, $N = 1024$ is the number of sites in the block, ϵ_{img} the loss per image, t_{round} the duration of the round, τ the vacuum lifetime, p_{move} the loss per move and n_{rep} the number of repair moves. For $\epsilon_{\mathrm{img}} = 10^{-4}$ and $\tau = 300$ s this gives about 0.14 defects, and a second round does not lower it (Table 27).

As can be seen from Table 27, a repair round brings poor transport (10^{-3} per move) up to the same 0.87 as good transport, provided the image is gentle. Once the loss per move is about 10^{-4}

Table 27: Probability of a defect-free 32×32 block after the cycle and after one and two image-and-repair rounds, and expected defects after one round (simulated, 40 loadings $\times$ 25 loss draws per row; all loss rates and lifetimes assumed).

loss per move	lifetime	loss per image	P(defect-free): cycle / 1 / 2 rounds	defects after 1 repair
10^{-3}	300 s	10^{-4}	0.27 / 0.87 / 0.87	0.14
10^{-4}	300 s	10^{-4}	0.84 / 0.87 / 0.87	0.14
10^{-3}	300 s	10^{-3}	0.27 / 0.35 / 0.33	1.04
10^{-4}	300 s	10^{-3}	0.84 / 0.35 / 0.33	1.03
10^{-3}	30 s	10^{-4}	0.14 / 0.59 / 0.62	0.53
10^{-4}	30 s	10^{-4}	0.48 / 0.62 / 0.63	0.48

or less, the probability of a defect-free block is set by the imaging loss and the vacuum lifetime, not by the transport. With good transport and an imaging loss of 10^{-3}, a repair round makes things worse: 0.84 becomes 0.35, because the image costs about one atom of the block. Imaging should then be skipped, or used only to post-select. For a 99% defect-free block each term of Eq. (1) must be near 10^{-5}: the loss per move, the loss per image, and t_{round}/τ. The continuous version of this balance, with periodic imaging and refilling of much larger arrays, is the subject of Sec. 8.

7.6 Limits

The scheduler is greedy and not optimised across stack positions. The loss per move is bounded by the transport simulations but not computed, and the values used here, like the imaging time, imaging loss and vacuum lifetimes, are assumptions. Detection is taken as perfect apart from the imaging-induced loss, with no false positives or negatives. The repair model moves one reservoir atom per hole. Photon scattering by the trapping light (estimated at a few photons per second at 0.3 to 0.5 mK) is not included, and neither is the possibility that an atom heated by its moves is lost during later moves. The heating per move comes from the one-dimensional quantum check and the three-dimensional classical runs of Sec. 6.4, with the axial focus offset held to ± 0.5 µm (Sec. 6.5).

8 Arrays of 10^4 to 10^6 sites

We ask what limits an array of 10^4, 10^5 or 10^6 sites at 3 to 4 µm pitch, in the order in which the limits appear, and what the transport hardware of the previous chapters must look like to assemble such an array and keep it filled. The record is `docs/megascale.md` (Secs. 1 to 10); the code is `scripts/m22_megascale.py` (budget, tiling), `picsim/megasort.py` with `scripts/m23_steady.py` (assembly and steady state), `scripts/m24_tiling_wave.py` and `scripts/m25_imaging.py`.

The short answer is as follows. At 4 µm pitch the transport side scales to 10^6 sites. An engine block of 32 rows $\times$ 4 engines (128 engines) assembles 10^6 sites from 85% loading to 10^{-3} defects in 1.0 to 1.1 s with fast handoff steps, and then holds the array at a time-averaged defect fraction of 1.8×10^{-4} (2.4×10^{-4} sampled at the image). Three walls come first: (i) the corrected field of the objective, (ii) the static trapping power, which is the first wall at 10^6 sites, and (iii) the loss rates (vacuum lifetime and loss per image), which set the steady-state floor at every array size. At 10^6 sites the transport adds a fourth constraint, the latency of a block that must sweep 252 positions to reach a hole; it is removed by fast steps, a larger block, or a slower imaging cadence. Table 28 collects the scale-dependent numbers.

Table 28: Scale of the problem (derived, `results/m22_budget.json`). Field radius $L/\sqrt{2}$ for a square of side L; resolution cell 1.06 µm at NA 0.5; static power at 4.5 mW per site per mK, 85% loading including the reservoir; block positions for the 32 × 4 block at 4 µm pitch; refills for $\tau = 3000$ s and $\epsilon_{\mathrm{img}} = 10^{-4}$ at 10 Hz imaging.

	10^4 sites	10^5 sites	10^6 sites
side at 4 / 3 µm pitch	0.40 / 0.30 mm	1.26 / 0.95 mm	4.0 / 3.0 mm
single objective: field radius (4 µm)	0.28 mm	0.89 mm	2.8 mm
resolution cells per side (4 µm)	372	1,187	3,754
zones of ±250 µm (objectives)	2	13	127
static power at the atoms, 0.1 / 1 mK	5.3 / 53 W	53 / 530 W	0.53 / 5.3 kW
32 × 4 block: positions	10	28	252
refills needed per second	13	134	1,335

8.1 The walls in order

Field and space-bandwidth. A square array of side L needs a corrected field of radius $L/\sqrt{2}$. At 4 µm pitch that is 372, 1,187 and 3,754 resolution cells per side for 10^4, 10^5 and 10^6 sites; the last corresponds to a space-bandwidth product of about 1.4×10^7. The ±100 µm field of the proposal [1] covers none of these. 10^4 sites need ±0.28 mm. 10^5 sites need ±0.9 mm, which is within the range of refractive NA 0.5 objectives (recalled, to be verified). 10^6 sites need a lithography-class field of ±2.8 mm at NA 0.5. Zones, i.e. independent arrays each with its own objective, static array and engine block, avoid the large field. A ±250 µm zone holds 7,700 sites at 4 µm and assembles like the 10^4 case. However, each zone needs its own objective, about 25 to 30 mm across at a 15 mm working distance, so the 127 zones needed for 10^6 sites are impractical. Zones therefore suit 10^4 (2 zones) to 10^5 sites (13 zones); at 10^6 a single large-field objective, or a few large zones, cannot be avoided.

Static trapping power. From the trap depth with $\alpha(1075\ \mathrm{nm}) = 1120$ au and a 0.85 µm waist, a tweezer needs 4.5 mW per site per mK at the atoms. The reservoir sites count as well. The budget of Table 28 uses the minimum reservoir fraction $1 - f$, i.e. 50% of all sites at 50% loading and 15% at 85%; the simulated 3:1 layout at 85% uses 25%, which adds 13% to these powers. At the 0.1 mK hold of the transport model (Sec. 6), 10^6 sites take 0.5 to 0.9 kW at the atoms (85% to 50% loading), and 5 to 9 kW at a 1 mK loading or imaging depth. With an assumed delivery efficiency of 30%, the laser must supply 1.8 to 3 kW at the hold and 18 to 30 kW at 1 mK. This is the first wall at 10^6 sites.

An optical lattice is the obvious alternative for the static potential, as in the continuously operated lattice array of Gyger et al. [11] (known to us from a search summary only). Table 29 compares the two (derived, `results/m22_lattice.json`). The lattice uses two orthogonal retro-reflected beams (incoherent sum, peak $8I$ per beam; coherent four-beam sum, $16I$), held within −20% of the peak over the array. A tweezer puts its power into about $\pi w_0^2/2 = 1.1$ µm^2 per site. A Gaussian lattice beam that keeps the corners within 20% spreads it over about $15a^2/G = 31$ µm^2. Hence a free-space lattice loses to tweezers by about 25 with Gaussian beams and by 2 to 4 with flat-top beams. Only an enhancement cavity with a buildup near 100 wins, by a factor of about 4, and that cavity would need a waist near 10 mm for 10^6 sites. Two levers do more than the choice of potential: the depth (a factor of 10 between 1 and 0.1 mK), and the delivery efficiency, which for the simpler lattice optics could gain up to about 2.7 (0.8 instead of 0.3). For 10^6 sites the static potential thus needs about 2 kW of laser power at the hold depth with tweezers, or about 0.5 kW with a buildup-100 cavity lattice, and loading and imaging must happen at a lower depth or in a smaller zone.

Table 29: Laser power for the static potential at 30% delivery (assumed), for N sites plus a 25% reservoir at 4 × 4 µm, at depths of 0.1 / 1 mK (derived, `results/m22_lattice.json`). Lattices: two orthogonal retro-reflected beams held within −20% of the peak; cavity buildup is a choice.

static potential	10^4 sites	10^5 sites	10^6 sites
tweezer array ($w_0 = 0.85$ µm)	20 / 200 W	0.2 / 2.0 kW	2.0 / 20 kW
free-space lattice, Gaussian beams	0.5 / 5.0 kW	5 / 50 kW	50 / 500 kW
free-space lattice, flat-top, incoherent	85 W / 0.85 kW	0.85 / 8.5 kW	8.5 / 85 kW
free-space lattice, flat-top, coherent	42 W / 0.42 kW	0.42 / 4.2 kW	4.2 / 42 kW
cavity lattice, Gaussian, buildup 10	50 W / 0.5 kW	0.5 / 5 kW	5 / 50 kW
cavity lattice, Gaussian, buildup 100	5 / 50 W	0.05 / 0.5 kW	0.5 / 5 kW

Engines, blocks and tiling along x. One engine (192 ports, 383 fast actuators, Sec. 3) serves a 180 µm row segment, i.e. 45 to 46 sites at 4 µm and 60 to 61 at 3 µm. Engines can be tiled in two directions. Along y, chips are stacked, one chip per row with its own collimator (Sec. 4). Along x, several engines share one chip edge: each gets its own x lenslet, and a common Fourier lens overlaps all of them in the pupil and maps their offsets to adjacent atom segments. The ABCD check (`m22_tiling`) gives a magnification of −60.1 for every engine and all chief rays at the pupil centre. The question is how close the engines can sit. A first-order argument gave a pessimistic answer: with 1.5 µm emitter pitch, each engine's first grating lobe lands 204 µm from its spot at the atoms, inside the neighbouring segment, so a shared slit plane would need opaque gaps as wide as the slit zones, and hence an engine pitch $P_e \geq 4.4$ mm (four engines per 18 mm edge).

The wave-optics check (`results/m24_tiling_wave.json`) supersedes this. We propagate the emitter near field of one D1 engine, forming a tweezer at $x_a = 0$, 60 and 90 µm (up to the segment edge), over $f_1 = 3.12$ mm to its lenslet (clear aperture equal to fill $\times P_e$), apply an ideal lens phase, propagate over f_1 to the shared slit plane Σ, and measure the intensity in the neighbouring engine's slit zone. Here, Σ is conjugate to the atoms, so this intensity is the ghost that a neighbouring segment sees. We keep only directions inside the pupil acceptance $A/(2f_1)$, because steeper light misses the objective, and an exact ray trace through the neighbouring lenslets checks the same question geometrically. Table 30 and Fig. 11 give the result. The first-order lobes are not the limit: the lenslet aperture and the pupil acceptance remove them, and no accepted ray reaches another slit zone at any pitch scanned. What sets the pitch is the lenslet aperture needed to pass the engine's own slit zone, $2f_1 \tan(\arcsin s_{\mathrm{cut}}) = 2.37$ mm. Below 2.6 mm the lenslet clips the engine's own beam and its diffraction tail enters the neighbouring zone. At $P_e = 3.0$ mm the accepted ghost is below 2×10^{-7} of a tweezer peak, under the coherent criterion of 10^{-6} (`docs/row_coherence.md`). Hence six engines fit on an 18 mm chip edge instead of four. Two scanner positions per row are still needed: the segment pitch at the atoms, $P_e f_{\mathrm{obj}}/f_c = 0.092\,P_e = 277$ µm at $P_e = 3.0$ mm ($f_c = 188$ mm, $f_{\mathrm{obj}} = 17.3$ mm), exceeds the 180 µm segment, so one block position addresses every other segment. The model is 1D (x only) with an ideal thin lens and no surface scatter; baffles between the lenslet channels remain the defence against scattered light, which it does not include.

A block of 32 rows (two 16-chip stacks, combined by polarisation or by the comb of Sec. 9) × 4 engines per row has 128 engines and 49,024 fast actuators. It holds 5,888 simultaneous tweezers, which need 85 W at 0.3 mK if all are lit; that happens at most during assembly, since in steady state only the few refilling tweezers are lit. A single 16-chip stack (a 16 × 1 block) is too small for 10^6 sites: it never finishes assembling.

Pitch. At 3 µm pitch in both directions the transport lane passes 1.5 µm from waiting atoms, and the moving tweezer pulls most of them out, so such a lattice would need empty corridor rows. The anisotropic lattice avoids this (Sec. 6): with 3 µm along the rows and rows 4 µm apart, the lane stays 2 µm from the rows and the lattice transports as cleanly as 4 × 4 µm. It has 33% more

Table 30: Ghost of one engine in its neighbour's slit zone versus engine pitch P_e (simulated, 1D, `results/m24_tiling_wave.json`). Worst case over $x_a = 0$, 60, 90 µm at lenslet fill 1 (the segment-edge tweezer, except at 3.5 mm where $x_a = 60$ µm is marginally worse). "Accepted" keeps only directions inside the pupil acceptance. A fill of 0.8 at 4.43 mm raises the accepted ghost to 2×10^{-9}. No accepted ray through a neighbouring lenslet reaches another slit zone at any pitch.

P_e	ghost peak / tweezer peak accepted	raw	ghost power / own slit power
8.0 mm	4.5×10^{-13}	1.7×10^{-9}	2.6×10^{-13}
4.43 mm	3.6×10^{-10}	2.5×10^{-7}	1.4×10^{-10}
3.5 mm	1.7×10^{-8}	3.1×10^{-6}	1.0×10^{-8}
3.0 mm	1.7×10^{-7}	7.7×10^{-6}	6.3×10^{-8}
2.6 mm	1.6×10^{-6}	1.3×10^{-5}	7.5×10^{-7}
2.4 mm	6.7×10^{-5}	5.6×10^{-5}	7.2×10^{-5}

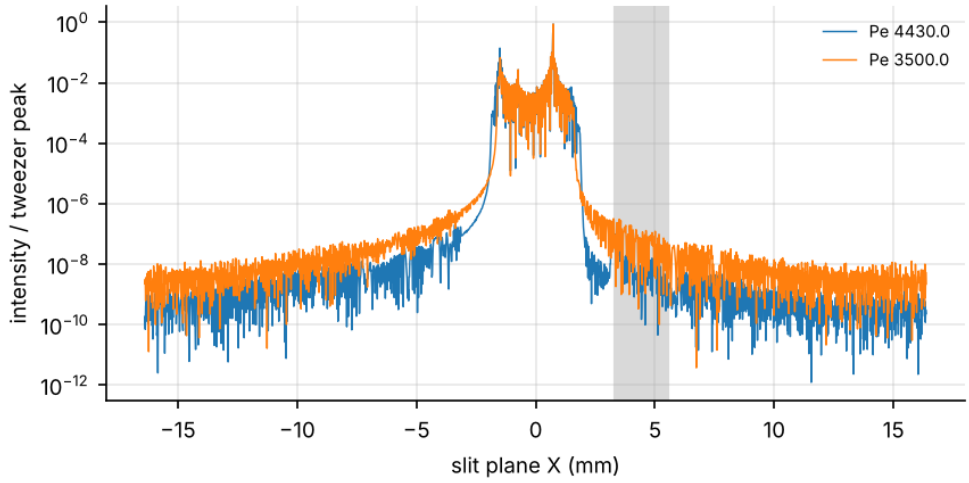


Figure 11: Intensity at the shared slit plane Σ, relative to the tweezer peak, for one D1 engine forming a tweezer at the segment edge ($x_a = 90$ µm), lenslet fill 1, engine pitch 4.43 mm (blue) and 3.5 mm (orange). 1D angular-spectrum propagation of the emitter near field over $f_1 = 3.12$ mm, ideal lens phase on the lenslet aperture, and propagation over f_1 (0.25 µm grid). The grey band is the neighbouring engine's slit zone at 4.43 mm pitch, $|X - P_e| < f_1 s_{\mathrm{cut}}$. The curves show the raw intensity, before the pupil-acceptance filter applied in Table 30.

sites per area (0.063 against 0.047 computation sites per µm^2 with a 3:1 reservoir layout). Below, "3 µm" means this 3 × 4 µm lattice.

8.2 Assembly and steady-state refilling

The simulation (`picsim/megasort.py`) represents the array as computation rows interleaved with reservoir rows (1:2 at 50% loading, 3:1 at 85%). The array is imaged every $T_{\mathrm{img}} = 0.1$ s for $t_{\mathrm{img}} = 10$ ms, with a loss $\epsilon_{\mathrm{img}} = 10^{-4}$ per image and 10^{-4} per move; the reservoir is refreshed every 1 s from an external supply. The engine block visits, round-robin, the tiles with known holes, and each visit runs up to four repair passes (a common row shift, an order-preserving x pass, and one move per refill). Two step timings come from the rearrangement chapter (Sec. 7): slow (200 µs ramps, 400 µs lane steps, 0.2 to 0.3 µK per move) and fast (50 µs ramps, 100 µs lane steps, 200 µs per 40 µm, about 1 to 10 µK per move). Table 31 and Fig. 12 summarise the runs at 4 µm pitch. Their steady-state values are sampled just after an image; Sec. 8.3 explains why the time average is lower.

Table 31: Assembly and steady state at 4 µm pitch (simulated, `results/m23_steady.json`). Time from a random loading to 10^{-2} and 10^{-3} defects (n.r.: not reached in 20 s), and the steady-state defect fraction sampled just after each image, in units of 10^{-4}, for $\tau = 3000$ / 300 s. Slow and fast step timings as in the text. At 10^6 sites with $\tau = 300$ s, slow steps do not reach 10^{-3}.

sites	loading	block	timing	to 10^{-2}	to 10^{-3}	steady state, τ 3000 / 300 s
10^4	50%	16×1	slow	0.34 s	0.34 s	2.7 / 5.6
10^4	50%	32×4	slow	0.12 s	0.12 s	2.6 / 5.6
10^4	85%	32×4	slow	0.12 s	0.12 s	1.8 / 5.7
10^5	50%	16×1	slow	1.8 s	1.8–2.8 s	2.3 / 6.0
10^5	50%	32×4	slow	0.46 s	0.46 s	2.5 / 6.1
10^5	85%	32×4	slow	0.24 s	0.24 s	2.5 / 5.8
10^5	85%	32×4	fast	0.12 s	0.12 s	2.5 (τ 3000 s)
10^6	50%	32×4	slow	2.2 s	2.8 s (τ 3000 s)	8.6 / 32
10^6	85%	32×4	slow	1.9 s	3.1 s (τ 3000 s)	9.3 / 38
10^6	85%	32×4	fast	0.79 s	1.1 s	2.4 (τ 3000 s)
10^6	50% / 85%	16×1	slow	14 s (τ 3000 s)	n.r.	70 to 290

Three findings follow. (i) Up to 10^5 sites the floor is set by the losses, and it is nearly the same for every block and timing (1.7 to 2.7×10^{-4} at image for $\tau = 3000$ s, the spread being run-to-run noise). (ii) At 10^6 sites latency matters: with slow steps a sweep of 252 block positions takes longer than the imaging period, new holes wait several sweeps, and the floor rises about four-fold. Fast steps restore the loss-limited floor. (iii) Each refill is a single move from a nearby reservoir row into the hole, so heating does not accumulate on any atom. In steady state at 10^6 sites the reservoir is consumed at about 1,300 atoms per second ($\tau = 3000$ s). Chiu et al. [12] (abstract read) report 30,000 initialized qubits per second from lattice conveyor belts, maintaining more than 3,000 atoms for over 2 hours; that rate would cover our demand with a margin of 20. For comparison with our loss parameters, Manetsch et al. [3] (abstract read) report a 23 min lifetime in a room-temperature apparatus and 99.98952% imaging survival ($\epsilon_{\mathrm{img}} = 1.05 \times 10^{-4}$) for a 6,100-atom Cs tweezer array, i.e. close to our nominal ϵ_{img} and a lifetime between our 300 s and 3000 s cases.

8.3 Three defect fractions

The values of Table 31 are sampled just after each image. That value counts imaging losses twice. An image removes a fraction ϵ_{img} of the atoms after detecting them, so these holes stay unseen for a full period, and at the next image they are still present when the new losses are added. The simulator therefore also records the value just before each image and the time average. The time

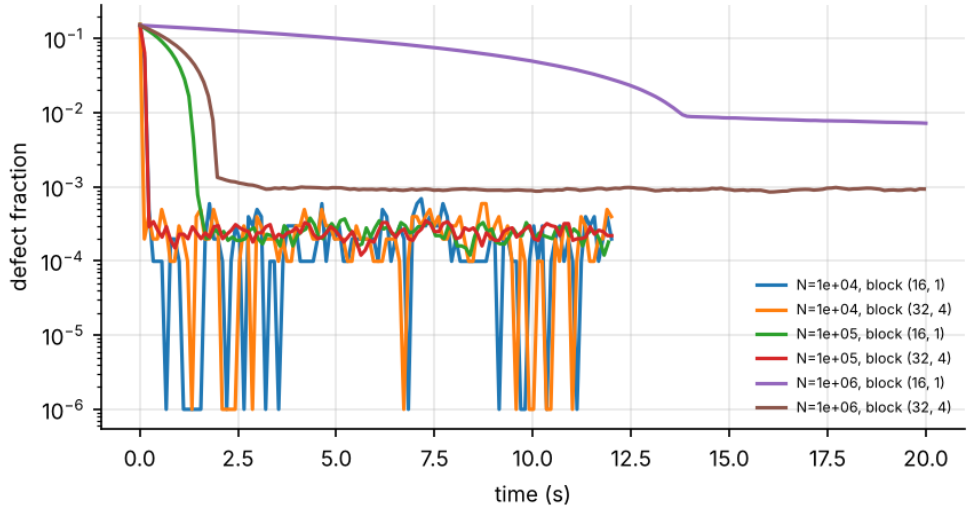


Figure 12: Defect fraction sampled just after each image versus time, from a random 85% loading (3:1 computation to reservoir rows) at 4 µm pitch, slow steps, $\tau = 3000$ s, $\epsilon_{\text{img}} = 10^{-4}$ per image and per move, imaging every 0.1 s, for blocks of 16 × 1 and 32 × 4 engines and 10^4 to 10^6 sites (`scripts/m23_steady.py`). Values of zero are drawn at 10^{-6}. With slow steps, 10^6 sites with the 32 × 4 block settle at the latency-limited 9×10^{-4}, and the 16 × 1 block does not reach 10^{-3}. Fast steps (not shown) restore the loss-limited floor.

average is a trapezoid sum over visit, image and idle intervals; it is exact in expectation for the vacuum losses, which are uniform in time, while each visit's fills are booked at its start, which biases the average low by at most one visit time (1 to 2 ms against $T_{\text{img}} = 100$ ms). The time average is what a computation started at a random moment sees. With fast steps at 10^6 sites the three values are 2.4×10^{-4} (at image), 1.4×10^{-4} (before image) and 1.8×10^{-4} (time average). The at-image value is close to $2\epsilon_{\text{img}} + (T_{\text{img}} + t_{\text{img}})/\tau = 2.4 \times 10^{-4}$, and the time average is close to $\epsilon_{\text{img}} + (T_{\text{img}} + t_{\text{img}})/(2\tau) = 1.2 \times 10^{-4}$ plus the repair latency. Thus the time average is about 25% below the at-image figures quoted in Table 31, and an earlier rule $\rho \approx (1.5 \text{ to } 1.8)(\epsilon_{\text{img}} + T_{\text{img}}/\tau)$, fitted to at-image values over a narrower range, is replaced by the fits of Sec. 8.5.

8.4 Remedies at 10^6 sites

At 10^6 sites the choice is between a larger block and faster, hotter steps. We scanned blocks of 128 to 256 engines and three step timings with heating per move of 10 µK (fast), 2 µK (intermediate) and 0.3 µK (slow), at 85% loading, $\tau = 3000$ s, $\epsilon_{\text{img}} = 10^{-4}$, $T_{\text{img}} = 100$ ms and rows 4 µm apart (`results/m23_remedy.json`, one seed per case). The intermediate timing (100 µs ramps, 200 µs lane steps) transports cleanly on the 3 × 4 µm lattice at 1 to 3 µK per move with no bystander lost or displaced (`results/m18_midlane.json`, 3,000 atoms per case; Sec. 6); we use 2 µK per move. Table 32 gives the results.

Four results follow. First, fast steps reach the loss floor with any block from 128 engines; more engines only shorten assembly (1.01 to 0.45 s at 4 µm). The 3 × 4 µm lattice holds 2.4×10^{-4} at image with the 32 × 4 block, as 4 × 4 µm does. An earlier value of 8.1×10^{-4} for this case was an artefact of the block placement: the last row of block positions ended short of the array edge, so the holes in the rows below it had no reservoir row in any block that reached them (563 of 716 remaining holes sat there); the last block now sits flush with the edge. Second, intermediate steps come within a factor of 1.3 to 1.6 of the floor with 256 engines (time average 2.4 to 2.9×10^{-4}), and within a factor of 2 to 2.3 with 128. Third, slow steps reach 10^{-3} within 1 to 3.6 s with every block, but hold 4.4 to 8.6×10^{-4}, a factor of 2.4 to 4.8 above the floor. Fourth, the 3 µm rows do slightly better than 4 µm rows with the slower timings. We attribute this to fewer block positions per sweep (a 180 µm segment holds 61 sites at 3 µm against 46 at 4 µm, so the 32 × 4 block needs

Table 32: Remedies at 10^6 sites (simulated, `results/m23_remedy.json`; 85% loading, $\tau = 3000$ s, $\epsilon_{\mathrm{img}} = 10^{-4}$, $T_{\mathrm{img}} = 100$ ms, rows 4 µm apart). Each cell: defect fraction at image / time average over the last 3 s of a 20 s run, in units of 10^{-4}; then the time from the first image to 10^{-3} at image. Heating per move 10 / 2 / 0.3 µK; loss per move 4×10^{-3} for fast steps (from the 3D transport model with 50 µs ramps) and 10^{-4} otherwise.

block (engines)	pitch along rows	fast	intermediate	slow
32×4 (128)	4 µm	2.5 / 2.0; 1.01 s	4.6 / 4.1; 1.25 s	9.1 / 8.6; 3.56 s
32×8 (256)	4 µm	2.4 / 1.8; 0.68 s	3.4 / 2.9; 0.92 s	6.4 / 5.9; 1.39 s
48×4 (192)	4 µm	2.5 / 1.9; 0.68 s	3.6 / 3.1; 0.92 s	6.7 / 6.2; 1.40 s
64×4 (256)	4 µm	2.4 / 1.8; 0.45 s	3.2 / 2.7; 0.69 s	5.5 / 4.9; 1.05 s
32×4 (128)	3 µm	2.4 / 1.9; 0.79 s	4.1 / 3.6; 1.14 s	7.9 / 7.4; 2.31 s
32×8 (256)	3 µm	2.4 / 1.8; 0.56 s	3.0 / 2.5; 0.80 s	5.0 / 4.5; 1.17 s
48×4 (192)	3 µm	2.4 / 1.8; 0.56 s	3.5 / 2.9; 0.80 s	5.9 / 5.3; 1.17 s
64×4 (256)	3 µm	2.4 / 1.8; 0.46 s	2.9 / 2.4; 0.58 s	4.9 / 4.4; 0.94 s

222 positions instead of 252), but we did not isolate this cause.

The heating cost is set per refilled atom, not per atom in the array. Every atom moves at most once, because refills come straight from a reservoir row; with the 32 × 4 block at 10^6 sites 17% of the atoms are moved, and the heating is 0.05 µK mean and 0.3 µK maximum with slow steps, against 1.7 and 10 µK with fast steps (`results/m23_mixed.json`). An adaptive schedule (fast until 10^{-2}, then slow) brings back the latency-limited floor of 9×10^{-4} at image: the refill load at 10^6 sites is too high for slow steps in steady state. Up to 10^5 sites, slow and cold steps reach the loss-limited floor.

The block shape is a hardware choice. A 32-row block is two 16-chip stacks; 48 or 64 rows need three or four stacks and a combining method beyond polarisation, which Sec. 9 provides (a slotted comb mirror, with a polarising beam splitter for 64 rows). A 32 × 8 block needs eight engines per chip edge, i.e. a 24 mm edge at the 3.0 mm engine pitch; at 2.6 mm (21 mm edge) the accepted ghost rises to 1.6×10^{-6}, just above the 10^{-6} criterion. The 64 × 4 block has the same 256 engines on 12 mm chip edges and does slightly better with intermediate steps (time average 2.7×10^{-4} on 4 × 4 and 2.4×10^{-4} on 3 × 4 µm), and Sec. 9 shows that it can be built. At the same time, Sec. 9 finds that the refractive collimator serves only half of each 180 µm segment, which doubles the number of engines per row at fixed coverage; the block counts above assume full segments.

8.5 Imaging cadence and the hole lifetime

Since the floor follows the losses, we scanned the imaging parameters at 10^5 sites (4 × 4 µm, 85% loading, 32 × 4 block, intermediate steps, $t_{\mathrm{img}} = 10$ ms, runs of 8 s or longer averaged over the second half, one seed per case; `results/m25_imaging.json`): $T_{\mathrm{img}} = 0.03$, 0.1, 0.3 and 1 s, $\epsilon_{\mathrm{img}} = 10^{-5}$, 10^{-4} and 10^{-3}, and $\tau = 300$, 3000 and 30000 s, with $t_{\mathrm{img}} = 3$ and 20 ms added, and three points per timing at 10^6 sites. The at-image values fit

$$\rho_{\mathrm{image}} = 2.24\,\epsilon_{\mathrm{img}} + 0.98\,\frac{T_{\mathrm{img}} + t_{\mathrm{img}}}{\tau} \tag{2}$$

with a median error of 10% (worst 40%), close to the $2\epsilon_{\mathrm{img}} + (T_{\mathrm{img}} + t_{\mathrm{img}})/\tau$ expected from the double counting (Fig. 13). Table 33 gives selected time averages.

The time average follows a hole-lifetime model. Let $T = T_{\mathrm{img}} + t_{\mathrm{img}}$. An imaging loss is unseen for one period T and is then repaired after a mean delay d; a vacuum loss waits $T/2$ on average for the next image, and then d. Hence

$$\rho_{\mathrm{mean}} = \epsilon_{\mathrm{img}}\,\frac{T + d}{T} + \frac{T/2 + d}{\tau}. \tag{3}$$

Table 33: Time-averaged defect fraction at 10^5 sites versus imaging period (simulated, `results/m25_imaging.json`; 4×4 µm, 85% loading, 32×4 block, intermediate steps, $t_{\mathrm{img}} = 10$ ms).

ϵ_{img}, τ	$T_{\mathrm{img}} = 0.03$ s	0.1 s	0.3 s	1 s
10^{-5}, 3000 s	2.2×10^{-5}	3.2×10^{-5}	7.0×10^{-5}	1.9×10^{-4}
10^{-4}, 300 s	2.9×10^{-4}	4.0×10^{-4}	7.6×10^{-4}	2.0×10^{-3}
10^{-4}, 3000 s	1.5×10^{-4}	1.4×10^{-4}	1.8×10^{-4}	2.3×10^{-4}
10^{-4}, 30000 s	1.3×10^{-4}	1.2×10^{-4}	0.9×10^{-4}	1.4×10^{-4}
10^{-3}, 3000 s	2.4×10^{-3}	1.4×10^{-3}	1.2×10^{-3}	1.2×10^{-3}

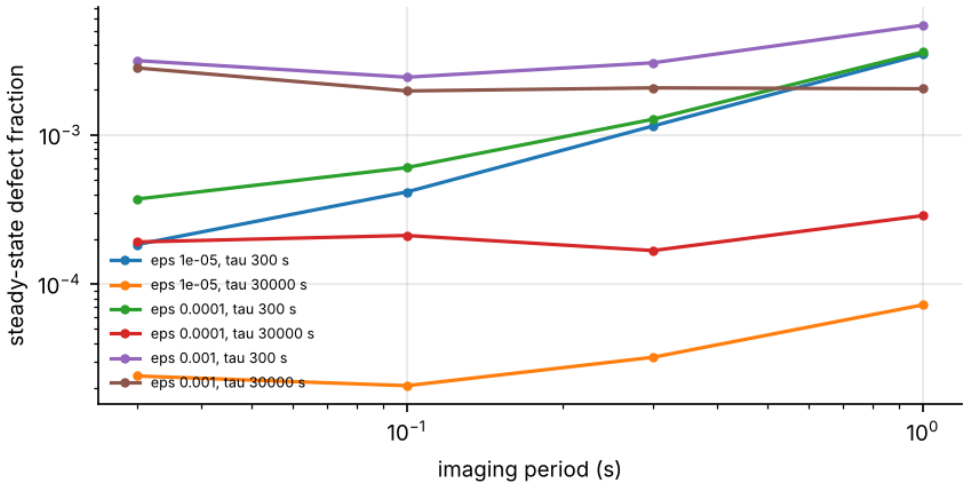


Figure 13: Steady-state defect fraction sampled just after each image versus imaging period, at 10^5 sites (4×4 µm, 85% loading, 32×4 block, intermediate steps, $t_{\mathrm{img}} = 10$ ms, loss 10^{-4} per move), for $\epsilon_{\mathrm{img}} = 10^{-5}$, 10^{-4}, 10^{-3} and $\tau = 300$ and 30000 s. Each point is one seed, averaged over the second half of a run of $\max(8\text{ s}, 8T_{\mathrm{img}})$. The full scan (with $\tau = 3000$ s) fits $\rho_{\mathrm{image}} = 2.24\epsilon_{\mathrm{img}} + 0.98(T_{\mathrm{img}} + t_{\mathrm{img}})/\tau$; time averages are in Table 33.

One delay, $d = 22$ ms, fits all 40 points at 10^5 sites with a median error of 9% ($d = 0$ underestimates by a median 16%). The worst point (41%) is $\epsilon_{\mathrm{img}} = 10^{-3}$ at $T_{\mathrm{img}} = 30$ ms, where the repair saturates. Minimising ρ_{mean} over T gives the optimum imaging period

$$T^* = \sqrt{2\,\epsilon_{\mathrm{img}}\,d\,\tau} \qquad (d \ll T). \tag{4}$$

For $d = 22$ ms and $\tau = 3000$ s this is 36 ms, 0.11 s and 0.36 s for $\epsilon_{\mathrm{img}} = 10^{-5}$, 10^{-4} and 10^{-3}, in agreement with the minima of the scan (0.03 s, 0.1 s, and a flat minimum between 0.3 and 1 s). At the optimum, $\rho_{\mathrm{mean}} \approx \epsilon_{\mathrm{img}} + \sqrt{2\epsilon_{\mathrm{img}} d/\tau} + d/\tau$; for $\epsilon_{\mathrm{img}} = 10^{-4}$ and $\tau = 3000$ s that is 1.5×10^{-4} (simulated 1.4×10^{-4} at $T_{\mathrm{img}} = 0.1$ s). Hence the loss per image dominates the floor once τ exceeds a few thousand seconds, and lowering ϵ_{img}, not imaging more often, is the lever. The imaging time matters only through the dead time at short periods: at $T_{\mathrm{img}} = 30$ ms, $t_{\mathrm{img}} = 3$, 10 and 20 ms give 1.28, 1.48 and 1.66×10^{-4}.

At 10^6 sites the delay is not a constant, because the block must sweep its positions to reach a hole (252 for a 32×4 block on 4×4 µm; a one-pass visit takes 0.9 to 1.0 ms with fast steps and 1.35 to 1.5 ms with intermediate steps). Table 34 gives the time averages for $\epsilon_{\mathrm{img}} = 10^{-4}$ and $\tau = 3000$ s. At $T_{\mathrm{img}} = 30$ ms both timings saturate. The loss rate, $\epsilon_{\mathrm{img}}/T + 1/\tau = 2.8 \times 10^{-3}$ per site per second, needs 2,800 fills per second. At $\rho = 2 \times 10^{-4}$ a block position holds about 0.8 holes, so a sweep of 252 visits (0.25 to 0.4 s) refills only about 200 atoms, while the imaging dead time takes 25% of the time; the holes pile up until each visit has enough work. By Little's law the mean hole lifetime is $\rho_{\mathrm{mean}}/(\epsilon_{\mathrm{img}}/T + 1/\tau)$: 0.15 s for fast steps at $T_{\mathrm{img}} = 0.1$ s, and 0.33 s for intermediate steps at both 0.1 and 0.3 s. Intermediate steps at $T_{\mathrm{img}} = 0.3$ s thus reach 2.2×10^{-4}, within 15% of fast steps at 0.1 s, with the 128-engine block and one fifth of the heating per refilled atom (2 against 10 µK). At 10^6 sites a slower imaging cadence is therefore an alternative to a larger block: the imaging period should match the sweep time of the block rather than be as short as possible.

Table 34: Time-averaged defect fraction at 10^6 sites with the 32×4 block, $\epsilon_{\mathrm{img}} = 10^{-4}$, $\tau = 3000$ s (simulated, `results/m25_imaging.json`).

timing	$T_{\mathrm{img}} = 0.03$ s	0.1 s	0.3 s
fast (10 µK per move)	8.7×10^{-4}	1.9×10^{-4}	2.0×10^{-4}
intermediate (2 µK per move)	1.5×10^{-3}	4.1×10^{-4}	2.2×10^{-4}

8.6 Limits

The steady-state model assumes perfect detection apart from the imaging loss, an external reservoir supply, a single objective whose field reaches every site, and one move per refill. It does not schedule the reservoir refresh physically (conveyor timing) and does not include photon scattering. The heating per move is taken from the 1D quantum and 3D classical transport results (Sec. 6), not computed within the steady-state model. The remedy and imaging scans use one seed per case. The tiling result is 1D, with ideal thin lenslets and no surface scatter, and the 24 mm chip edge for a 32×8 block is not designed. The static-potential comparison is a first-order budget with an assumed delivery efficiency and cavity buildup. Recommendations follow directly. For 10^4 sites, one large-field objective or two zones and one 16-chip stack suffice. For 10^5 sites, a 32×4 block, a field of ± 0.9 mm or about 13 zones, and about 85% loading, which cuts the reservoir from two thirds of the sites to a quarter and the static power by 40%. For 10^6 sites the transport side is feasible with a block of 128 to 256 engines, but the static potential needs a different source (a cavity lattice or much lower depths), the field needs a lithography-class objective, and the loss rates must improve to reach 10^{-4} defects.

9 Combining stacks and the real collimator

The engine blocks of Sec. 8.4 need 32 to 64 rows, i.e. two to four 16-chip stacks of the kind described in Sec. 4. A polarising beam splitter combines two stacks; we show here how to combine stacks in space with a slotted mirror at the collimator plane, and then repeat the analysis with the real, traced collimator. The second step changes little for the combiner, but it exposes a limit of the whole stacked relay: a cylindrical collimator at NA 0.9 defocuses the light of tweezers away from the row centre, so a row is usable over only half of its 180 µm segment. The record is `docs/megascale.md` (Secs. 11 and 12) and `scripts/m26_stack_combine.py`.

9.1 A slotted comb at the collimator plane

The plane S, formed by the back focal planes of the per-chip collimators, is conjugate to the atoms with magnification $1/D$. Here, $D = 125$ for rows 4 µm apart (Sec. 4). At S, each chip's beam is a stripe of $\pm f_{\mathrm{FAC}}\mathrm{NA}_{\mathrm{FAC}} = \pm 158$ µm on the 0.5 mm chip pitch, so neighbouring beams leave gaps of 184 µm. A mirror with slots, tilted at 45° (a "comb"), placed in these gaps interleaves two stacks built at twice the chip pitch. Stack A passes through the slots; stack B, turned by 90°, is reflected by the teeth. Each beam then sees a window of half-width $a \leq 250$ µm. The comb uses no polarisation, and the downstream relay is the same as for a single stack of all the rows. In addition, it doubles the chip pitch to 1.0 mm, which leaves room for the fibre and electrical access that the stacked relay left open (383 fast actuators per engine, 1,532 lines per four-engine chip).

Our choice of a 45° comb is a compromise between two considerations. First, it keeps the optical path of every row of stack B equal to that of stack A: with B's facets in the plane $y = -z_c$, the path to any plane $z = Z$ is Z for all rows. However, the comb then spans the combined stack height along z, so the rows at the far end meet it farther from S, where the beams have spread. The question is therefore how far from S a window of half-width a can sit.

9.2 What sets the comb distance

We propagate the field of one chip in y from the collimator exit (aperture f_{FAC} before S, sine-condition lens, pupil fill $\beta = 0.7$ µm, $\mathrm{NA}_{\mathrm{FAC}} = 0.9$) to the comb at a distance z from S, apply the window, propagate back to S, and continue to the atoms through the pupil (`picsim/opa/stack.RelayY`, `results/m26_edge.json`). Without a comb this reproduces the earlier stacked-relay values (transmission 0.973, crosstalk at 4 µm 1.09×10^{-4} against 1.06×10^{-4}). Table 35 and Fig. 14 give the penalty for a single tweezer.

Table 35: Comb penalty versus distance z from S, single tweezer, 1D in y, ideal thin collimator with a hard edge, slot window of half-width a (simulated, `results/m26_edge.json`). Loss is the power clipped by the comb (%); peak is relative to no comb; crosstalk (xt) to the neighbouring row at 4 and 3 µm, in units of 10^{-4}.

	$a = 250$ µm				$a = 225$ µm		
z from S	loss	peak	xt 4 µm	xt 3 µm	loss	peak	xt 4 µm
no comb	0	1	1.1	2.9	0	1	1.1
10 mm	$< 0.1\%$	0.998	1.0	2.8	0.06%	0.992	1.1
20 mm	0.25%	0.992	1.5	4.0	0.72%	0.985	1.8
30 mm	0.85%	0.987	2.7	7.1	1.6%	0.981	3.1
45 mm	1.9%	0.973	6.0	17	3.4%	0.956	7.9
60 mm	3.6%	0.953	13	29	5.3%	0.931	16

The hard edge of the collimator is not the problem. Its Fresnel tail spreads into the gap, but it consists of directions steeper than the pupil accepts, and clipping it costs nothing. A Fresnel-tail estimate that ignores the pupil predicts 0.36% loss at 10 mm; the simulated loss there is below

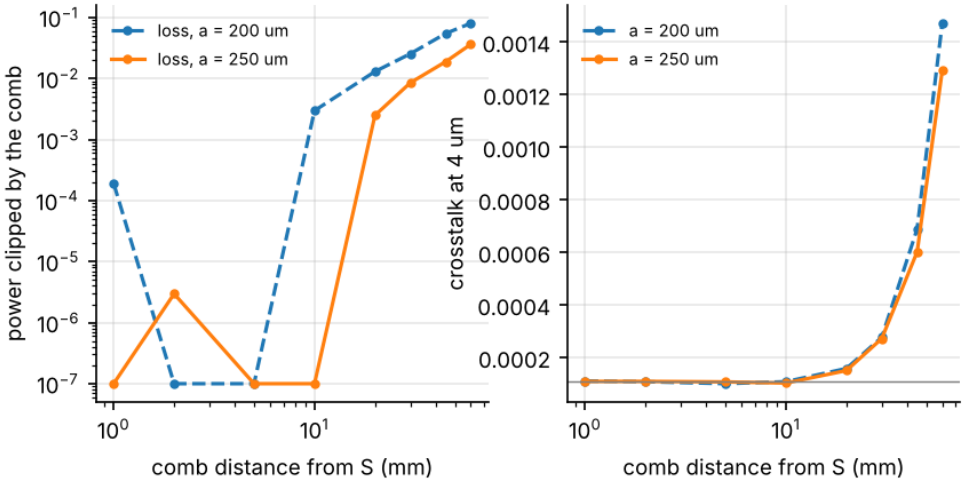


Figure 14: Comb penalty versus distance of the comb from S, single tweezer, 1D wave optics in y (ideal thin sine-condition collimator, $\beta = 0.7$ µm, $\mathrm{NA_{FAC}} = 0.9$, hard edge, NA 0.5 pupil), for slot windows of half-width $a = 200$ µm (dashed) and 250 µm (solid). Left: power clipped by the comb (values below 10^{-7} drawn at 10^{-7}). Right: crosstalk to the neighbouring row at 4 µm; grey line, without comb. The rise sets in near $z_{\max} = (a - f_{\mathrm{FAC}}\mathrm{NA_{FAC}})D/\mathrm{NA}$.

0.1%. A graded collimator edge (raised cosine from 0.8 to 0.9 of the aperture) therefore does not help: it raises the baseline crosstalk to 1.5×10^{-4} and gains no distance. What matters is the light inside the pupil acceptance, which leaves S at half-angle $\mathrm{NA}/D = 4$ mrad. It reaches the window edge at

$$z_{\max} = \frac{(a - f_{\mathrm{FAC}}\mathrm{NA_{FAC}})\, D}{\mathrm{NA}}, \tag{5}$$

which is 23 mm for $a = 250$ µm and 17 mm for $a = 225$ µm. This agrees with the scan: at $z = 20$ mm and $a = 250$ µm the worst row loses 0.25% and the crosstalk at 4 µm rises from 1.1×10^{-4} to 1.5×10^{-4}, while at 30 mm it reaches 2.7×10^{-4}. A single edge, as used for block combining (rows 0 to 15 from one stack and 16 to 31 from the other), behaves similarly. It also shifts the boundary row by 1.7 nm at $z = 20$ mm ($a = 250$ µm) and by 12 nm at $a = 200$ µm, a static offset to calibrate. The rows are mutually incoherent (`docs/row_coherence.md`), so this crosstalk adds in intensity.

9.3 Layouts for 32 to 64 rows

Table 36 lists the layouts (`results/m26_layout.json`). We place the first comb point at $z_c = 7.0$ mm, 0.5 mm (a choice) beyond the x slit plane of the traced x lens and collimator (Sec. 9.4), and the worst row is the one at the far end of the comb. Two 16-chip stacks at 1.0 mm pitch need a comb from 7 to 23 mm, which ends at $z_{\max}$; their worst row loses 0.45% of its power, with peak 0.991 and crosstalk 1.8×10^{-4} at 4 µm. Two such pairs, combined by a polarising beam splitter (PBS), give 64 rows at the same penalty. Combining only by comb with longer stacks puts the far rows beyond $z_{\max}$ (7 to 31 mm for 24-chip stacks, 7 to 39 mm for 32-chip stacks) and costs up to 1.3% and 3.9×10^{-4}. A second comb level (stacks at 2.0 mm, combined pairwise and then again) would put the second comb beyond $z_{\max}$, so we do not use it. With $a = 225$ µm, which is about what a 50 µm tooth plate ($t/(2\sqrt{2}) = 18$ µm per edge) and ±10 µm edge placement leave, the worst rows of the two preferred layouts lose 0.86% and reach 2.0×10^{-4}.

A single taller stack (32 or 64 chips at 0.5 mm) needs no combiner. In first-order optics it only enlarges the y-relay field, to ±8 or ±16 mm at the first telescope lens and 3.7 or 7.4 mrad at the pupil. Its costs are mechanical: 16 to 32 mm of facets flush to about 100 µm, and all the input and output lines at 0.5 mm pitch.

Table 36: Layouts for 32 to 64 rows (simulated and derived, `results/m26_layout.json`; first comb point at $z_c = 7$ mm). Worst row at $a = 250$ µm: power loss, peak and crosstalk at 4 µm (units of 10^{-4}). The last column states whether the objective must accept both polarisations. Bold: preferred layouts.

rows	layout	chips per stack, pitch	comb z range	worst row loss	worst row peak	worst row xt 4 µm	pol.-insensitive objective
32	one stack	32, 0.5 mm			(no comb)		no
32	2 stacks, PBS	16, 0.5 mm			(no comb)		yes
32	**2 stacks, comb**	**16, 1.0 mm**	**7–23 mm**	**0.45%**	**0.991**	**1.8**	**no**
48	4 stacks, comb + PBS	12, 1.0 mm	7–19 mm	0.21%	0.992	1.4	yes
48	2 stacks, comb	24, 1.0 mm	7–31 mm	0.93%	0.987	2.9	no
64	**4 stacks, comb + PBS**	**16, 1.0 mm**	**7–23 mm**	**0.45%**	**0.991**	**1.8**	**yes**
64	2 stacks, comb	32, 1.0 mm	7–39 mm	1.3%	0.984	3.9	no
64	2 stacks, PBS	32, 0.5 mm			(no comb)		yes

2D check. The 1D model leaves out the circular pupil, which couples x and y, and the full rows. We therefore pass the comb-windowed y pupil of both chips through the 2D model of the stacked relay (D1 x response, circular NA 0.5 pupil; `results/m26_twod.json`). Chip m forms a full row of 41 tweezers at 3 µm, chip $m + 1$ forms 11 tweezers offset by 1.5 µm, and one tweezer of chip m then moves by 0.25 µm. Without a comb this reproduces the earlier 2D values. Table 37 shows that 2D does not change the conclusions. At the worst row the full-row crosstalk rises by a factor 1.3, and the effect of a moving tweezer on the next row by a factor 1.55 to 1.65 (to 3.9×10^{-5} in depth at 4 µm, 9.6×10^{-5} at 3 µm). This is still more than an order of magnitude below the in-row figures of D1 (0.11%, 3.5 nm). Tweezers at $x = 0$ and at the row end (60 µm) give the same values to 0.1%, because the comb acts on y alone and the circular pupil couples the two axes only weakly. The 1D single-tweezer crosstalk (1.8×10^{-4} at $z = 23$ mm) slightly overstated the 2D one (1.5×10^{-4}). The block-combining edge moves the boundary row by 7 nm in 2D (peak position), a static offset to calibrate. With coherent rows the comb would raise the static depth change at 4 µm from 2.6% to 2.9%; rows must stay incoherent, as they must without a comb.

Table 37: 2D check of the comb (simulated, `results/m26_twod.json`; both chips windowed, rows incoherent). Crosstalk (xt) and the relative change of the neighbour row's trap depth in units of 10^{-4}; neighbour row at 4 µm, at 3 µm in brackets.

comb	single tweezer peak	single tweezer xt 4 µm	full row xt 4 µm, max over x	neighbour depth change static	neighbour depth change moving tweezer
none	1	1.1	1.8	1.7 (2.0)	0.24 (0.62)
z 15 mm, a 250 µm	0.995	1.1	1.8	1.8 (2.2)	0.29 (0.74)
z 23 mm, a 250 µm (worst row)	0.990	1.5	2.3	2.1 (2.6)	0.39 (0.96)
z 23 mm, a 225 µm	0.982	1.6	2.4	2.2 (2.2)	0.39 (0.80)
z 30 mm, a 250 µm ($> z_{\max}$)	0.984	2.1	2.9	2.5 (3.1)	0.53 (1.2)
single edge, z 23 mm, a 250 µm	0.995	1.3	2.0	1.9 (2.2)	0.34 (0.84)

Tolerances. The tolerances follow from the conjugations and the decentre results of the stacked relay (derived). A comb tilt θ about x deflects stack B by 2θ at S. This shifts its facet image in the pupil by $2\theta F_y$, with $F_y = 2.17$ m, which gives 1% peak loss at $\theta = 0.16$ mrad and 3% at 0.28 mrad. The same tilt moves B's rows by $2\theta z/D$: 0.32 nm per µrad at $z = 20$ mm, and 50 nm at 0.16 mrad. This static offset alternates between rows, because A and B rows interleave, so it must be calibrated once against the static array; keeping the alternating pitch error below 3 nm then needs a tilt stability of about 10 µrad (drift, not alignment; mount stability not yet checked against a source [REF?]). Moving the comb by $\mathrm{d}z$ along z changes B's path by $\mathrm{d}z$, which tolerates

100 µm (chip piston, peak ≥ 0.992); it also shifts the windows by $\mathrm{d}z$ in y, which should stay within about ± 10 µm. Light that misses a tooth leaves at 90° to a dump; light from stack A that hits the back of a tooth is reflected towards stack B and needs an absorbing back face.

Comb roll. A comb edge that is not parallel to x makes the window depend on x. Beyond the slit plane the x relay is telecentric, so each tweezer meets the comb at its slit-plane position $X = f_1 s$, and four engines at 3.0 mm pitch span $X = \pm 5.6$ mm. One tweezer's footprint at the comb is at most ± 0.78 mm (17 mm beyond the slit plane, half-angle $(A/2)/f_1 = 0.046$). Hence, for a roll of 2 mrad, the edge moves by only 3 µm across one tweezer, against the tens of µm over which the result changes, and we treat each tweezer with its local window $a(X)$. In 2D at $z = 23$ mm the peak falls from 0.989 to 0.972 as a goes from 250 to 205 µm (about 0.04% per µm), while the crosstalk at 4 µm stays at 1.4 to 1.6×10^{-4}. A roll therefore mainly makes the depth vary along the row, by about $\pm 0.4\%$ for a ± 10 µm change of a, which a 1.8 mrad roll produces over ± 5.6 mm. The engines set each tweezer's power in any case, so this is a calibration, not a limit; the same holds for an edge straightness of ± 10 µm over 12 mm.

Thus the blocks of Sec. 8 can be built as follows. (i) For 32 rows, two 16-chip stacks at 1.0 mm pitch and one comb. This needs no polarisation optics, and since alternate rows come from different stacks, the passive orthogonal-polarisation decorrelation of `docs/row_coherence.md` comes for free if the objective accepts it. (ii) For 64 rows, two such comb pairs combined by a PBS. Four 16-chip stacks with four engines per chip at 3.0 mm pitch (12 mm edge) then form the 64 × 4 block, with 256 engines and 98,048 fast actuators. With intermediate steps at 10^6 sites it reaches a time average of 2.9×10^{-4} on 4 × 4 µm and 2.7×10^{-4} on 3 × 4 µm, without the 24 mm chip edge that 32 × 8 needs. The worst rows lose 0.9 to 1.5% of their peak depth to the comb ($a = 250$ / 225 µm), which the per-tweezer power set by the engines recovers. The edge scan is 1D in y and the 2D check uses the same windowed pupil; both take the window of each slot at one distance z, whereas the 45° tilt places the two edges of a 0.5 mm slot 0.25 mm before and after that distance, which changes z by at most 4% at $z_c = 7$ mm. Tooth-edge roughness, scatter from the comb and a lens design of the larger y-relay field are not included.

9.4 The comb with the real collimator

The analysis above used an ideal thin collimator in air. The adopted collimator is a fused-silica Weierstrass block glued to the facet, followed by a 10 µm gap and a 250 µm aplanat of index 1.8 (`docs/collimator.md`). We repeat the comb study with this traced lens (`results/m26_lens.json`). The exit field is built from the traced lens map (exit height, wavefront and amplitude per ray) at a plane 60 µm beyond the last vertex, and S lies at the traced back focal plane, 0.44 mm from the facet. Without a comb this reproduces the wave model of the traced collimator: crosstalk 8.0×10^{-5} at 4 µm against 8.4×10^{-5}, and 3.8×10^{-4} against 4.0×10^{-4} at 3 µm. The real geometry also moves the comb. The traced x lens puts the x slit plane 6.76 mm from the facet, the collimator glass adds 0.15 mm in x ($t(1-1/n)$ per section), and we keep 0.5 mm of clearance; the comb therefore starts 7.0 mm from S, and the far row of a 32-row comb sits at 23.0 mm.

Table 38: Comb penalty with the real (traced) collimator against the ideal thin collimator, slot window $a = 250$ µm, 1D (simulated, `results/m26_lens.json`). Values for $a = 225$ µm are given in the text.

	real collimator			thin collimator		
z from S	loss	peak	xt 4 µm	loss	peak	xt 4 µm
15 mm	5×10^{-4}	0.994	7.6×10^{-5}	5×10^{-4}	0.994	1.2×10^{-4}
20 mm	2.6×10^{-3}	0.991	9.2×10^{-5}	2.5×10^{-3}	0.992	1.5×10^{-4}
23.0 mm (far row)	5.0×10^{-3}	0.991	1.1×10^{-4}	4.5×10^{-3}	0.991	1.8×10^{-4}
30 mm	9.0×10^{-3}	0.988	1.9×10^{-4}	8.5×10^{-3}	0.987	2.7×10^{-4}

As can be seen from Table 38, the immersed facet puts more light near the edge of the beam at S, but loss and peak follow the thin-lens values within 12%, and the crosstalk is lower. With $a = 225$ µm the real collimator gives loss / peak / crosstalk at 4 µm of 3.4×10^{-3} / 0.986 / 7.5×10^{-5} at 15 mm, 7.6×10^{-3} / 0.985 / 9.7×10^{-5} at 20 mm, 9.3×10^{-3} / 0.986 / 1.2×10^{-4} at the far row and 1.7×10^{-2} / 0.983 / 2.0×10^{-4} at 30 mm. Equation (5) therefore still holds. In 2D at the far row ($z = 23.0$ mm, $a = 250$ µm), the full-row crosstalk at 4 µm is 1.6×10^{-4} (1.75×10^{-4} without the comb). A moving tweezer changes the next row by 2.8×10^{-5} in depth at 4 µm (1.2×10^{-5} without the comb) and by 6.9×10^{-5} at 3 µm (3.6×10^{-5}). Hence the comb conclusions hold for the real collimator in the real geometry.

9.5 Skew rays: light that leaves the chip at an angle along x

A tweezer at position x_a along the row is formed by light that leaves the emitters with x direction sine $s_x = x_a/x_{\text{scale}}$. Here, $x_{\text{scale}} = 288$ µm for D1, so s_x reaches 0.31 at the end of a 180 µm segment and 0.355 at the slit edge. The collimator is cylindrical: its surface normals have no x component, so k_x is conserved, and the y-z problem is exactly the 2D trace with every index n replaced by the effective index

$$n_{\text{eff}} = \sqrt{n^2 - s_x^2}. \tag{6}$$

A unit test checks this against 3D vector refraction. All earlier y models (the stacked relay, the traced collimator, and the comb study above) used $s_x = 0$. For $s_x > 0$ the effective index ratios of the lens rise, from 1.45 to 1.50 for the block and from 1.80 to 1.89 for the aplanat at $s_x = 0.355$, and the lens is no longer aplanatic for the facet (`results/m26_skew.json`). Table 39 and Fig. 15 give the result.

Table 39: Tweezer at x_a with the adopted collimator, relative to an ideal collimator at the same s_x (simulated, `results/m26_skew.json`; effective-index trace, downstream relay ideal, no comb; with the comb at the far row the crosstalk changes by a factor 0.9 to 1.6). Peak in 1D (2D in brackets), equivalent facet focus error, residual wavefront after refocusing the facet for this tweezer, and crosstalk at 4 µm.

x_a (s_x)	peak	equivalent facet defocus	residual after refocusing	xt 4 µm
0	1.000 (1.000)	0	0	8.4×10^{-5}
15 µm (0.052)	0.999 (0.997)	−0.27 µm	0.001 waves	2.2×10^{-4}
30 µm (0.104)	0.971 (0.942)	−1.10 µm	0.005 waves	3.0×10^{-3}
45 µm (0.156)	0.794 (0.725)	−2.46 µm	0.014 waves	8.2×10^{-3}
60 µm (0.208)	0.481 (0.440)	−4.36 µm	0.029 waves	6.9×10^{-3}
90 µm (0.312)	0.231	−9.67 µm	0.096 waves	9.0×10^{-3}

The error is almost purely a focus shift of the facet,

$$\Delta z \approx -99\, s_x^2 \text{ µm}. \tag{7}$$

A thin glass lens in air would give $-f(n+1)s_x^2/(2n) = -148\, s_x^2$ µm, the same order. Refocusing the facet for one tweezer restores its peak fully; the residual after refocusing stays below 0.1 waves up to 90 µm. However, all tweezers of a row share the facet and the collimator. The phase error sits at S, which is conjugate to the atoms, so refocusing the atom plane does not help either. In 2D the best atom-plane focus recovers the peak up to $x_a = 30$ µm (1.02), but with a 1.5 µm y-focus shift, outside the ±0.5 µm axial registration of the transport model (Sec. 6), and not beyond. Beyond about 60 µm the 2D spot breaks up (peak 0.11 at 75 µm), because the y pupil is then filled to its rim and the circular pupil cuts it differently for each x pupil column.

What a common focus offset recovers. A single facet offset for the whole row, i.e. the length of the glued block set at fabrication, trades the centre against the ends (`results/m26_skew_offset.json`, first-order focus phase, 1D). With $\Delta z = -1.25$ µm the peak stays at or

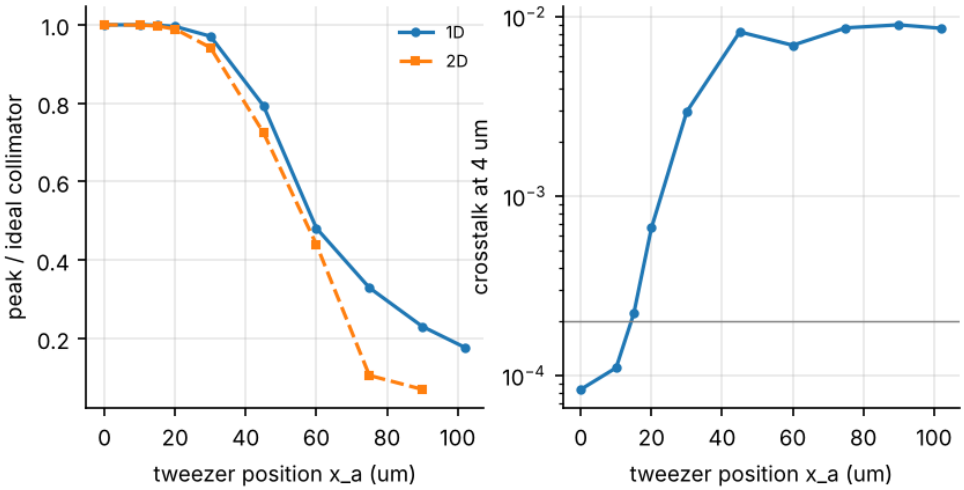


Figure 15: Effect of the x direction sine $s_x = x_a/288$ µm on the adopted cylindrical collimator (glued fused-silica Weierstrass block and 250 µm aplanat of index 1.8, NA 0.9), traced with effective indices $\sqrt{n^2 - s_x^2}$, with an ideal downstream relay, no comb and no facet focus offset. Left: peak of a tweezer at x_a relative to an ideal collimator at the same s_x, 1D (solid) and 2D with the circular NA 0.5 pupil (dashed). Right: crosstalk at 4 µm to the neighbouring row (1D); grey line at 2×10^{-4} for reference.

above 0.96 out to $|x_a| = 45$ µm, and the static crosstalk at 4 µm stays at or below 3.6×10^{-3}. That is within the criteria of `docs/row_coherence.md` (others' depth change below 3×10^{-3} when a tweezer moves), since for incoherent rows the moving-tweezer effect was about 0.13 of the static crosstalk in the earlier stacked-relay model; we did not recompute it for skewed tweezers. Beyond 50 µm no offset helps: the peak is 0.90 at 50 µm and 0.66 at 60 µm, and no offset brings the worst peak over $|x_a| \leq 90$ µm above 0.43.

Consequences. A refractive cylindrical collimator at NA 0.9 serves about half of the 180 µm segment ($|x_a| \leq 45$ µm, about 23 sites at 4 µm), not all of it. Every power and crosstalk figure of the stacked relay, of the traced collimator and of the comb study above holds near the row centre and is optimistic for the outer half. For the large arrays of Sec. 8 this doubles the number of engines per row at fixed coverage (90 µm segments instead of 180 µm). The alternative is a collimator that is insensitive to s_x. Serving $|x_a| \leq 90$ µm within the ± 0.5 µm focus tolerance, with the best common offset, needs $|\Delta z / s_x^2| \leq 10$ µm, ten times less than now.

We see four routes, none of them designed here. (i) An all-reflective collimator in one medium. The law of reflection and a uniform medium are unchanged by the effective-index map, so the wavefront error vanishes for every s_x, and a flat exit face normal to the collimated beam keeps it so (derived). However, such a collimator does not obey the sine condition and needs a folded geometry within the 0.5 mm chip pitch. (ii) A cemented refractive design. Glass-glass interfaces are less sensitive than glass-air ones (relative power change $0.41\, s_x^2$ for a 1.45 to 2.0 interface against $0.84\, s_x^2$ for 1.45 to air, derived), but a factor of two is not enough on its own. (iii) A corrector at the x slit plane, where x encodes s_x. This fails at large s_x: at $x_a = 90$ µm the error acts at S like a y lens of 4 mm focal length (2.9 waves at 158 µm), so the beam has passed a focus before it reaches the slit plane, 6.5 mm later. (iv) A larger vertical facet mode, i.e. a lower facet NA. This helps by less than a factor of two, because the optimal pupil fill, and with it f_{FAC}, grows with the mode. Of these, only the reflective route removes the error in principle. The downstream cylindrical lenses of the y relay see x angles below about 0.05 and are not modelled; the skew trace is exact for cylindrical surfaces, but the downstream relay is taken as ideal, and the common-offset result uses a first-order focus phase in 1D.

10 Discussion and conclusions

10.1 What was shown

Within the models used, the study establishes five points. (i) The passive SiN elements meet the proposal's targets on the 400 nm film, and two targets need restating: the steering update rate should be $\geq$ 200 kHz for moves at 1.2 mK, and position noise must be specified as a spectral density at the trap frequency rather than as an rms value (Section 2). (ii) Independent motion of about 20 tweezers along a row, the integrated equivalent of the acousto-optic deflector, is possible with a sliding-window focal-plane OPA. In its optimised form (design D1: 128 ports, 192 arms and emitters at 1.5 µm pitch), a 1.2 mK tweezer needs about 57 mW once the facet loss and the glued collimator are included (Sections 3 and 4). (iii) Rows of such engines can be stacked and relayed onto a 4 µm lattice. The rows do not need one frequency shifter per chip: common phase noise on the seed (about 3 rad rms, 1 to 2 GHz) with different fibre delays makes them mutually incoherent. Its conversion to intensity noise stays 16 dB inside the heating criterion if no filter narrower than about 100 pm and no pair of reflections with $R_1 R_2 > 1.7 \times 10^{-3}$ lies in the path (Section 4). (iv) Transport with handoff in three dimensions delivers 3,000 of 3,000 simulated atoms when the moves run along lanes between rows, the moving depth is at least three times the hold depth, and the hold depth is about $20\,k_B T$. With 200 µs ramps and 400 µs lane steps the heating per handoff falls to 0.12 µK (about 0.3 µK per move including the x move) at a moving depth of only 0.3 mK, and the axial registration of the two foci must be held to ±0.5 µm (Section 6). (v) At 10^6 sites, a block of 128 engines with fast steps holds a time-averaged defect fraction of 1.8×10^{-4} to 2.0×10^{-4}, set by vacuum and imaging losses rather than by the repair. Slower steps reach a similar floor if the imaging period is raised to match the sweep time of the block (Section 8).

10.2 What is model-dependent

Several results rest on models whose limits matter. The x response of the engine comes from a 2D beam propagation calibrated to an effective index, and the facet transmission from a 2D FDTD. The y axis of the relay is scalar, with ideal relay lenses downstream of the collimator. The transport model is classical in three dimensions, checked quantum mechanically in one dimension per axis. The steady-state simulator draws losses at assumed rates (vacuum lifetime, loss per image and per move) from a single random seed per case.

The most consequential model dependence was found last. Every y-axis result before Section 9 assumed light without an x component. A tweezer at x_a is formed by light with x direction sine $s_x = x_a / x_{\mathrm{scale}}$. The cylindrical collimator, which conserves k_x, then acts with effective indices $\sqrt{n^2 - s_x^2}$ and defocuses the facet by about $-99\,s_x^2$ µm. This keeps the earlier y figures valid near the row centre only. With the best common focus offset a row is usable over $|x_a| \leq 45$ µm, half of the 180 µm segment. The transport, power and crosstalk numbers for tweezers in the outer half of a segment are therefore optimistic until the collimator is redesigned.

In addition, a review of the whole project for errors corrected several earlier results, all reflected in this report: (i) the steady-state simulator skipped block positions after each visit, which had inflated every assembly time and the slow-step floors at 10^6 sites; (ii) the phase-noise option for row decorrelation scaled away the incoherent part of the crosstalk and looked better than fully incoherent rows; (iii) the 2×2 MMI design offset of the switches was treated as random spread; (iv) the field-mean laser power of the first engine was evaluated in the wrong units; (v) the arm-phase aberration of the first star coupler was mis-evaluated by phase unwrapping; (vi) the first x relay lens did not cover the slit edge; and (vii) the position error of the 20-port star-coupler engine depends on the relative port phases, which an earlier text treated as fixed. Some figures still differ between documents for stated reasons: the heating per slow step is 0.12 µK per handoff and about 0.3 µK per move including the x move (the 3D classical run gives 0.16 µK); the greedy scheduler's pass count depends on the set of random loadings (7.8 or 8.05 passes); the steering half-range at 128 channels is 67 µm by the usable-range criterion and 68 µm by the formula $N\lambda/(4\,\mathrm{NA})$; the other

tweezers shift by 1.5 nm in the paraxial engine and up to 3.5 nm in D1; and the time-averaged floor at 10^6 sites with fast steps is 1.8 to 2.0×10^{-4} across runs with different seeds. Defect fractions quoted at the image instant count imaging losses twice and exceed the time average by about 25% at the nominal point.

10.3 What is extrapolated

The large-array chapters extrapolate in three respects. (i) The optics for 10^6 sites assume a corrected field of about ±2.8 mm at NA 0.5, which we did not design. (ii) The static potential for 10^6 sites needs of the order of kilowatts of laser power, from tweezers or from a cavity lattice, which we priced but did not design. (iii) The comb that combines four 16-chip stacks into a 64-row block, and the reflective collimator that would restore full segments, exist only as first-order concepts with wave-optics checks of their key penalties.

10.4 Conclusions

We have shown, in simulation, that SiN circuits can steer independent transport tweezers along rows of a 4 µm lattice, that stacked rows can be made independent without per-chip frequency shifters, and that handoff transport is clean and cold with slow steps and lanes between rows. The main limitation is the cylindrical collimator, which in its present refractive form serves only the central half of each engine's segment. The next step is the design of a collimator insensitive to the x angle, for example all-reflective in one medium, followed by measurements on fabricated D1 engines of the quantities the models assume: phase accuracy and update rate of the actuators, facet transmission, and the intensity noise of the decorrelated laser chain.

References

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A Parameter ledger

Inputs of every simulation, with their provenance, as recorded in the `inputs` of `results/*.json` and collected in `docs/parameters.md` by `scripts/m7_budget.py`. Rows of type SIM (outputs of one module used as inputs of another) are omitted here (5 rows); they are listed in Volume II. USER: proposal text; SOURCE: read in session; DERIVED: calculation; CHOICE: stated design or sweep choice; RECALLED: not verified, to be checked before use.

value	unit	type	pointer	used in
4.0	um	CHOICE	buried thermal oxide; checked against substrate leakage in scripts/m1_platform.py	m5_heaters
2.5	um	CHOICE	lensed/tapered fibre spot, typical catalogue range 2-3 um	m2_couplers
[0.2, 0.3, 0.4]	um	CHOICE	thickness sweep agreed with PI; 200 nm covers foundry runs	m1_width_sweep
5.60228947368421	V	DERIVED	2.8 V x 1064.4/532 assuming the same coil length and stress-optic coefficient; real value depends on mode overlap	m5_actuators
886.0600542866746	um	DERIVED	R = N p_a p_port / lambda_s (one port per resolvable spot)	m9_E_window
15.0	mm	DERIVED	focal length taken as the 15 mm working distance (order of magnitude)	m9_routes
84000.0	Hz	DERIVED	radial, 1.2 mK, w0 = 1.04 um (m3_update_rate)	m9_routes
5000.0	m/s	RECALLED	AlN surface-acoustic-wave velocity, order of magnitude; verify for the AlN/SiN/SiO2 stack	m9_routes
6.2	um	RECALLED	Corning HI1060 MFD at 1060 nm approx. 6.2 um; verify datasheet	m2_couplers
2.5e-19	m^2/W	RECALLED	Kerr index of LPCVD Si3N4, approx. 2.4-2.5e-19 m^2/W (Ikeda 2008); verify	m1_design_modes
20.0	W/(m K)	RECALLED	LPCVD SiN film, literature spread ~10-30 W/(m K); verify	m5_heaters
5.3	um	RECALLED	PM780-HP MFD at 850 nm approx. 5.3 um; verify datasheet	m2_couplers
20.0	W/(m K)	RECALLED	TiN/NiCr heater film order of magnitude; verify	m5_heaters
3.55	-	RECALLED	crystalline Si near 1.06 um, approx. 3.55 + 0.0001i; verify (Green 2008)	m1_box_leakage
1628000	J/(m^3 K)	RECALLED	density 2200 kg/m^3 x c_p 740 J/(kg K)	m5_heaters
1631000	J/(m^3 K)	RECALLED	density 2330 kg/m^3 x c_p 700 J/(kg K)	m5_heaters
1.38	W/(m K)	RECALLED	fused/thermal silica thermal conductivity; verify	m5_heaters
148.0	W/(m K)	RECALLED	silicon thermal conductivity at 300 K; verify	m5_heaters
(model)	-	SOURCE	Luke et al., Opt. Lett. 40, 4823 (2015), doi:10.1364/OL.40.004823; Sellmeier via refractiveindex.info Si3N4/Luke (340 nm LPCVD film), read 2026-10-31	m1_width_sweep

value	unit	type	pointer	used in
(model)	-	SOURCE	Malitson, JOSA 55, 1205 (1965), doi:10.1364/JOSA.55.001205; via refractiveindex.info SiO2/Malitson, read 2026-10-31	m1_width_sweep
400000.0	Hz	SOURCE	Montifiore et al., arXiv:2601.15695 (abstract, read 2026-10-31): coil MZM 3-dB bandwidth 0.4 MHz	m3_update_rate, m5_actuators
2.8	V	SOURCE	Montifiore et al., arXiv:2601.15695 (abstract, read 2026-10-31): coil MZM V_pi 2.8 V at 532 nm	m5_actuators
2.6e-08	s	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): 10-90% rise 26 +/- 7 ns	m4_dynamics, m5_actuators
200.0	V	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): V_pi 200 V at 1013 nm	m4_dynamics, m5_actuators
74.7	V	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): single-ended V_pi 74.7 +/- 3.7 V at 795 nm	m4_dynamics, m5_actuators
2.45e-05	1/K	SOURCE	arXiv:2606.05673, introduction (values at 193 THz; ~7% dispersion to visible), read 2026-10-31	m1_design_modes
1e-05	1/K	SOURCE	arXiv:2606.05673, introduction (values at 193 THz; ~7% dispersion to visible), read 2026-10-31	m1_design_modes
(model)	-	SOURCE	heating rates: O'Hara et al., arXiv:physics/0003049 (PRL 82, 4204 (1999)), Gamma = pi^2 nu^2 S_I(2 nu), Q_dot = 4 pi^4 M nu^4 S_x(nu); read 2026-11-01	m7_budget
1160.0	a.u.	USER	ANRF/ARG/2026/009937 draft v97, Sec. 3 citing Le Kien et al. EPJD 67, 92 (2013)	m3_update_rate, m7_budget, m8_power, m9_E_power
4.0	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 3: 'about five sites at 4 um pitch'	m3_field_tradeoff, m3_multibeam, m7_budget, m8_final, m8_star_design
1.04	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4 error budget: 1.04 um waist	m3_spot, m3_update_rate, m7_budget, m8_power, m9_E_power
0.01	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4 error budget: <= 10 nm rms	m3_noise, m7_budget
0.0012	K	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): transport tweezers ~1.2 mK	m3_update_rate, m7_budget, m8_power, m9_E_power
1.4	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~1.4 W on chip for 21 tweezers	m1_design_modes, m6_power
10.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~10 dB of chip and OPA loss	m3_loss_budget

value	unit	type	pointer	used in
0.0066	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~6.6 mW per tweezer at the atoms	m3_loss_budget, m7_budget
16	-	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: 16 channels on the 895 nm router	m4_router
1.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: 5-6 MMI stages, ~1 dB	m2_mmi
[1.0, 2.0]	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: ~1-2 dB per facet	m2_couplers
[1e-06, 2e-05]	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: ~1-20 uW per site at the atoms	m4_router, m8_power
0.4	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 7.4: CoE-CPPICS stoichiometric LPCVD SiN up to 400 nm	m2_mmi
50.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Table 1: >= 50 dB at 895 nm	m4_extinction, m4_polarisation
100.0	um	USER	ANRF/ARG/2026/009937 draft v97, Table 1: Strehl >= 0.8 over >= +/-100 um	m3_field_tradeoff
0.5	-	USER	ANRF/ARG/2026/009937 draft v97, Table 1: meta-objective NA >= 0.5	m3_field_tradeoff, m3_multibeam, m3_noise, m3_spot, m7_budget, m8_star_design, m9_E_window
1.064435	um	USER	ANRF/ARG/2026/009937 draft v97, Table 2: side-B tweezers at 1064.4350 nm (45S1/2 double magic)	m1_width_sweep, m3_field_tradeoff
0.8946	um	USER	ANRF/ARG/2026/009937 draft v97: 895 nm Raman addressing (Cs D1 line, 894.6 nm)	m1_width_sweep, m7_budget

B Reproduction map

Each module, its script and its main outputs (from `README.md`). Every script writes `results/<name>.json` with its inputs (with provenance), outputs and the git revision; `make test` runs the test suite.

Module	Script	Main outputs
1 Platform	`scripts/m1_platform.py`	design widths, n_eff, n_g, dn/dT, bend loss, substrate leakage, coupling length
2 Couplers, MMI	`scripts/m2_couplers_mmi.py`	taper overlap and tolerance; MMI loss, bandwidth, split-ratio statistics
3 OPA	`scripts/m3_opa.py`, `scripts/m3_grating_fdtd.py`	range vs channels, pitch, gratings (FDTD), multi-tweezer synthesis, noise, update rate, loss
4 Switches, router	`scripts/m4_switches.py`	extinction Monte Carlo, polarisation, actuator dynamics, 16-channel router
5 Heaters	`scripts/m5_heaters.py`	P_pi, rise time, thermal crosstalk; actuator comparison
6 Power	`scripts/m6_power.py`	self-heating, facet intensity, Kerr phase

Module	Script	Main outputs
7 Budget	`scripts/m7_budget.py`	error-budget links, `docs/parameters.md`
8 Twenty tweezers	`scripts/m8_twenty.py`	star-coupler OPA with 20 switched ports vs phase synthesis; `docs/twenty_tweezers.md`
9 Independent steering	`scripts/m9_independent.py`	sliding-window focal-plane OPA vs AOD-emulation routes; `docs/independent_steering.md`
10 Star-coupler BPM	`scripts/m10_fpr.py`, `scripts/m10_fpr_meep.py`	calibrated effective index, wide-angle BPM of the dense-port coupler, FDTD check, least-squares synthesis; `docs/fpr_design.md`
11 Geometry optimisation	`scripts/m11_optimise.py`	oversampling, emitter facet, slab widths, window width
12 Scaling	`scripts/m12_scaling.py`	tweezers per row (packing), aberration vs size, scaling table; `docs/scaling.md`
13 Stacked relay	`scripts/m13_stack.py`	stacked-chip anamorphic relay: facet far field, y fill, 2D row crosstalk (coherent / incoherent), collimator tolerances, layout; `docs/stacked_relay.md`
14 Collimator	`scripts/m14_collimator.py`, `scripts/m14_facet_meep.py`	ray-traced fast-axis collimators (Weierstrass + aplanat adopted), facet FDTD, tolerances, form errors, first x lens vs calibration, power and frequency plan; `docs/collimator.md`
15 Row coherence	`scripts/m15_coherence.py`	independent rows without per-chip AOMs: common phase-noise modulation with fibre delays, alternating polarisation (vector focusing), passive apodiser; `docs/row_coherence.md`
16 PM-to-AM	`scripts/m16_pm_am.py`	intensity noise from filtering the phase-noise-modulated light (gain tilt, band-pass, multipath, PM fibre, dispersion, chain); `docs/pm_am.md`
17 Scaling II	`scripts/m17_scale.py`	transport depth vs move time (classical Monte Carlo), row select, configuration table; `docs/scaling_beyond.md`
18 3D transport	`scripts/m18_transport3d.py`	3D classical transport with handoff between the static array and the moving tweezer (realistic 3D field), lanes, depth ratios, temperature, bystanders, tolerances; `docs/transport3d.md`
19 Quantum check	`scripts/m19_quantum.py`	1D split-step propagation of handoff and move from the ground state, Wigner-classical comparison; `docs/rearrangement.md`
20 Rearrangement	`picsim/rearrange.py`, `scripts/m20_rearrange.py`	greedy pass scheduler for a stack with row select and lanes: passes, cycle time, moves per atom, defects; `docs/rearrangement.md`
21 Repair	`scripts/m21_repair.py`	stochastic losses, image-and-repair rounds, scheduler pass-selection modes; `docs/repair.md`
22 Megascale budget	`scripts/m22_megascale.py`	1e4-1e6 sites at 3-4 um: field, zones, static power, engine blocks, x-tiling optics, loss rates; `docs/megascale.md`

Module	Script	Main outputs
23 Steady state	`picsim/megasort.py`, `scripts/m23_steady.py`	assembly and steady-state refilling of 1e4-1e6 sites with engine blocks, reservoir bands and losses; `docs/megascale.md`
24 Tiling wave optics	`scripts/m24_tiling_wave.py`	ghost light from one engine in its neighbour's slit zone (shared slit plane): wave propagation through the lenslet with pupil acceptance, ray trace through neighbouring lenslets, engine-pitch scan; `docs/megascale.md`
25 Imaging cadence	`scripts/m25_imaging.py`	steady-state defect fraction (at image, before image, time average) against imaging period, loss per image, imaging time and vacuum lifetime at 1e5 and 1e6 sites; `docs/megascale.md`
26 Stack combining	`scripts/m26_stack_combine.py`	interleaving chip stacks with a slotted 45 deg comb mirror at the collimator plane: 1D wave optics of the window vs distance, layouts for 32-64 rows with PBS, tilt and path tolerances, 2D check; the real (traced) collimator, and its detuning for tweezers away from the row centre (skew rays, effective indices); `docs/megascale.md`

path	content
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`picsim/params.py`	every physical input, with provenance (USER, SOURCE, DERIVED, SIM, CHOICE, RECALLED)
`picsim/materials.py`	Sellmeier models: LPCVD Si3N4 (Luke 2015), SiO2 (Malitson 1965), AlN
`picsim/modes.py`	vector FEM mode solver wrapper (femwell): straight, bent, coupled and leaky guides
`picsim/slab.py`	analytic and finite-difference slab modes
`picsim/couplers.py`	inverse-taper edge couplers (mode overlap with Gaussian fibre modes)
`picsim/mmi.py`	MMI couplers by effective index + eigenmode expansion
`picsim/circuits.py`	MZI transfer matrices, Monte Carlo extinction, actuator dynamics, routing tree
`picsim/heaters.py`	2D thermal FEM of heaters: P_pi, rise time, crosstalk
`picsim/opa/`	OPA near field, Fourier-plane slit, pupil relay, multi-tweezer synthesis, phase noise, star coupler
`picsim/transport.py`	motional excitation of a moved atom (harmonic trap, staircase trajectories)
`picsim/budget.py`	error-budget links from optical specs to atom metrics
`picsim/bpm.py`	2D wide-angle FD-BPM and the calibrated lateral effective-index model
`scripts/m*_*.py`	one script per module; each writes `results/<name>.json` (+ `.png`)
`tests/`	unit tests against analytic limits

C Open placeholders and recalled values

Values marked [VALUE?] or [REF?] and values recalled but not verified, collected from the verification notes of each document. The full provenance ledgers (every quoted value with its type and pointer) are at the end of each chapter of Volume II.

`docs/collimator.md`

- Open placeholders: [REF?] UV-adhesive shrinkage and creep at the 0.1 um level; [REF?] placement accuracy of two-photon-printed micro-optics on chip facets; [VALUE?] CW power handling of acousto-optic shifters at 3 W, 1064 nm; [VALUE?] available single-frequency 1064 nm power at 40 to 55 W.
- RECALLED: CTE Si 2.6e-6/K, fused silica 0.55e-6/K (Sec. 7); verify

`docs/findings.md`

- **Recalled, not verified (check before use).** Weighted Gerchberg-Saxton: R. Di Leonardo et al., Opt. Express 15, 1913 (2007) (publisher page blocked); fibre MFDs (HI1060 6.2 um, PM780 5.3 um); Si index 3.55; Kerr index 2.5e-19 m^2/W; thermal conductivities and heat capacities; Cs mass.
- **Open placeholders.** [VALUE?] CW facet-damage threshold of SiN/SiO2 at 1 um; [VALUE?] PZT hysteresis magnitude; [VALUE?] AlN actuator f0 and Q at 894.6 nm; [VALUE?] measured V_pi at 894.6 nm; [REF?] verified citation for weighted Gerchberg-Saxton.

`docs/independent_steering.md`

- Open placeholders: [REF?] AOD multi-tone independence and intermodulation; [VALUE?] PZT hysteresis and long-term drift after closed-loop calibration; [VALUE?] dense-port FPR excess loss from EME/BPM.
- RECALLED: v_SAW ~5 km/s (Sec. 3); verify for an AlN/SiN/SiO2 stack

`docs/megascale.md`

- RECALLED: refractive NA 0.5 objective fields (Sec. 2); verify

`docs/parameters.md`

- RECALLED: 5000.0 (m/s); AlN surface-acoustic-wave velocity, order of magnitude; verify for the AlN/SiN/SiO2 stack
- RECALLED: 6.2 (um); Corning HI1060 MFD at 1060 nm approx. 6.2 um; verify datasheet
- RECALLED: 2.5e-19 (m^2/W); Kerr index of LPCVD Si3N4, approx. 2.4-2.5e-19 m^2/W (Ikeda 2008); verify
- RECALLED: 20.0 (W/(m K)); LPCVD SiN film, literature spread ~10-30 W/(m K); verify
- RECALLED: 5.3 (um); PM780-HP MFD at 850 nm approx. 5.3 um; verify datasheet
- RECALLED: 20.0 (W/(m K)); TiN/NiCr heater film order of magnitude; verify
- RECALLED: 3.55 (-); crystalline Si near 1.06 um, approx. 3.55 + 0.0001i; verify (Green 2008)
- RECALLED: 1628000 (J/(m^3 K)); density 2200 kg/m^3 x c_p 740 J/(kg K)
- RECALLED: 1631000 (J/(m^3 K)); density 2330 kg/m^3 x c_p 700 J/(kg K)
- RECALLED: 1.38 (W/(m K)); fused/thermal silica thermal conductivity; verify
- RECALLED: 148.0 (W/(m K)); silicon thermal conductivity at 300 K; verify

`docs/pm_am.md`

- RECALLED: PM-fibre DGD 1.2 ps/m, beta2 20 psˆ2/km (Sec. 2); verify for the chosen fibre

docs/row_coherence.md

- Open placeholders: [REF?] SBS suppression by phase modulation in high-power 1064 nm fibre amplifiers; [VALUE?] AOM insertion loss at 3 W; [VALUE?] second-order differential light shift of the double-magic condition over a 5 pm spectrum; vector polarisability of Cs at 1064 nm from a primary source.
- RECALLED: vector-to-scalar ratio 0.09 g_F m_F (Sec. 4); far-detuned formula (Grimm et al., arXiv:physics/9902072), not read this session; counter-rotating terms neglected

docs/scaling_beyond.md

- RECALLED: galvanometer repeatability, objective fields (Sec. 4, 5); verify

docs/stacked_relay.md

- Open placeholders: [REF?] achievable wavefront error of NA 0.9 micro-collimators (commercial acylinders, two-photon-printed lenses); [REF?] alignment accuracy and long-term stability of actively aligned, bonded collimators; [VALUE?] power handling of fibre-coupled or free-space frequency shifters at 2 to 3 W, 1064 nm.
- RECALLED: Si CTE 2.6e-6/K (Sec. 6); verify

docs/twenty_tweezers.md

- Open placeholders: [VALUE?] PZT shifter footprint and hysteresis for 128 arms; [VALUE?] AlN resonance and Q at 894.6 nm; [VALUE?] measured arm phase errors after fabrication (calibration range).